



Making Figures Accessible

May 2024 v.2 | Journals Edition

Contents

Resources	03
-----------	----

The Principles

Use of Colour	04
Colour Contrast	05
Colour Palette	09
Text Accessibility	10

In Practice

Line Graphs	12
Grouped Bar Graphs	18
Scatter Plots	23
Pie Charts	28
Diagrams	33
Alt Text	39

Quick Links to Cheat Sheets

Line Graphs	16
Grouped Bar Graphs	22
Scatter Plots	27
Pie Charts	32
Diagrams	38

This guide seeks to help all those involved with creating and publishing content to improve the accessibility of any figures produced. Individual projects may still face constraints that mean that not all guidance can be adhered to.

For example, the space available within a print format may mean the size of the labelling cannot be as large as is recommended. As guidance, research, and understanding of accessibility develops, we also intend to update and improve this guidance.

If you have any feedback on these guidelines, or suggestions for improvements, please get in touch with us at the address below:

Charlotte.Lloyd@oup.com
Head of Design, Academic

Useful Links

Accessible Fonts

- [FS Me Regular / FontSmith website](#)
- [Source Sans Pro Regular / Google Fonts](#)
- [Open Sans Regular / Google Fonts](#)

Colour Blindness Simulator

[Coblis](#)

Colour Contrast Checkers

- [WebAIM Colour Contrast Checker](#)
- [Snook checker](#)

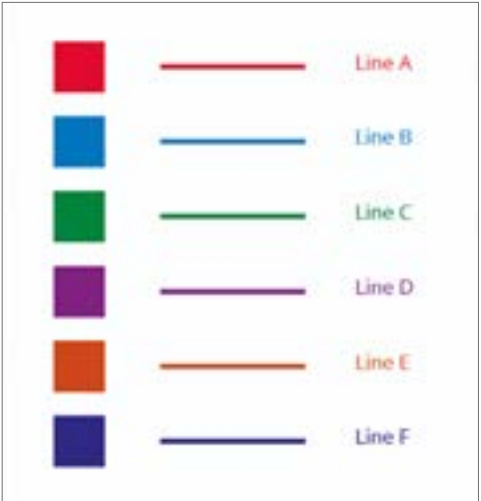
Web Content Accessibility Guidelines (WCAG)

- [Introduction and Overview](#)
- [Use of Colour](#)
- [Minimum Colour Contrast Requirements](#)

Does making colours accessible just mean avoiding red and green?

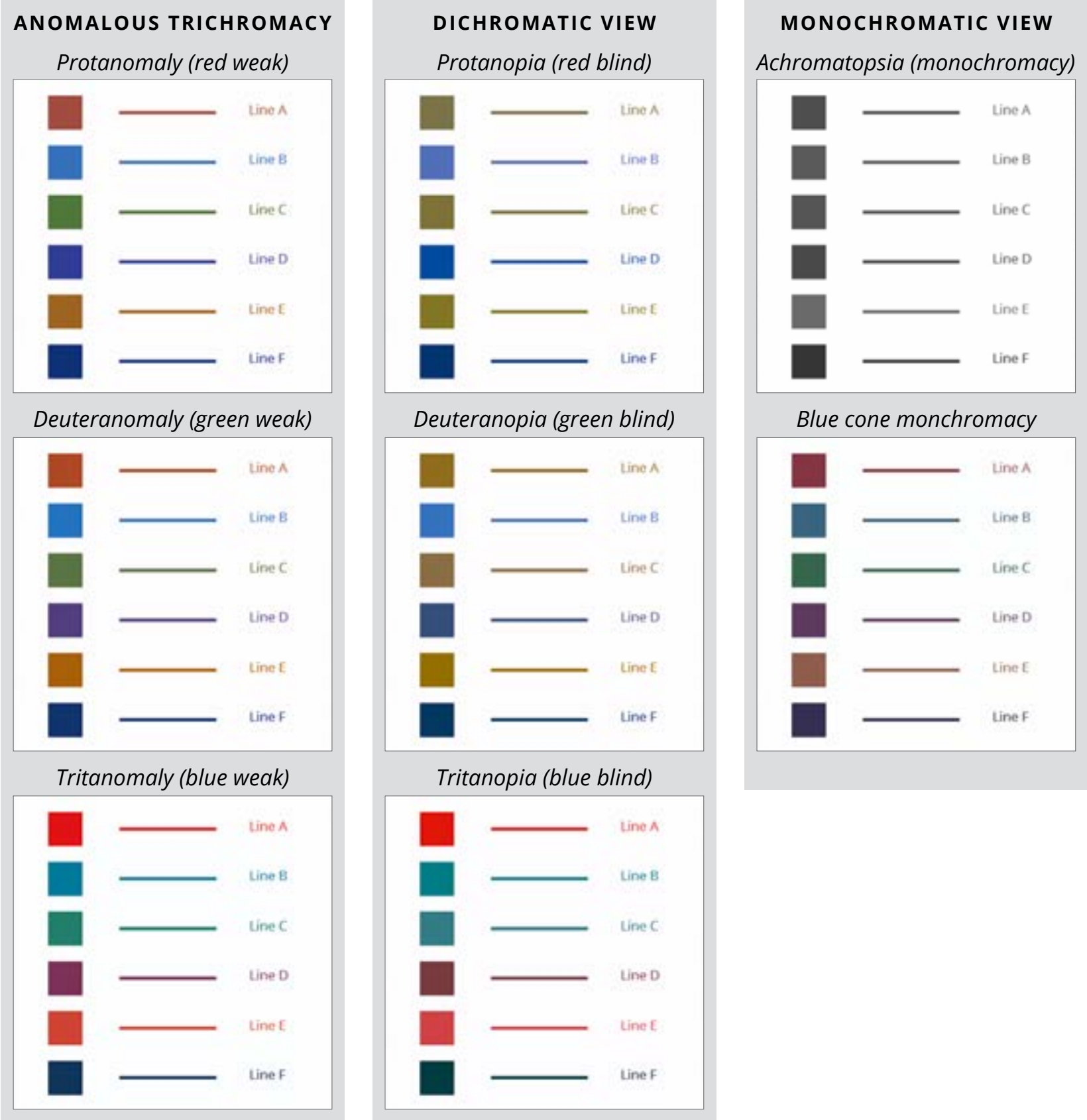
No—there are many forms of colour blindness, meaning that colour can be perceived in many different ways.

For example, a typical colour palette is shown below, and the images on the right demonstrate how this palette may be perceived by readers with various forms of colour blindness.



If a manuscript refers to the ‘red line’ or ‘red area’ in a figure, a reader with protanopia would not be able to perceive which line/area is being referred to. Therefore, in order to be accessible:

- meaning should not be communicated only through colour – avoid referring to colour in the manuscript as the only cue to meaning
- use other cues such as text labels, varying pattern and/or shapes etc



What is colour contrast?

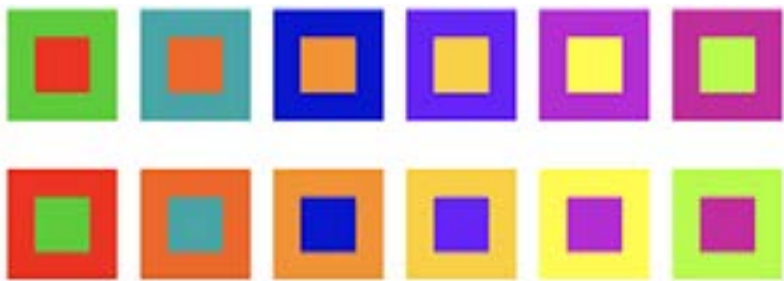
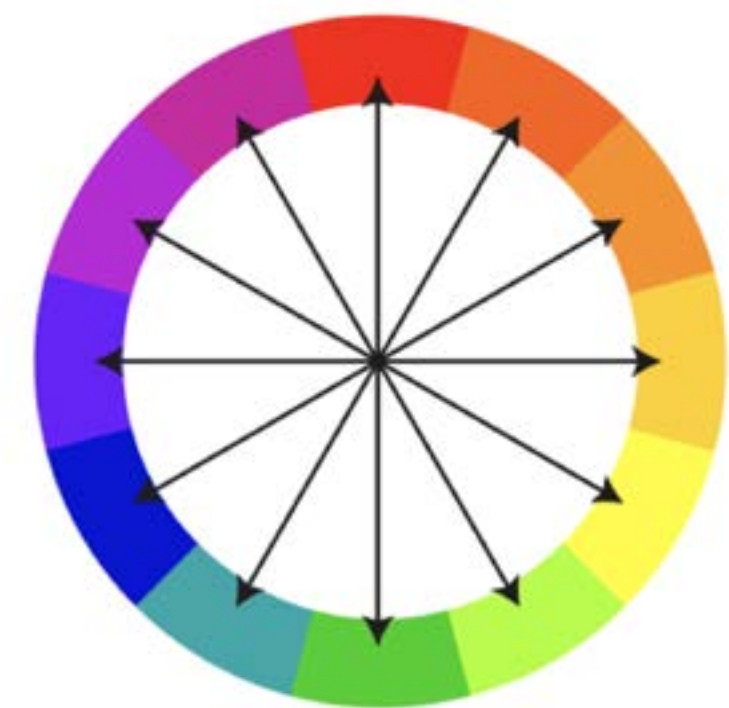
Contrasting colours are colours that differ from one another. Levels of contrast vary from high to low, depending on their position on the colour wheel. For example, colours that are directly opposite one another on the colour wheel have the highest contrast possible, while colours next to one another have a low contrast.

Partial sight, aging, and congenital colour deficits all produce changes in perception that reduce the visual effectiveness of certain colour combinations. Two colours that contrast sharply to someone with normal vision may be far less distinguishable to someone with a visual disorder. It is important to appreciate that it is the contrast of colours against another that makes them more or less discernible rather than the individual colours themselves.

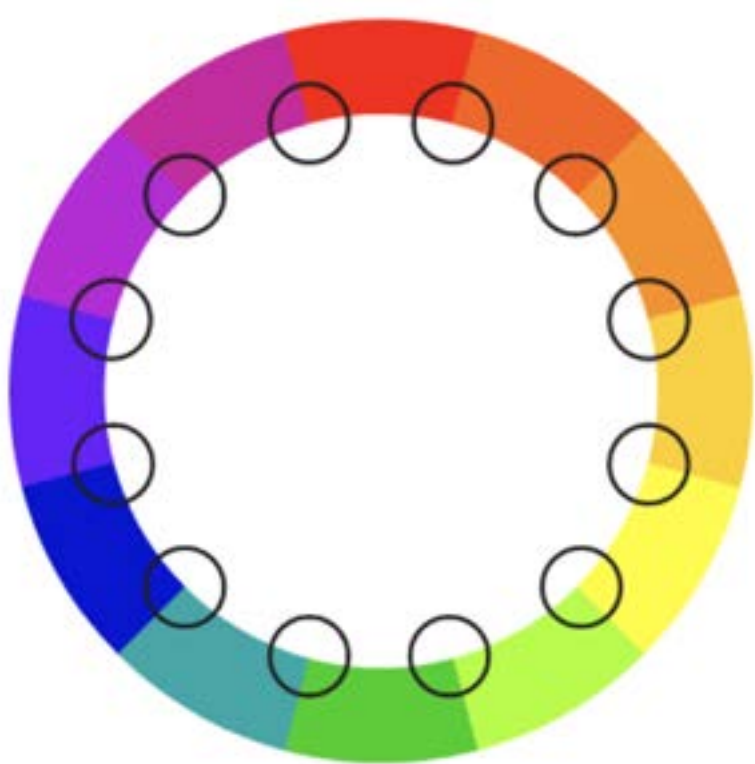
Here are three rules for making effective colour choices:

- Exaggerate lightness differences between foreground and background colours
- Avoid contrasting hues from adjacent parts of the circle
- Use a colour contrast checker to validate the success of the colour combinations within the Minimum (AA) WCAG Contrast and Colour Requirements

HIGH CONTRAST COLOURS

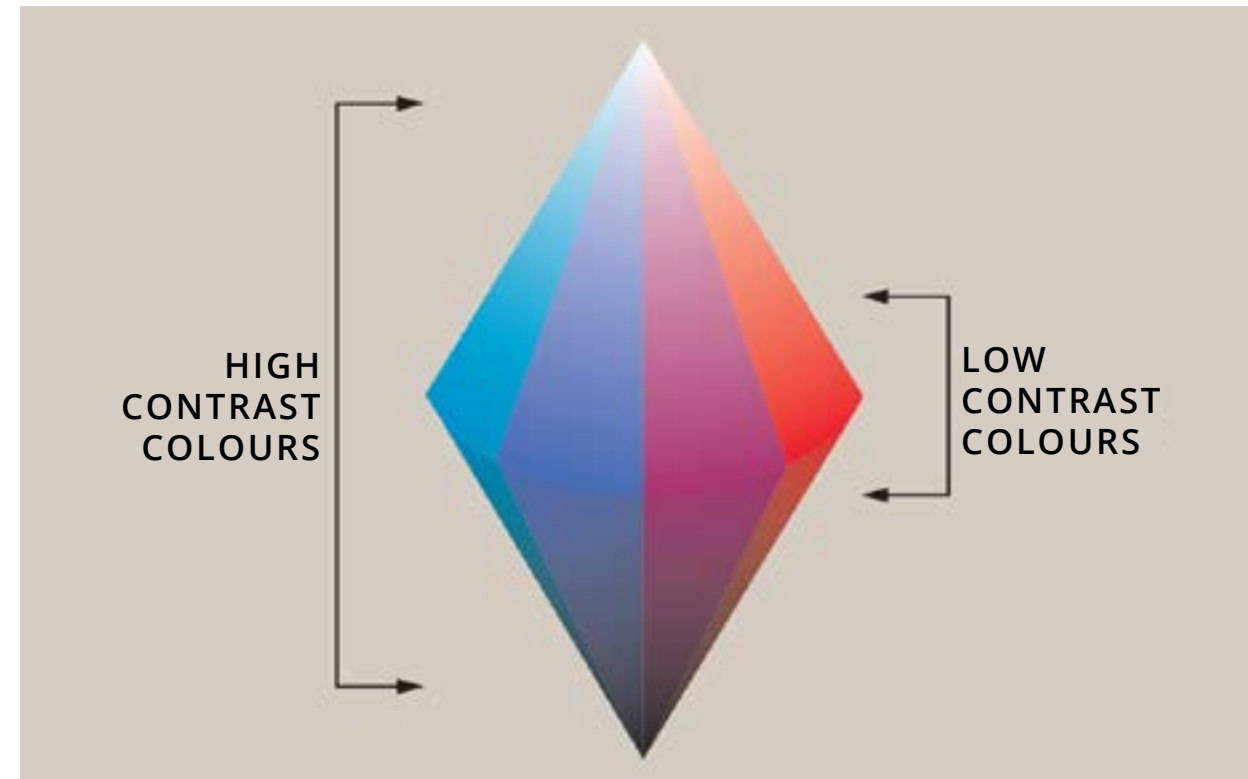


LOW CONTRAST COLOURS

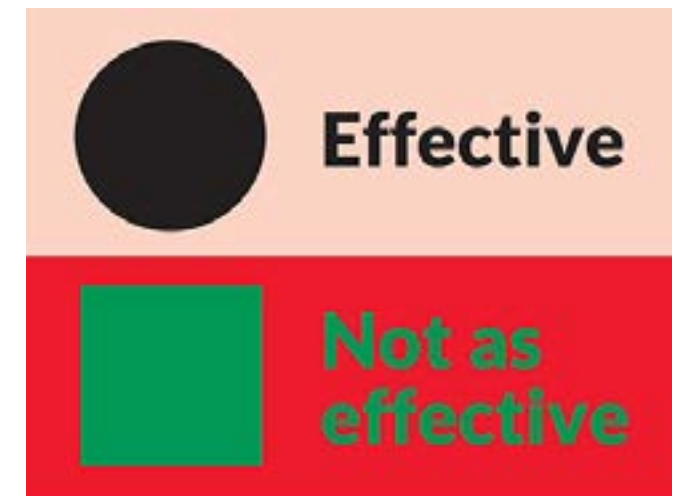


1. Exaggerate lightness differences between foreground and background colours

Don't assume that the lightness you perceive will be the same as the lightness perceived by people with colour deficits. If you lighten your light colours and darken your dark colours, you will increase the visual accessibility of your design.

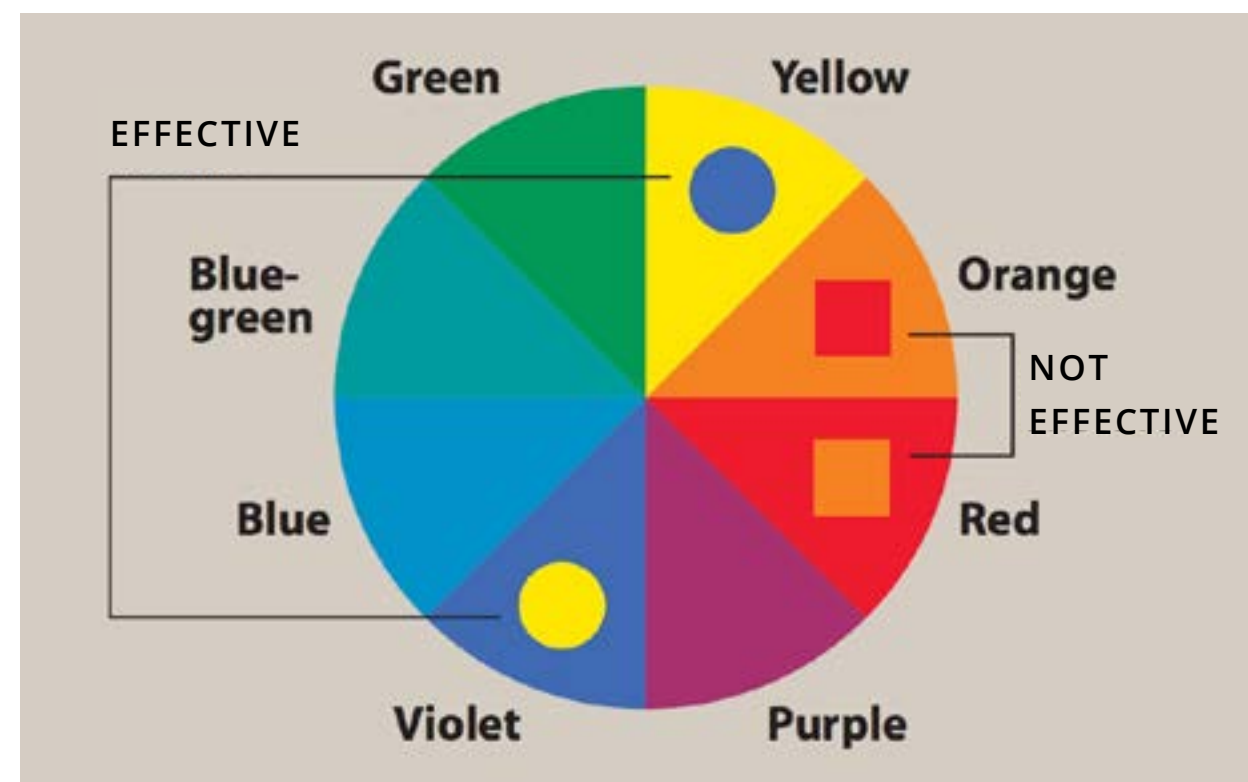


Example 1



2. Avoid using colours of similar lightness adjacent to one another, even if they differ in saturation or hue

Hue enables us to identify basic colours, such as blue, green, yellow, red and purple. People with normal colour vision report that hues follow a natural sequence based on their similarity to one another. With most colour deficits, the ability to discriminate between colours on the basis of hue is diminished, and it is difficult to discriminate between colours of similar hue.



Example 2



Understanding the WCAG “Contrast Ratio”

OUP recommends that colour used in our publications meet the WCAG minimum (AA) contrast ratios. In WCAG, contrast is a measure of the difference in perceived “luminance” or brightness between two colours. This brightness difference is expressed as a ratio ranging from 1:1 (e.g. white on white) to 21:1 (e.g., black on a white). There are different requirements for non-text and text contrast.

WCAG 2.1 SUCCESS CRITERION 1.4.11 NON-TEXT CONTRAST

The visual presentation of Graphical (non-text) Objects should have a contrast ratio of at least **3:1** against adjacent colour(s). If a graphic is needed to understand the content then it should be perceivable by people with low vision or other impairments without the need for contrast-enhancing assistive technology.

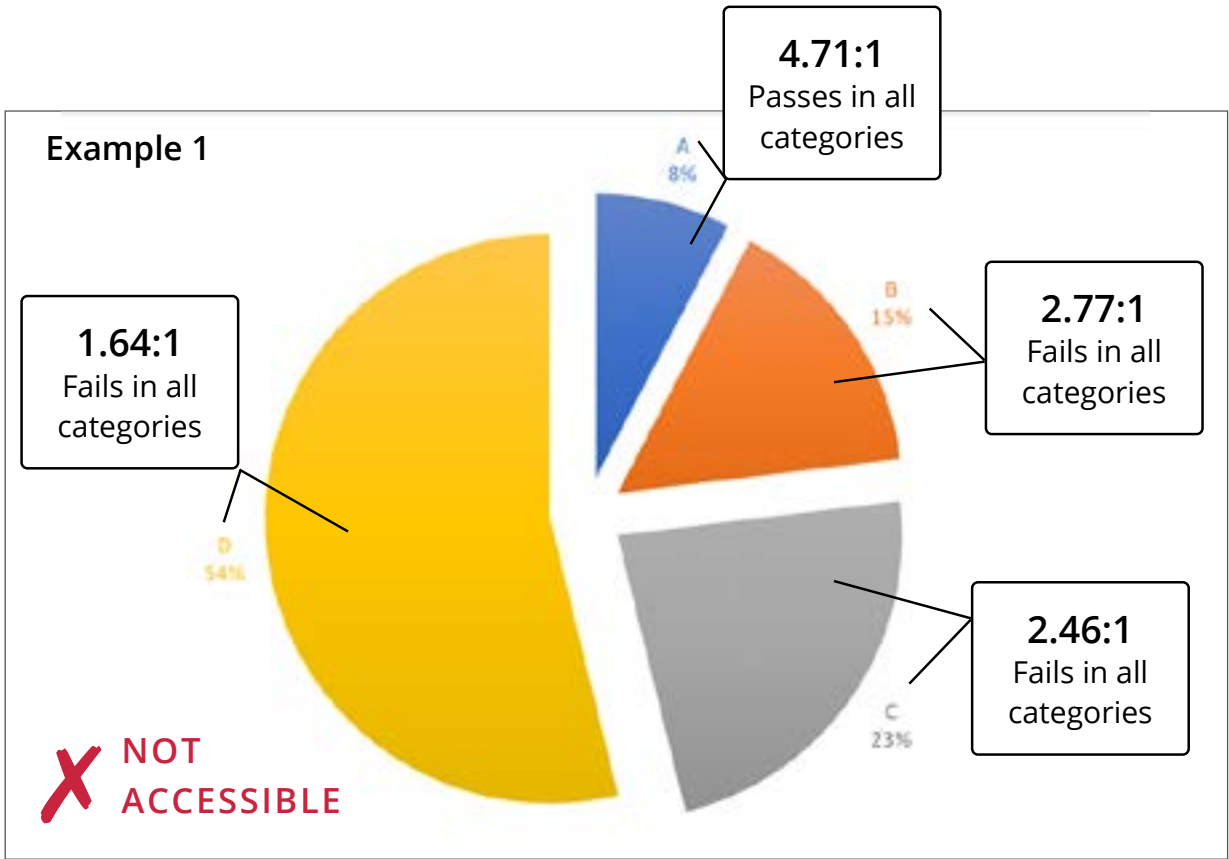
WCAG 2.1 SUCCESS CRITERION 1.4.11 TEXT CONTRAST

The presentation of text should have a contrast ratio of at least **4.5:1** for normal text and **3:1** for large text. Large text is defined as 14 point and bold or larger, or 18 point regular or larger. To give a frame of reference, on a white background:

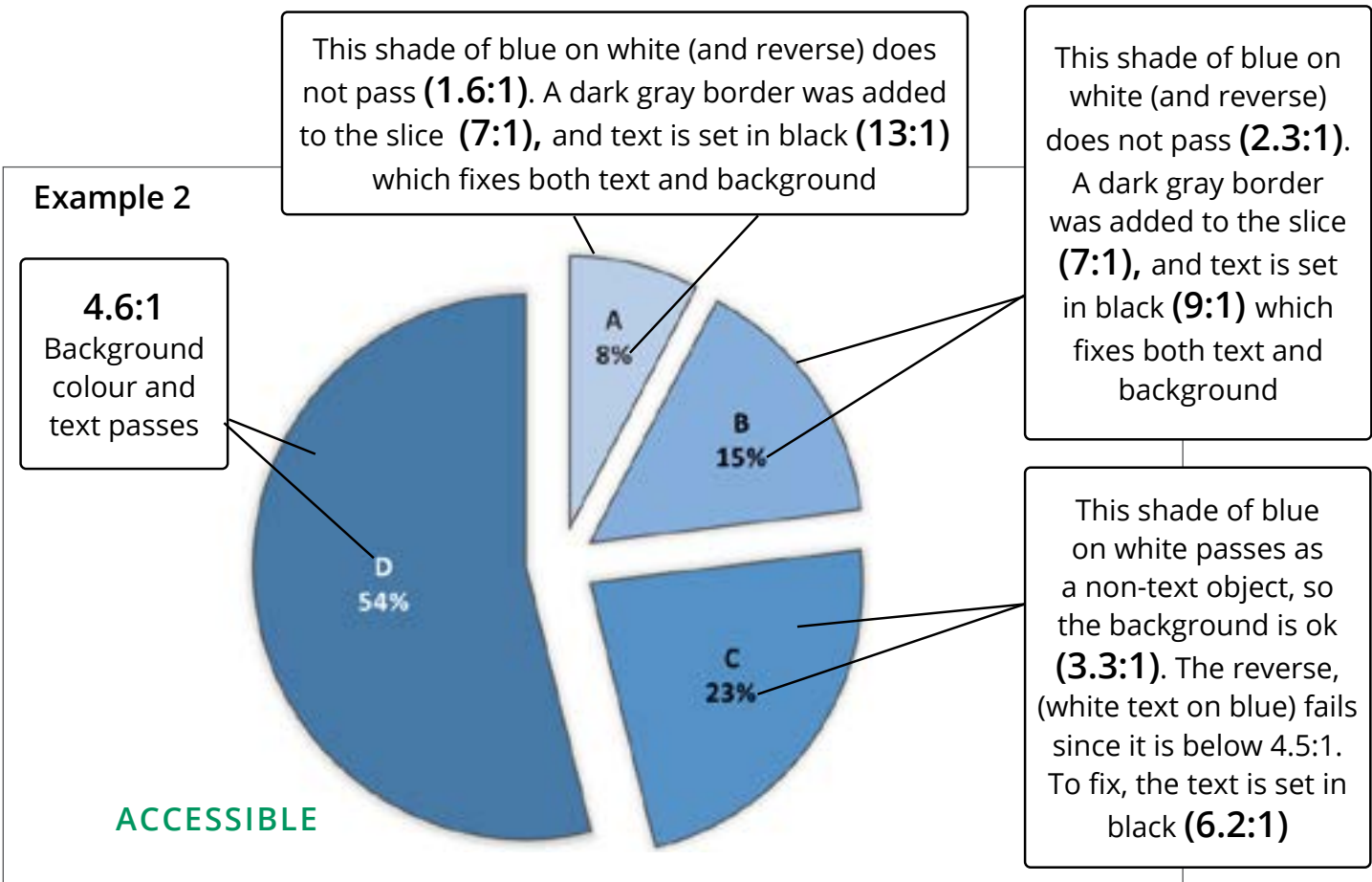
Pure red (#FF0000) has a ratio of 4:1:
SAMPLE TEXT AT 14 PT

Pure green (#00FF00) has a very low ratio of 1.4:1
SAMPLE TEXT AT 14 PT

Pure blue (#0000FF) has a contrast ratio of 8.6:1
SAMPLE TEXT AT 14 PT



In **Example 1**, only the blue on white passes the WCAG requirements for colour contrast. The other colours do not pass both with text or non-text objects. See below for one way this could be fixed.



In **Example 2**, variations in layout are added (a dark rule around light non-text background colours) as well as alternating black and white text as needed. It is okay to treat labels in different ways to ensure they meet contrast requirements.

To create a uniform look, the gray border is applied to all slices, even though it isn't required on Slices C and D.

Using an online Contrast Checker

There are many resources online that can help you to validate the success of colour combinations, such as [WebAim.org](#) (See also the [Useful Links](#) section of this document). Enter a foreground and background colour in RGB hexadecimal format (e.g., #F7DA39). The Lightness slider can be used to adjust the selected colour.

Based on the values entered, you will be told if you pass or fail a Colour Contrast check using the WCAG requirements for:

- Normal Text (4.5:1)
- Large Text (3:1)
- Graphical Objects/non-text (3:1)

In the examples, the yellow on white test has failed in all categories, as the contrast is well below the 3:1 or 4.5:1 requirements. This combination should not be used for either text or non-text objects.

To the right of that example, the lightness in the yellow has been decreased, so it appears more as a dark gold. The gold on white passes in all categories, as the contrast is above 3:1 and 4.5:1. This combination would work well for both text or non-text objects.

Foreground Color

#F7DA39

Lightness

Background Color

#FFFFFF

Lightness

Contrast Ratio

1.39:1

[permalink](#)

Normal Text

WCAG AA: Fail

WCAG AAA: Fail

The five boxing wizards jump quickly.

Large Text

WCAG AA: Fail

WCAG AAA: Fail

The five boxing wizards jump quickly.

Graphical Objects and User Interface Components

WCAG AA: Fail

Text Input

Foreground Color

#7F6D06

Lightness

Background Color

#FFFFFF

Lightness

Contrast Ratio

5.13:1

[permalink](#)

Normal Text

WCAG AA: Pass

WCAG AAA: Fail

The five boxing wizards jump quickly.

Large Text

WCAG AA: Pass

WCAG AAA: Pass

The five boxing wizards jump quickly.

Graphical Objects and User Interface Components

WCAG AA: Pass

Text Input

X NOT ACCESSIBLE

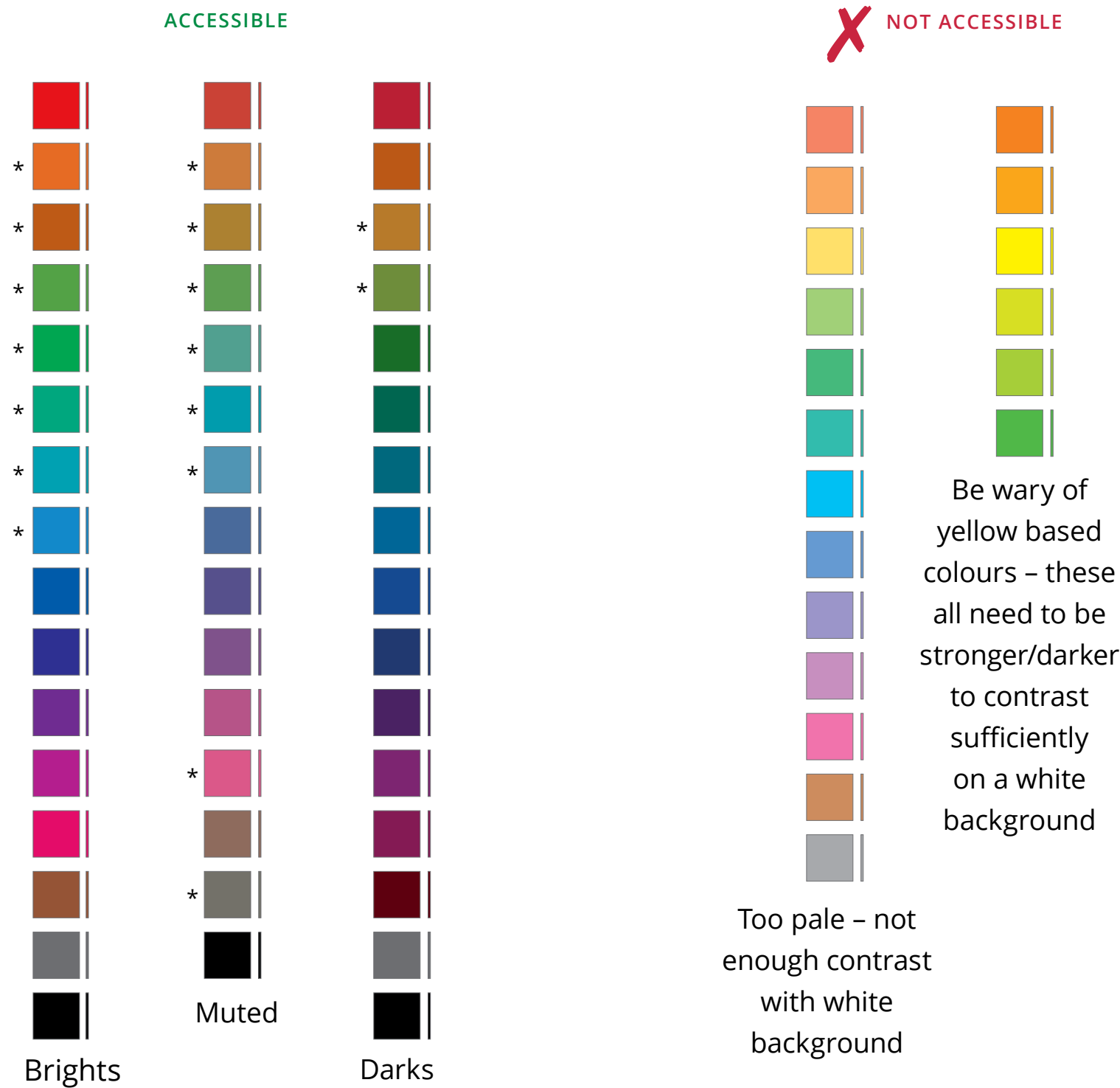
ACCESSIBLE

1. Example colours tested for non-text contrast

The colour boxes to the right are colours tested for non-text contrast (i.e. graphical objects) against white to AA Accessibility standard.

- Avoid using colour as the only cue to convey meaning, and do not refer to the object by colour
- Colours should be strong and contrast well with surrounding areas, and each other when being used to differentiate
- Where tints are used, use an outline to define the shape so that it contrasts with the surrounding background
- Do not use colour for typography unless absolutely necessary – it is best to use black text on a white background (not all colours in the figure specification may be suitable for text use – those with an * on the right accessible palettes are not)
- Yellow should also be avoided for graphical objects – it needs to be dark enough to contrast with the background, meaning it becomes more gold, like the examples on the right

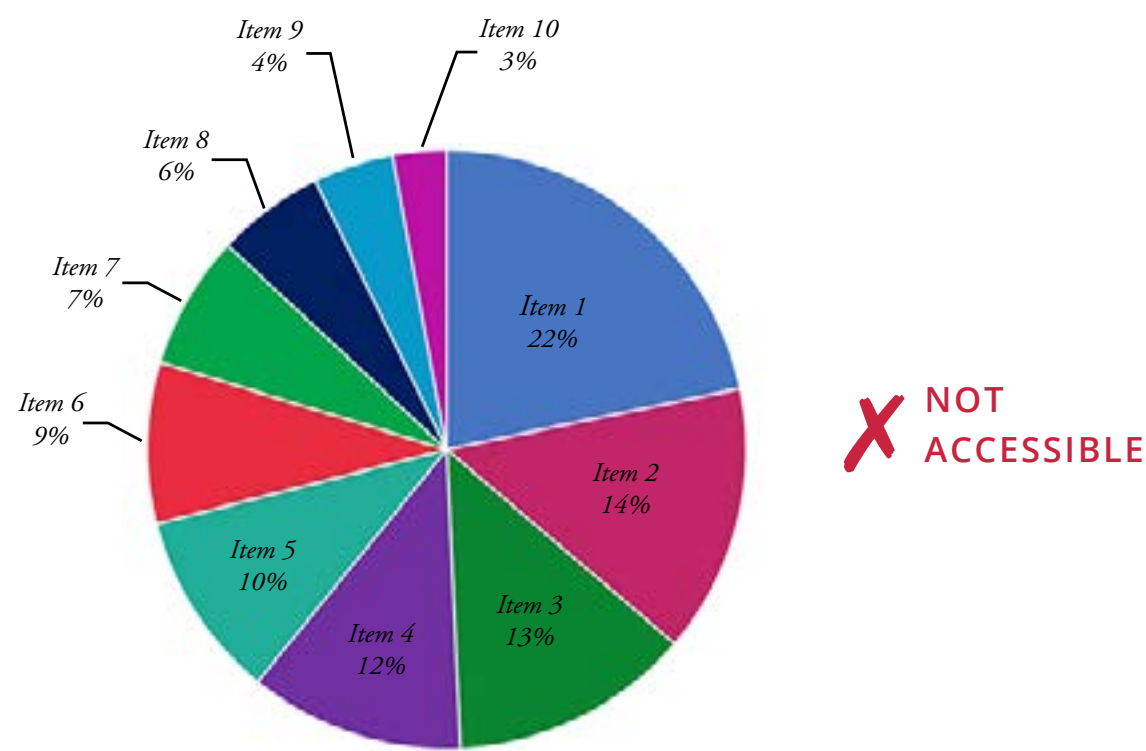
Example colours tested for non-text contrast



What do we need to consider for text accessibility?

Font type and presentation has a significant impact on readability of people with dyslexia. Studies have shown that sans serif, roman and monospaced font types increased the reading performance of dyslexic readers. A combination of principles regarding typography, readability, legibility, and colour will have a significant effect on text accessibility. Therefore, in order to be accessible, if possible please:

- Use a font size within figures set to 12-14pt or equivalent
- Limit the use of font variations such as italic, bold, ALL CAPS or other styles that may make the content difficult to read
- Do not use underlines for items that are not links
- The use of colour should adhere to general colour accessibility principles. Ideally, place text on single colour backgrounds. Avoid background patterns or pictures and distracting surrounds.

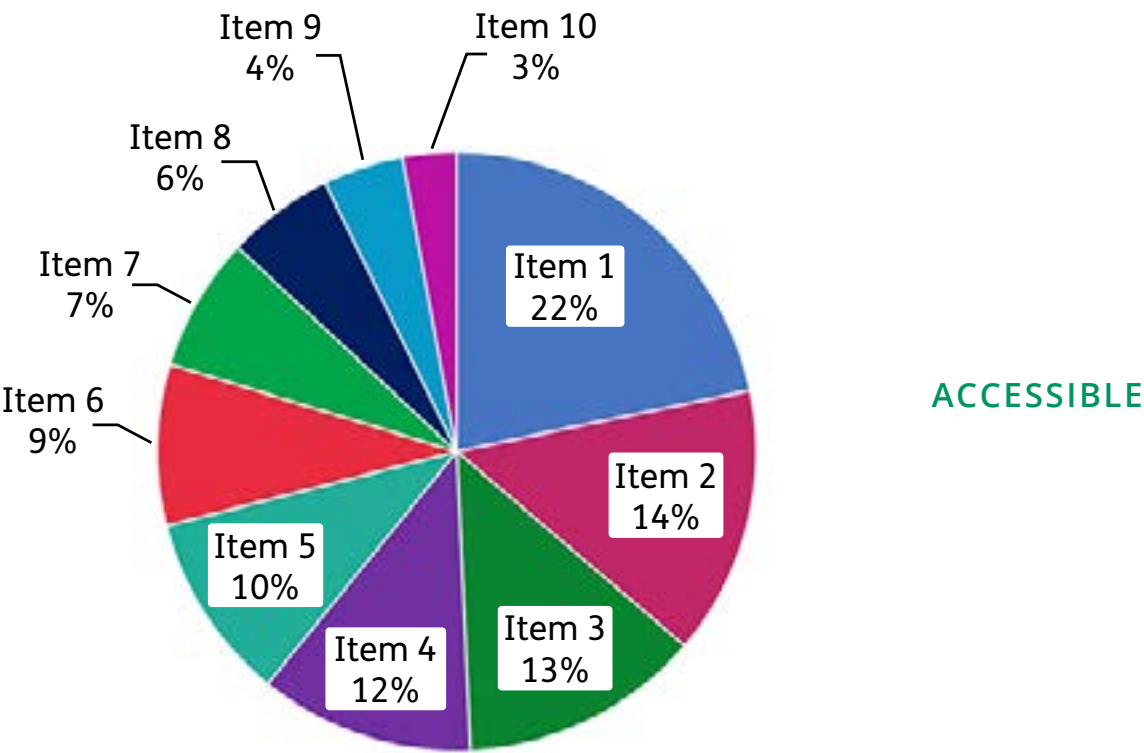


- ✗ Avoid using a serif-font
- ✗ Using a font-size under 12pt or equivalent is not recommended
- ✗ Avoid font variations such as italic, bold, ALL CAPS, or other styles when possible
- ✗ Avoid placing text on top of a colour or pattern

Accessible Fonts

OUP recommends using one of the following approved fonts for your charts and figures. Please find licensing information the [Useful Links](#) section of this document.

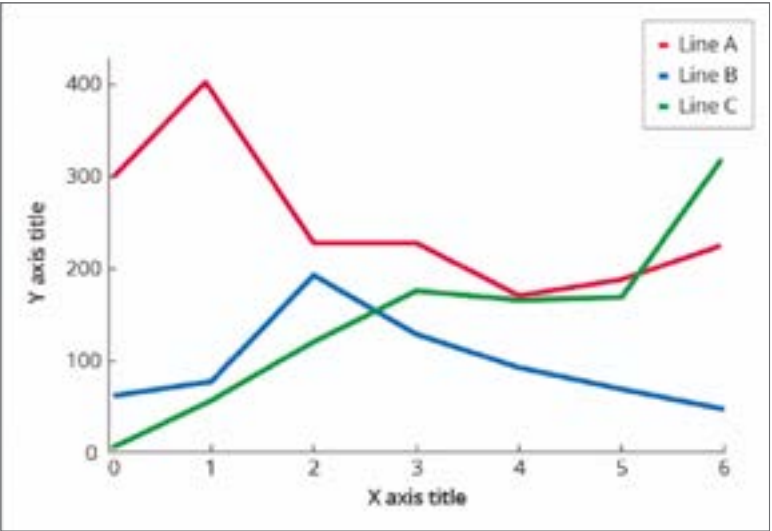
- ✓ **FS Me Regular:** designed specifically to improve legibility for people with learning disabilities. Every letter of FS Me was tested for its appeal and readability with a range of learning disability groups across the UK.
- ✓ **Source Sans Pro Regular:** A Google Font. These fonts are licensed under the Open Font License. You can use them freely in your products & projects - print or digital, commercial or otherwise.
- ✓ **Open Sans Regular:** A Google Font. These fonts are licensed under the Apache License, Version 2.0. You can use them freely in your products & projects - print or digital, commercial or otherwise.



- ✓ Uses FS Me Regular
- ✓ Labels are set at 12.5pt font size
- ✓ Does not have any styling outside of font type and size
- ✓ Type is placed on a background that meets colour contrast guidelines, and/or is placed in a white box.

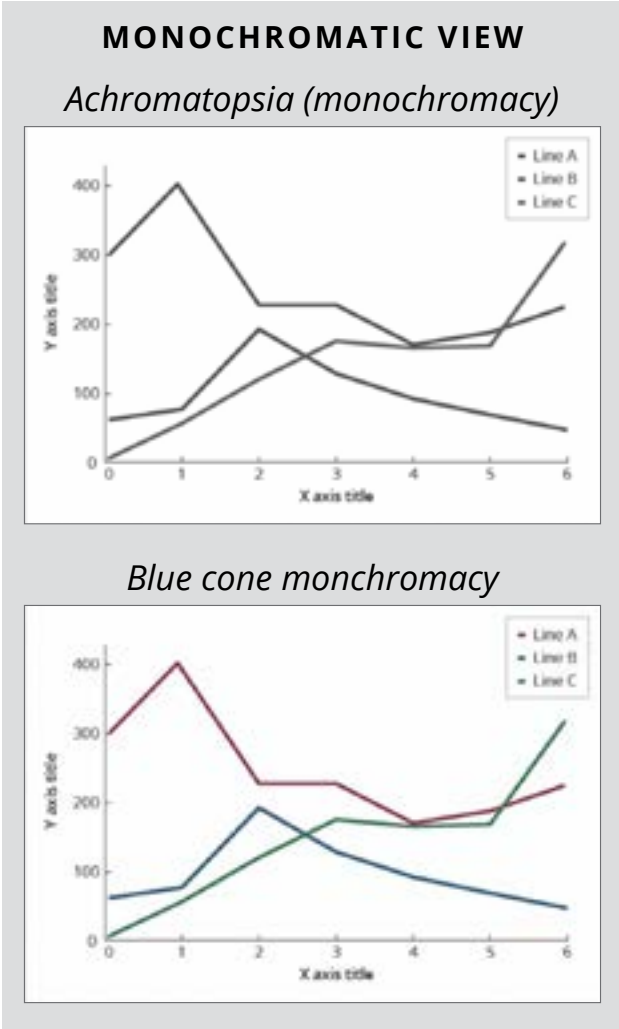
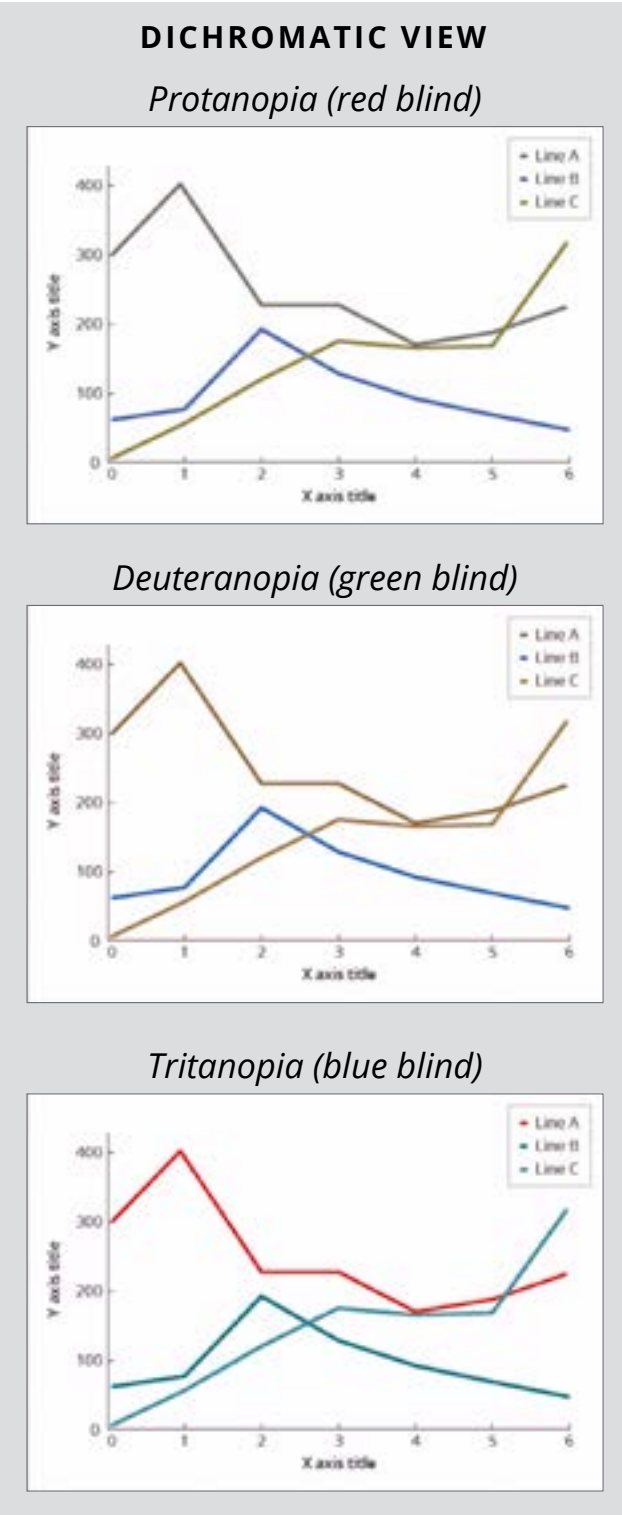
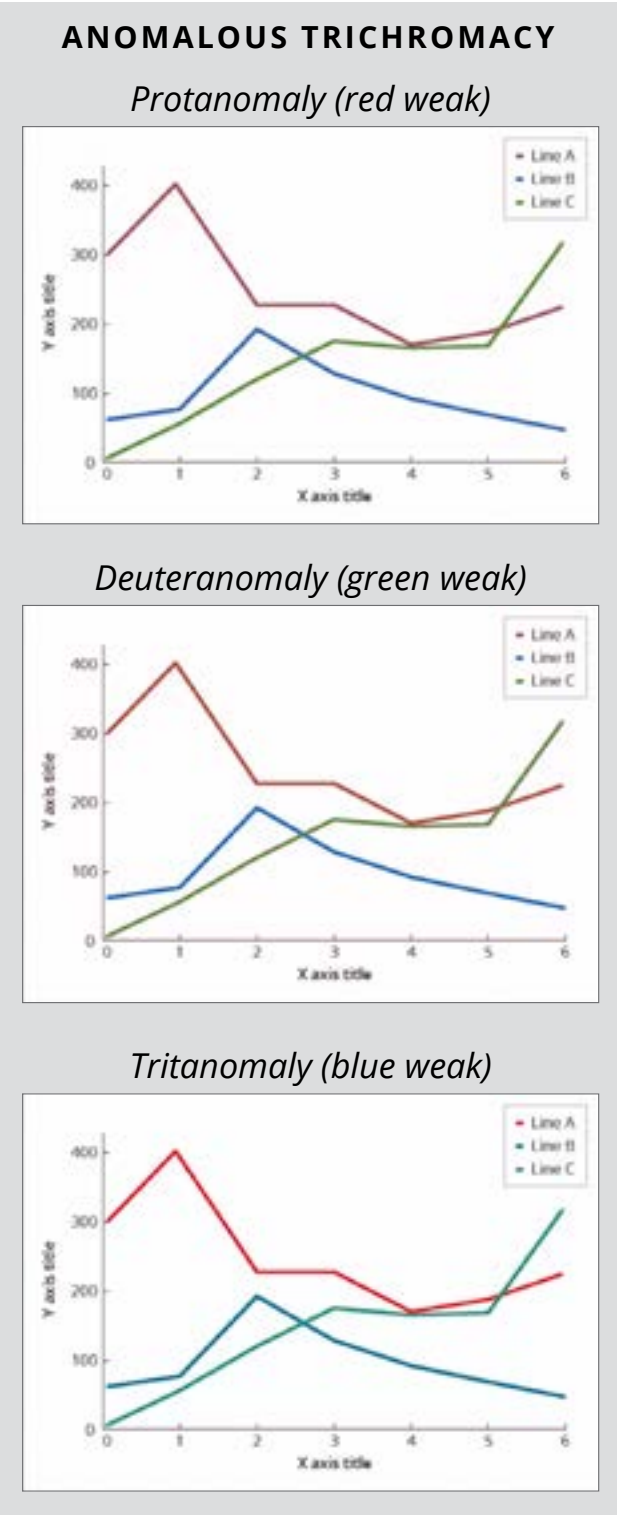
What do we need to consider for line graphs?

A simple line graph is shown below, and the images on the right demonstrate how this graph may be perceived by readers with various forms of colour blindness.



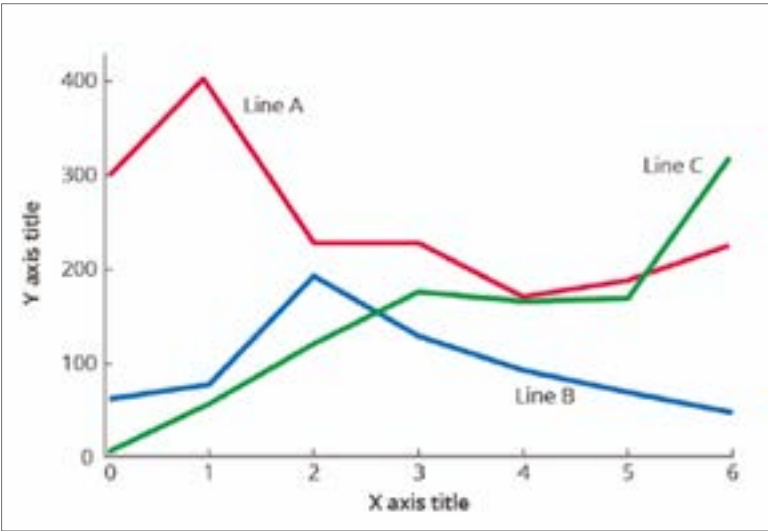
In most cases the reader may be able to match the colour in the legend to the colour of the plot line, even if not perceiving the colour as intended. But, there are instances where the colour isn't easily discernible, e.g. deuteranopia. Therefore, in order to be accessible, the plot lines on a line graph need to either:

- have a label next to the plot line (with or without leader line, as necessary)
- use different stroke styles (e.g. solid, dashed, dotted)
- use different symbols for data points

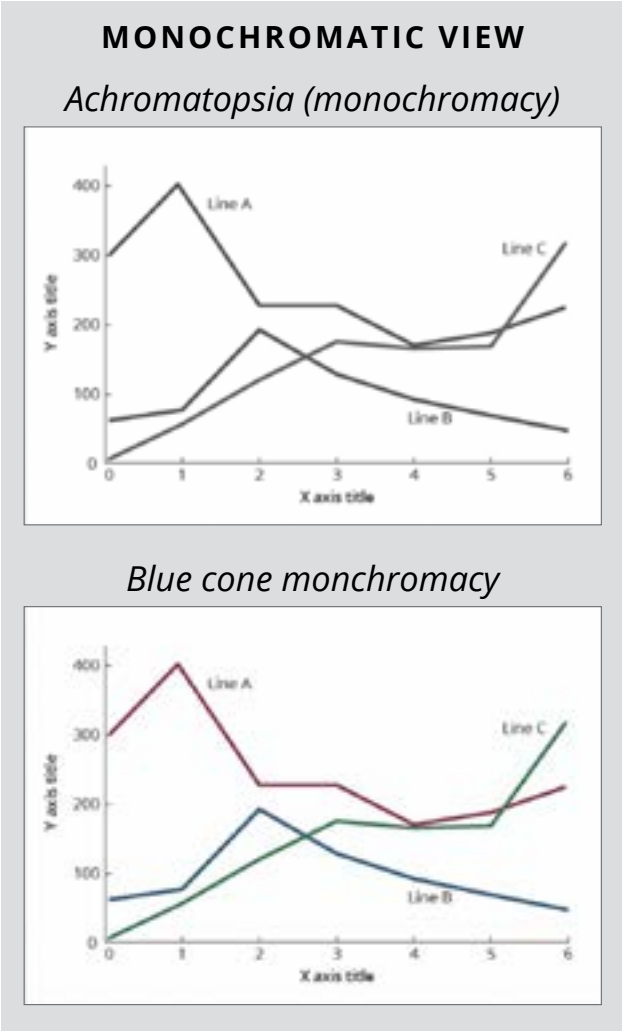
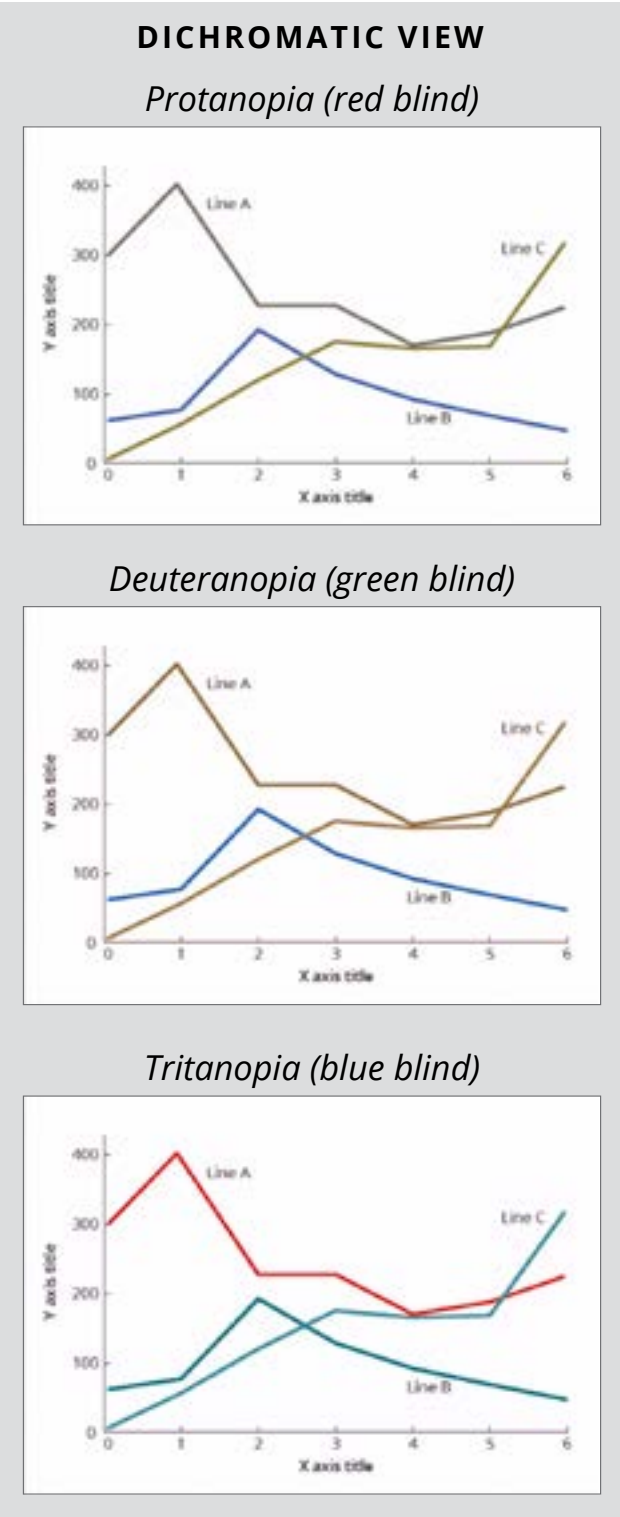
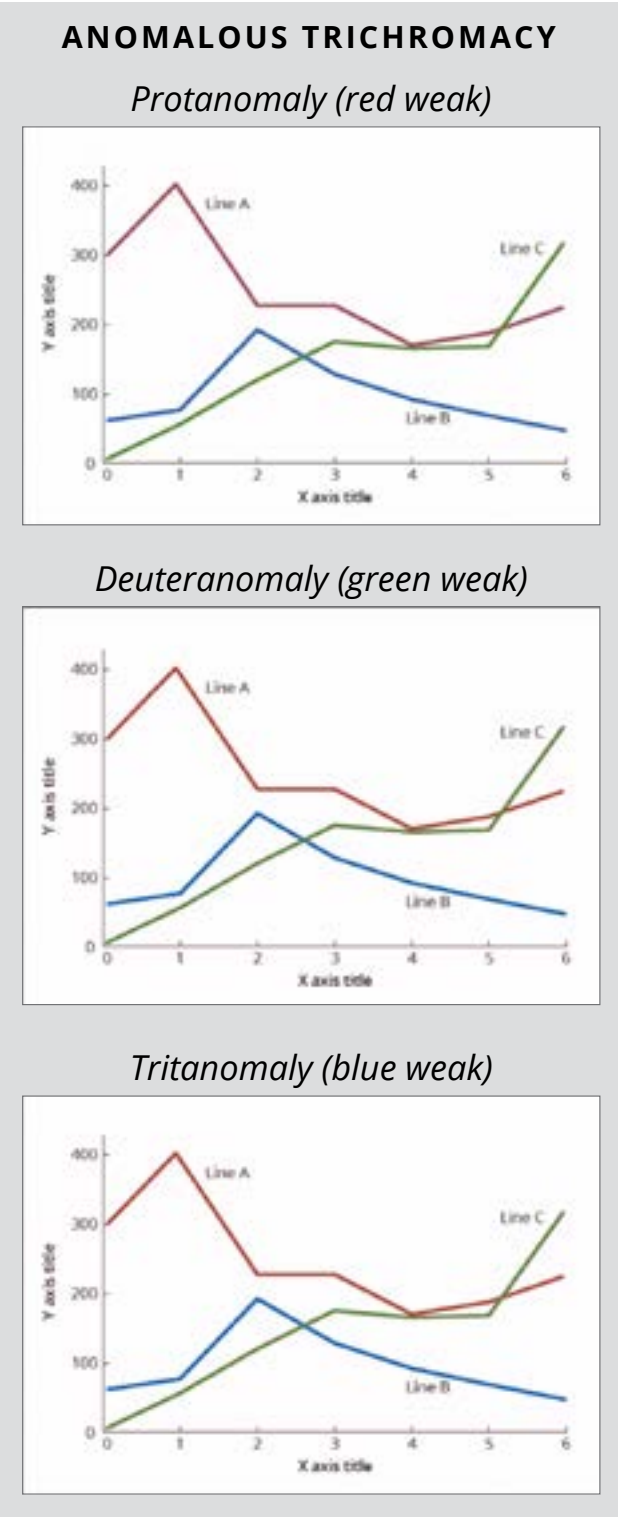
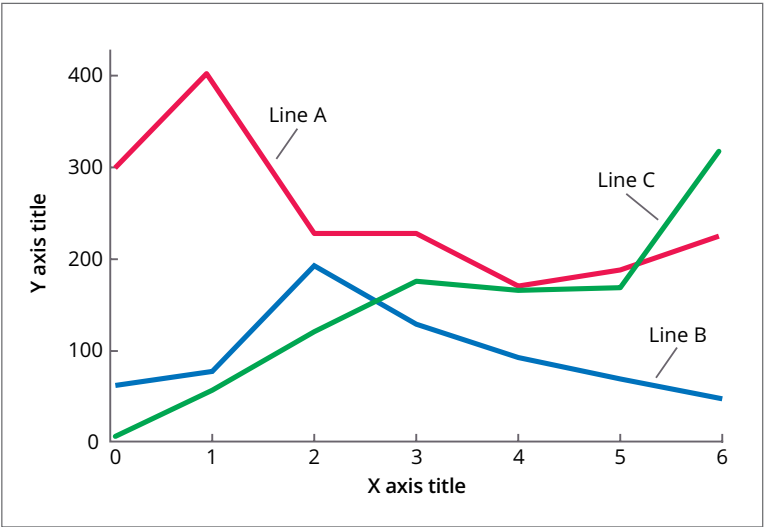


Option 1:
Labels next to the plot lines

Here the line graph has a label next to each plot line, which ensures the colour isn't necessary to identify which line is which – the colour is supplementary. A legend isn't required.

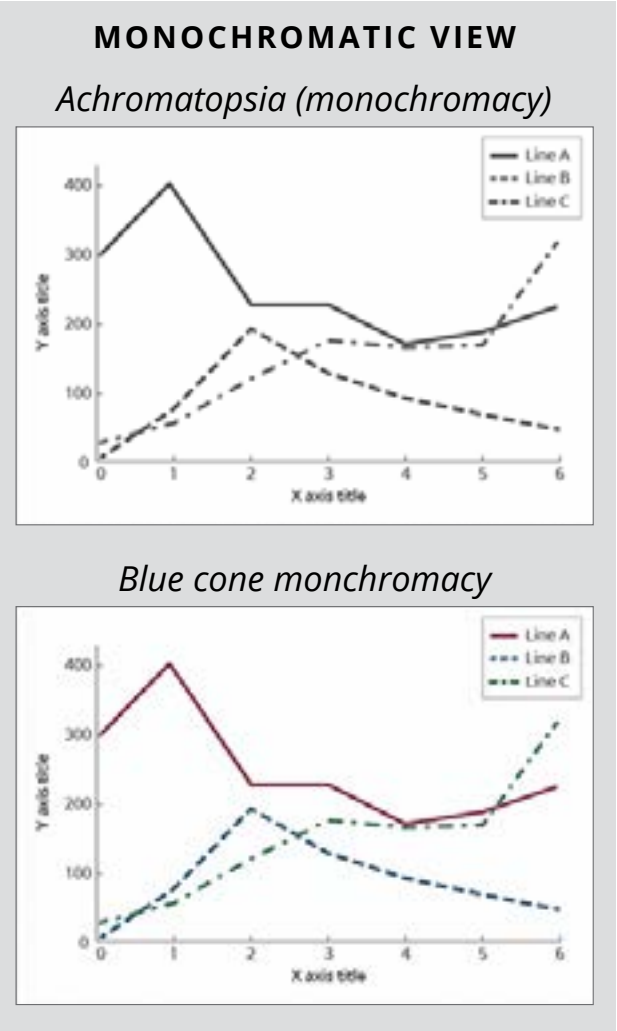
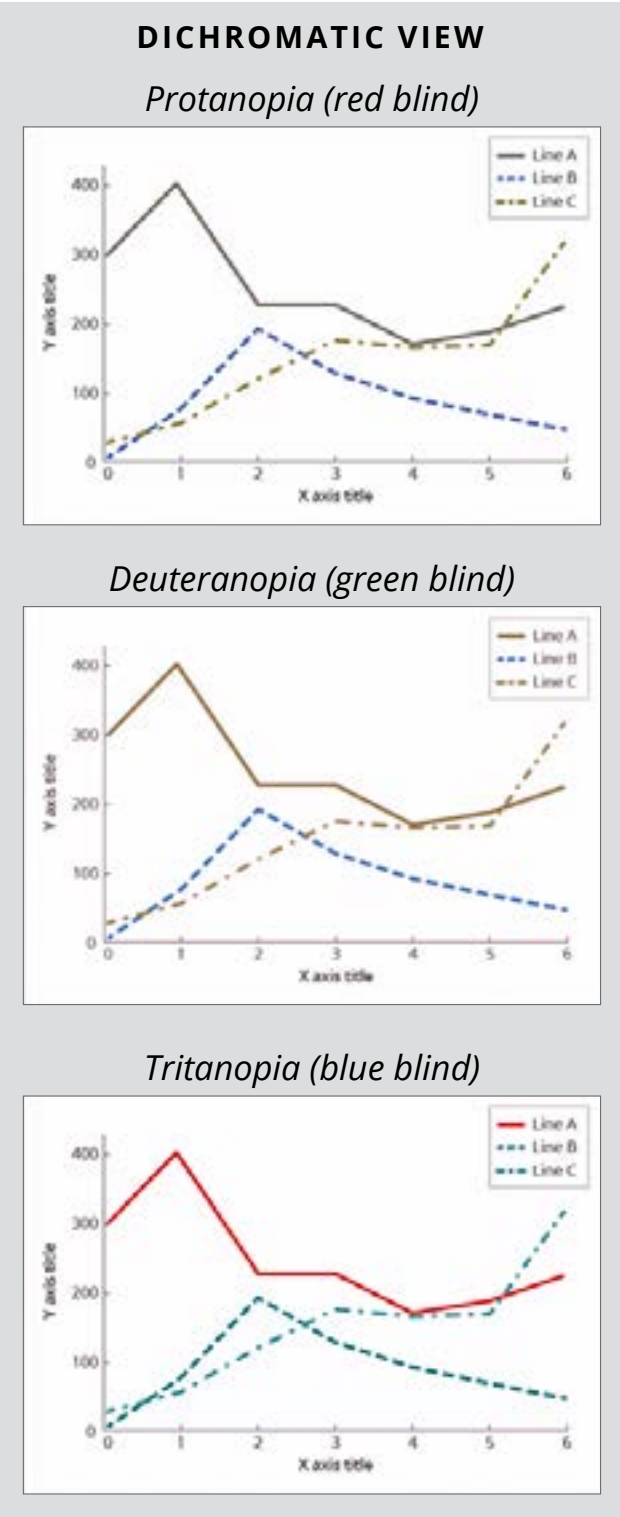
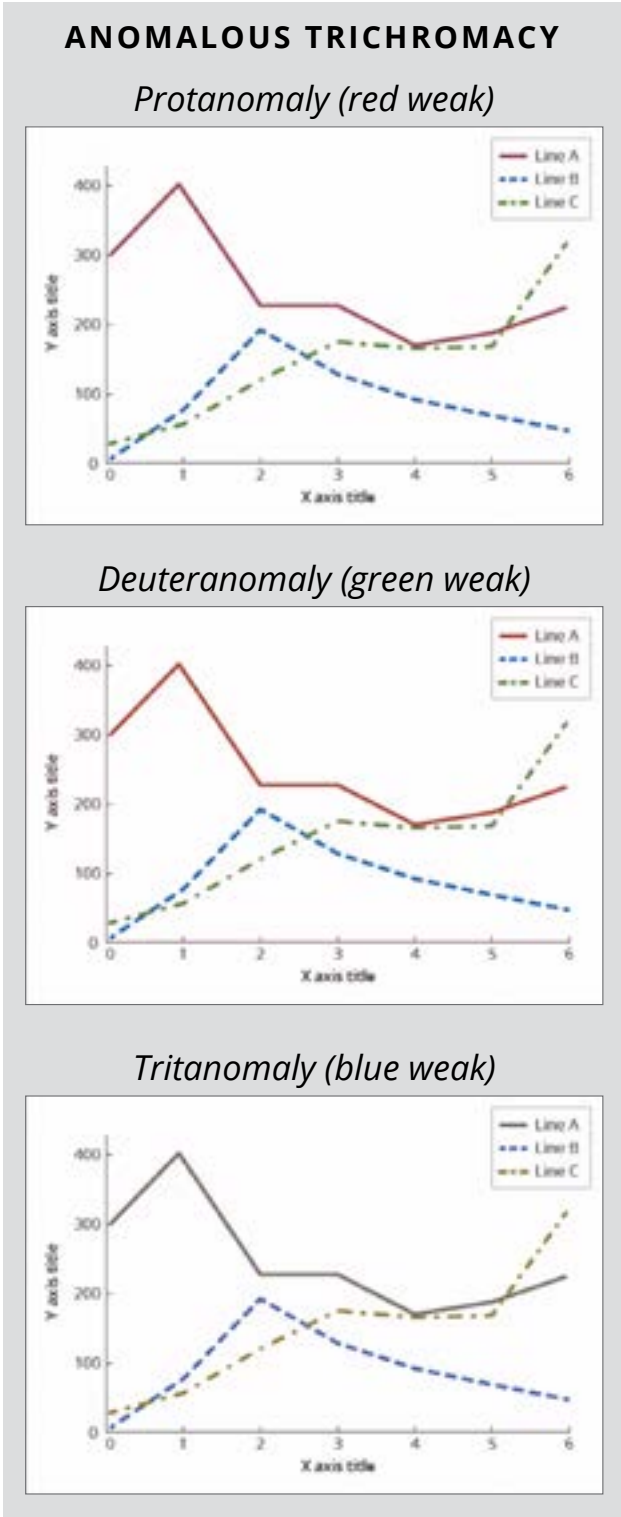
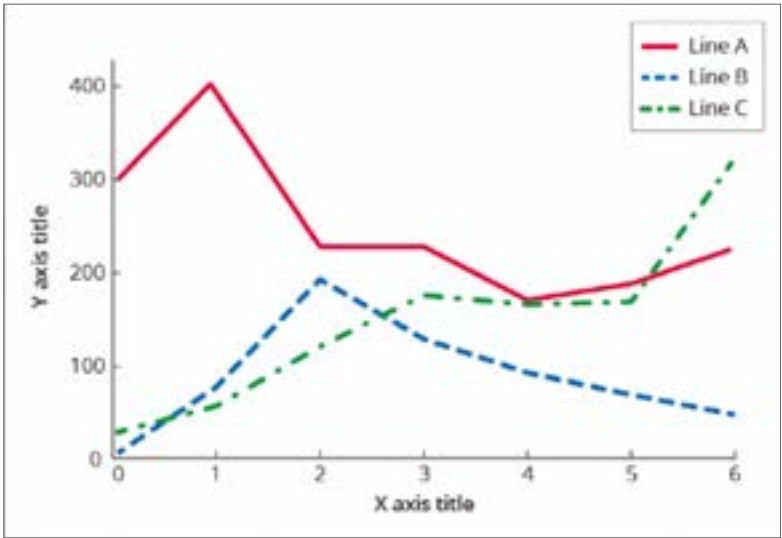


Where graphs are more complex, and/or placing a label nearby is difficult, a leader line could be used for clarity:



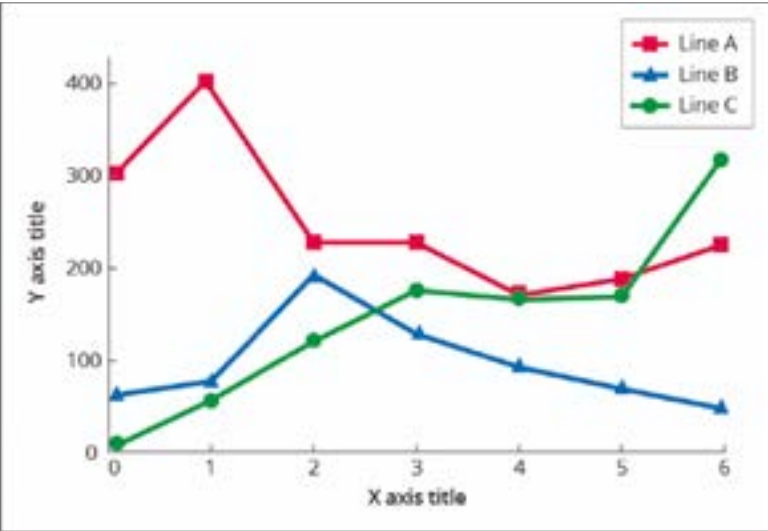
Option 2:
Different stroke styles for plot lines

These plot lines each have a different stroke/line style, which ensures the plot line can be matched to the legend without relying on colour. Colour is a supplementary aid.

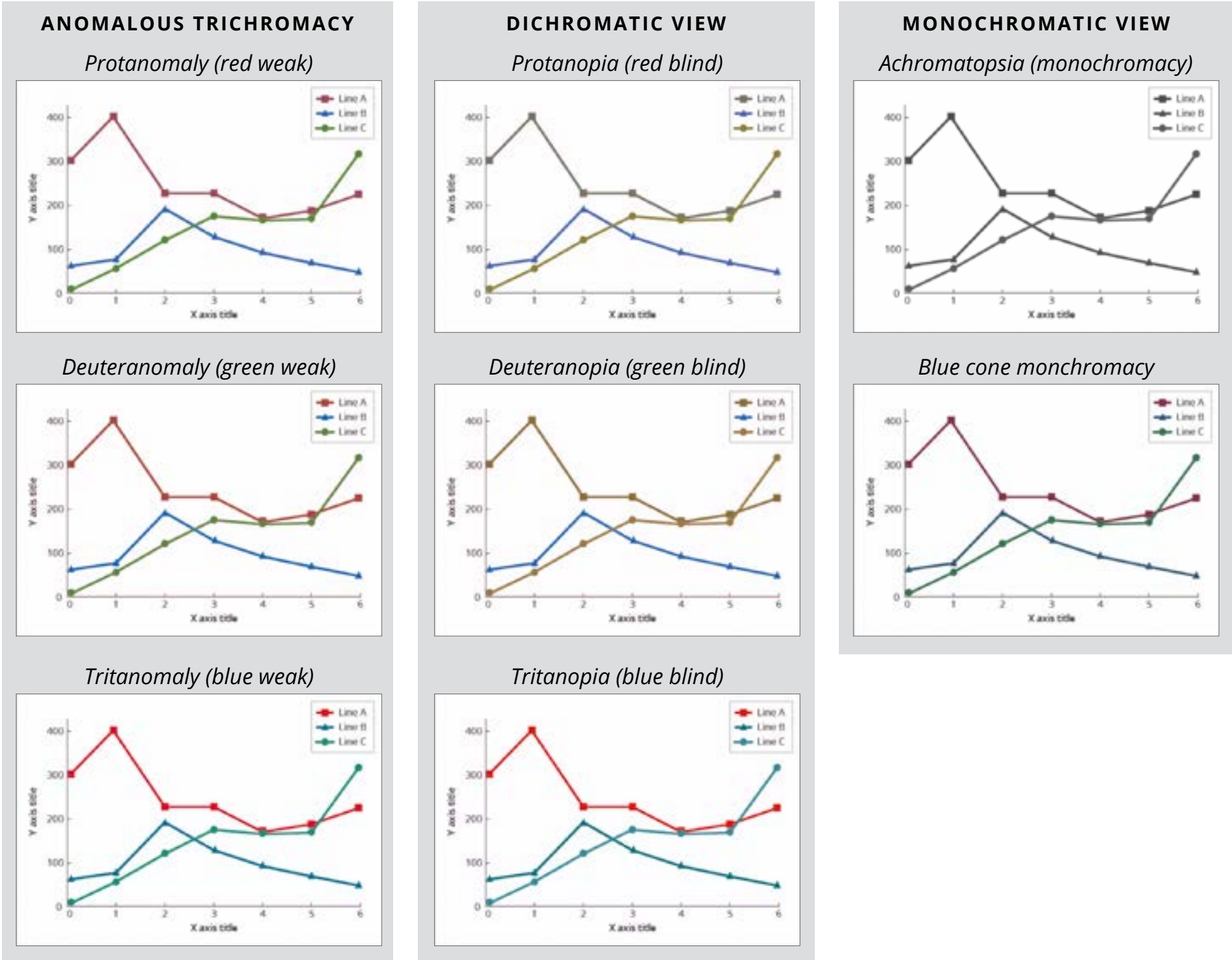


Option 3:
Different symbols for data points on
plot lines

The plot lines each have a different symbol to identify data points, which also aids recognition of which line is which without relying on colour. Colour is a supplementary aid.

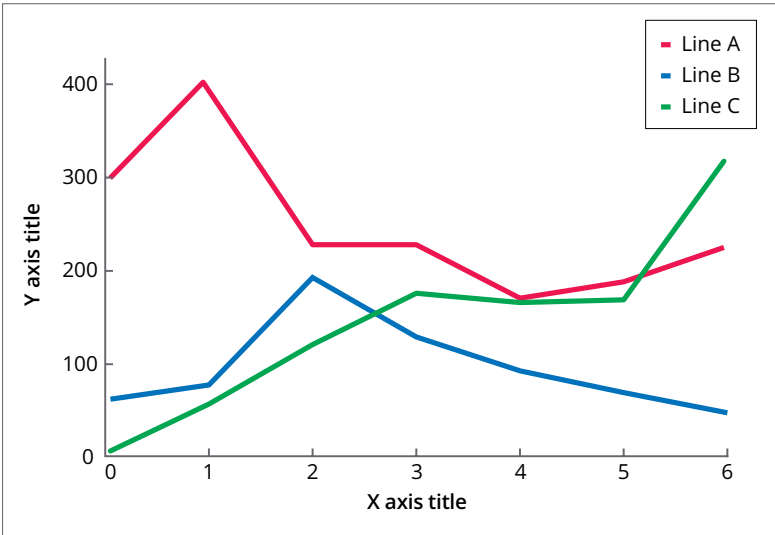


N.B. Where graphs are very complex, adding data points may make the graph overcrowded.



Cheat sheet

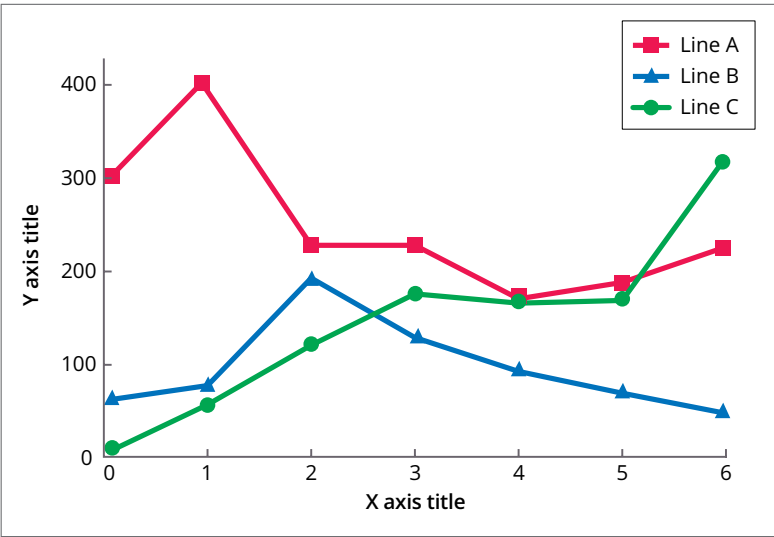
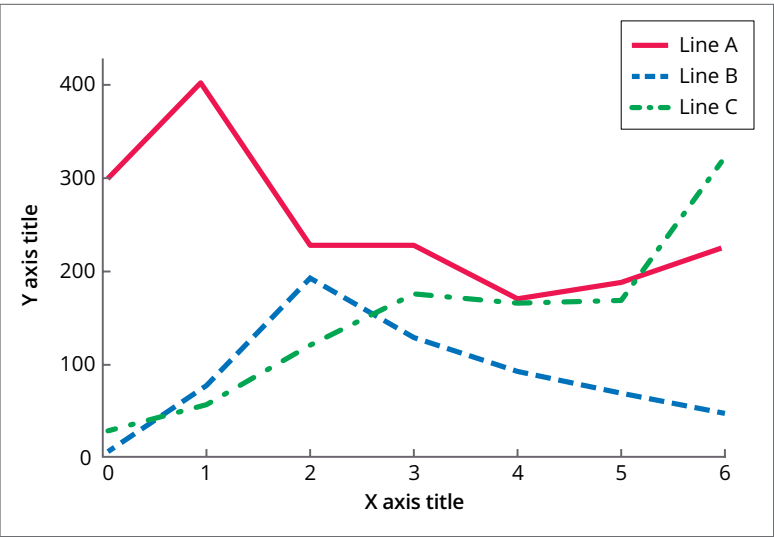
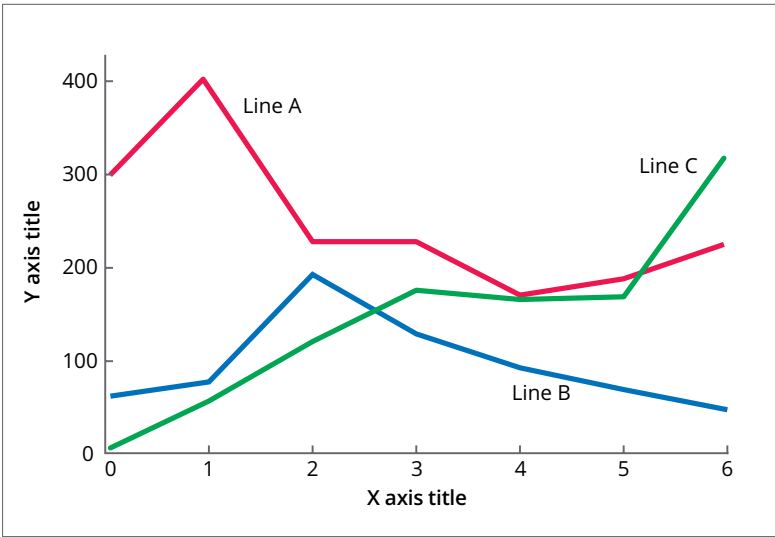
✗ Do not solely rely on colour to identify the plot line from the legend



✗ NOT ACCESSIBLE

The plot lines on a line graph need to either:

- ✓ have a label next to the plot line (with or without leader line, as necessary)
- ✓ use different stroke styles (e.g. solid, dashed, dotted)
- ✓ use different symbols for data points

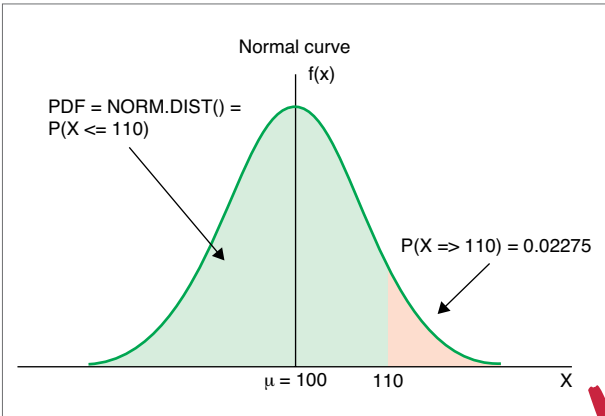


ACCESSIBLE

Other elements to consider

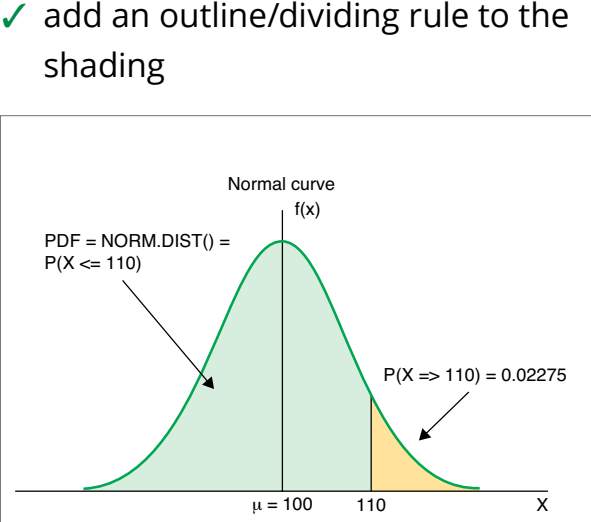
1. Highlighting areas

Any areas to be highlighted need to be easily identifiable, by having sufficient colour contrast and outline as necessary, and/or pattern. A few examples are shown:

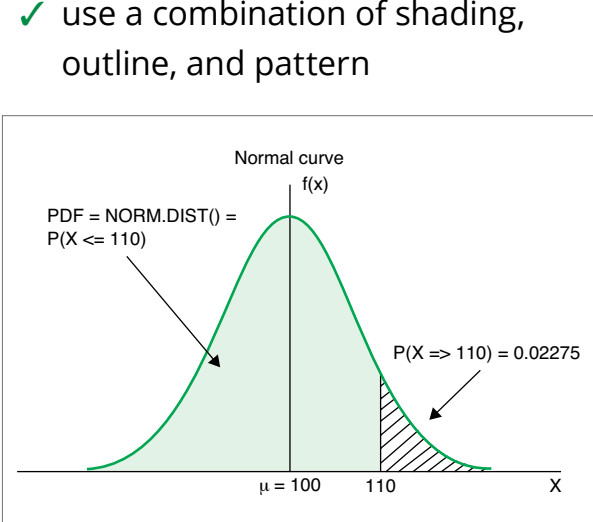


✗ The shaded areas do not contrast sufficiently

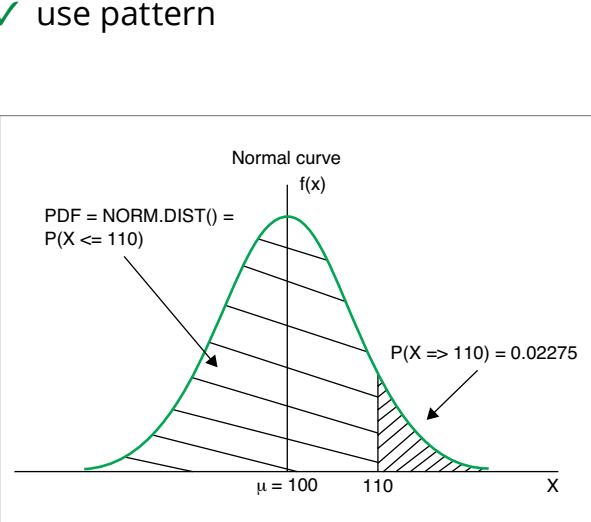
✗ NOT ACCESSIBLE



✓ add an outline/dividing rule to the shading



✓ use a combination of shading, outline, and pattern

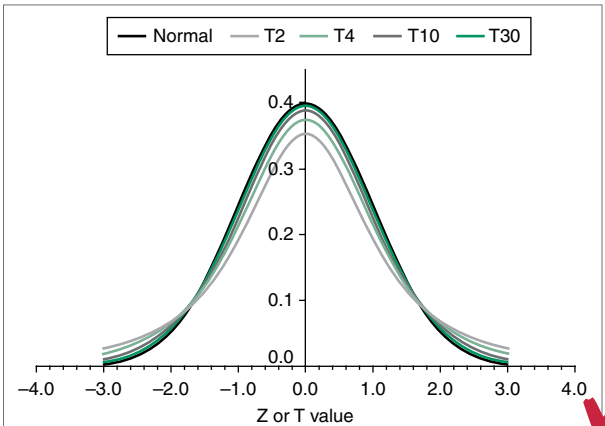


✓ use pattern

ACCESSIBLE

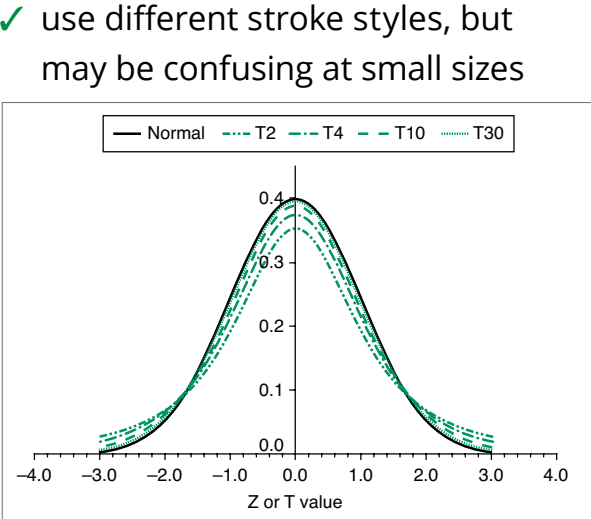
2. Complex graphs/multiple lines

Where a graph is complex, containing multiple plot lines overlaid, it may become more confusing to use stroke styles for each plot line, and data points and adjacent labels not practical. Therefore you may need to consider other ways to show multiple plots so that they are not overlaid, if detail is required.

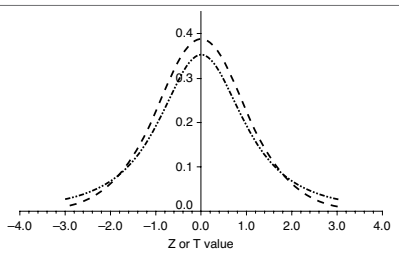
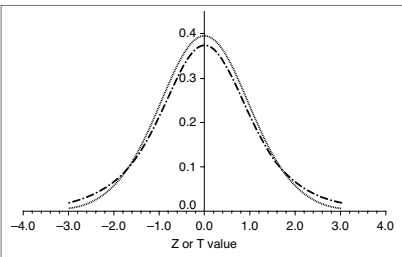
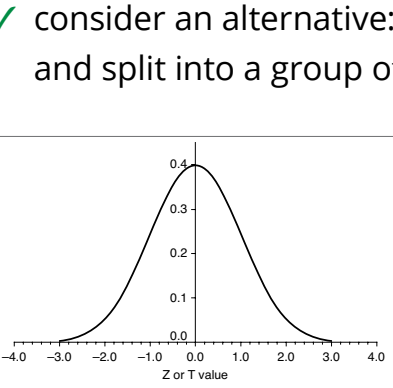


✗ relies solely on colour, without sufficient colour contrast of plot lines to background

✗ NOT ACCESSIBLE



✓ use different stroke styles, but may be confusing at small sizes

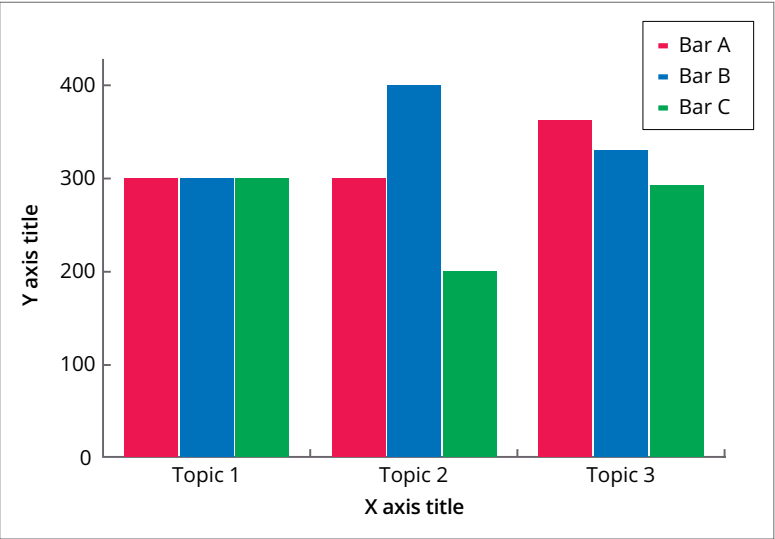


✓ consider an alternative: reducing the number of plots on one graph, and split into a group of graphs if detail is required

ACCESSIBLE

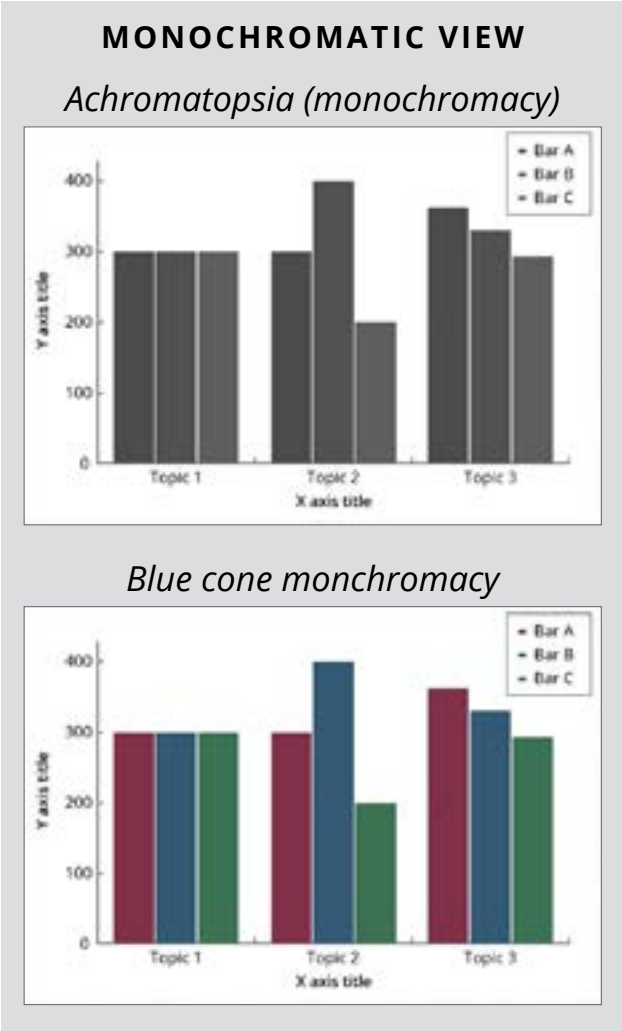
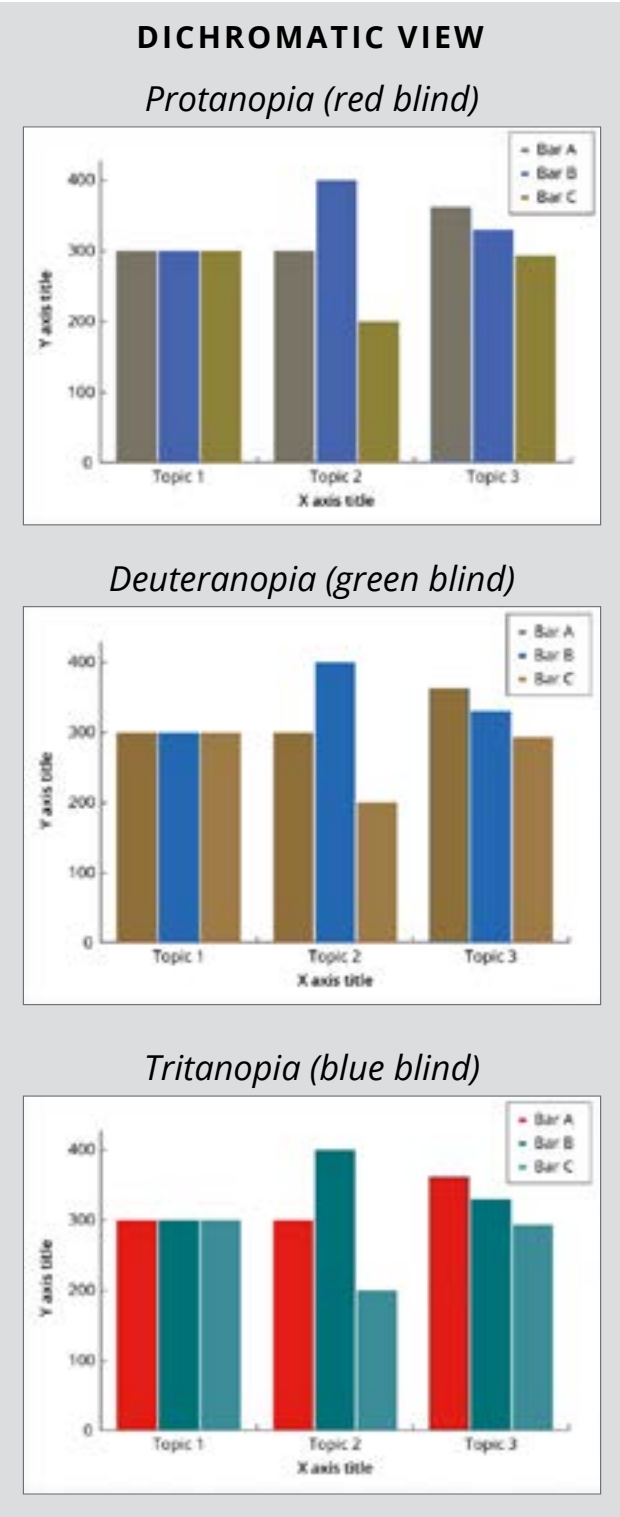
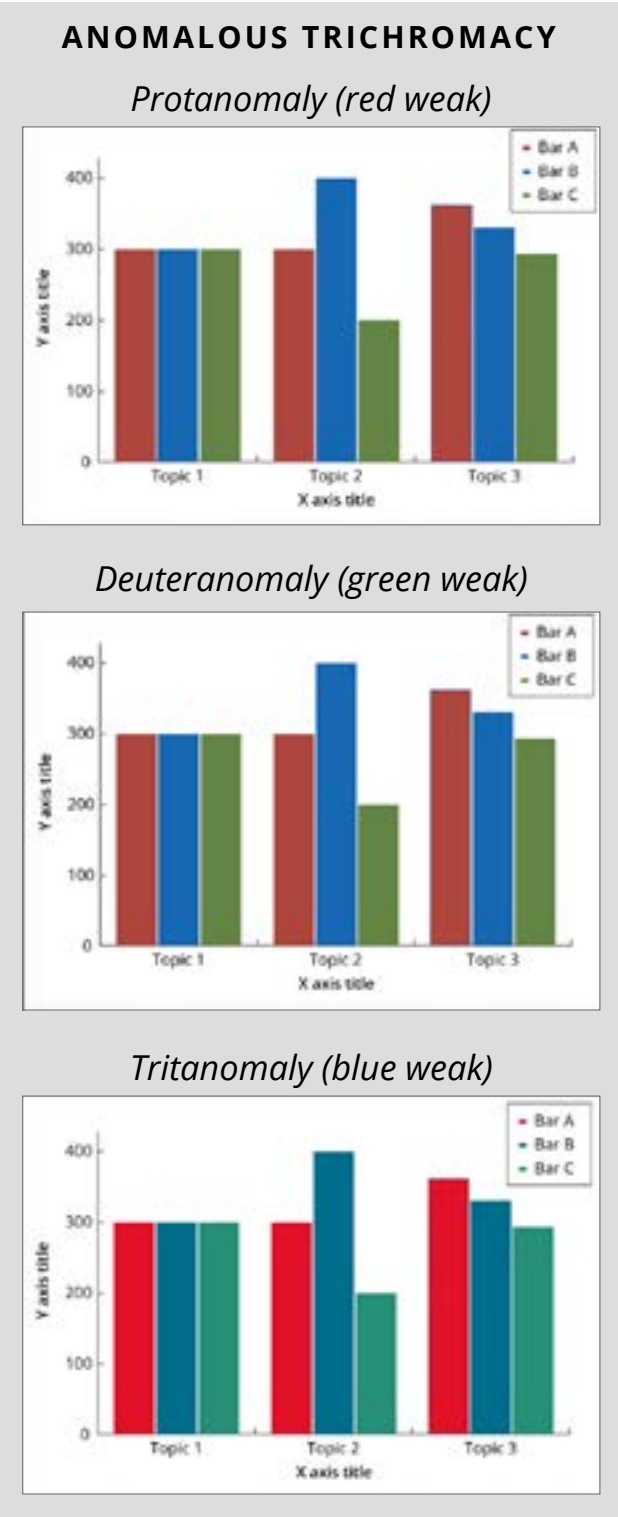
What do we need to consider for grouped bar graphs?

Readers with various forms of vision impairment may have difficulty telling areas of the grouped bar graph apart.



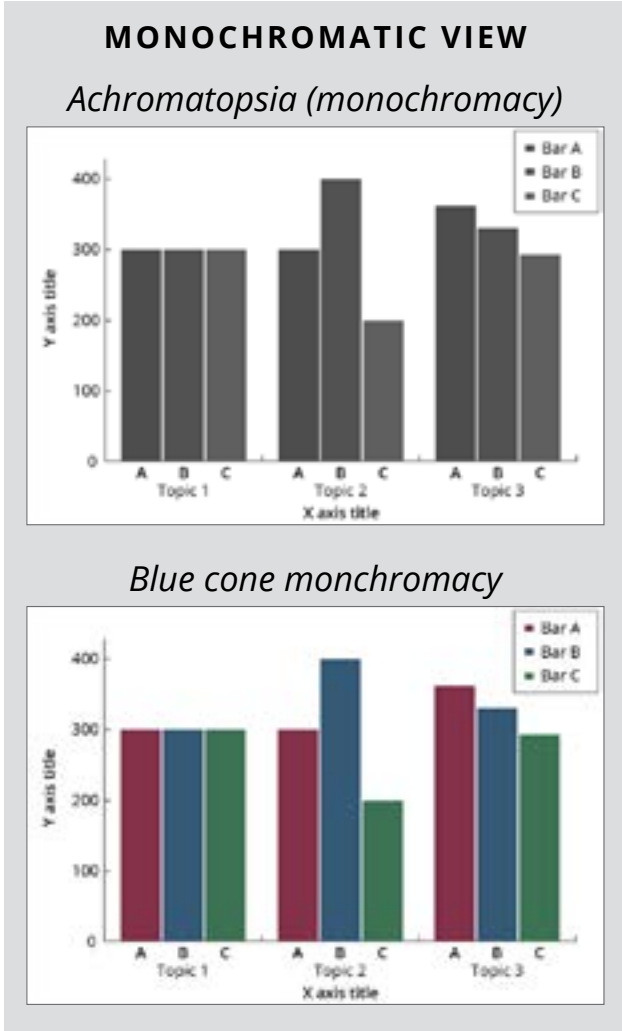
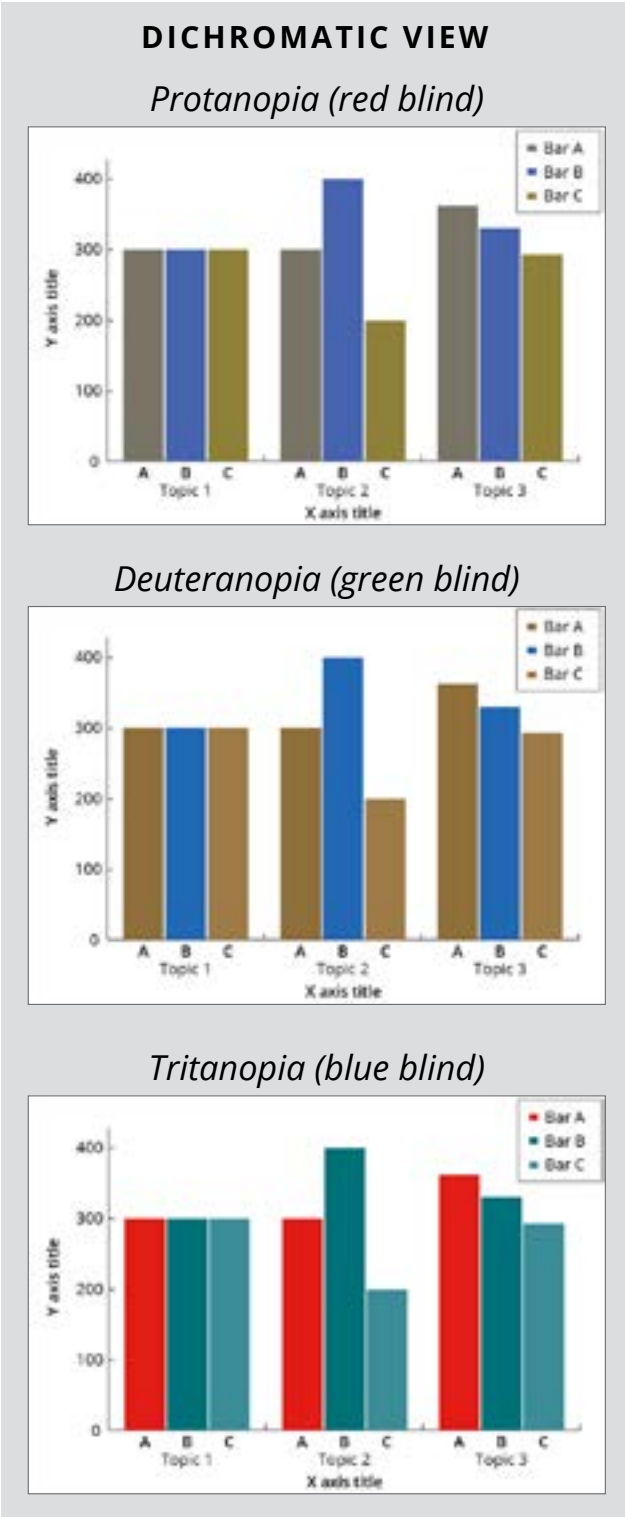
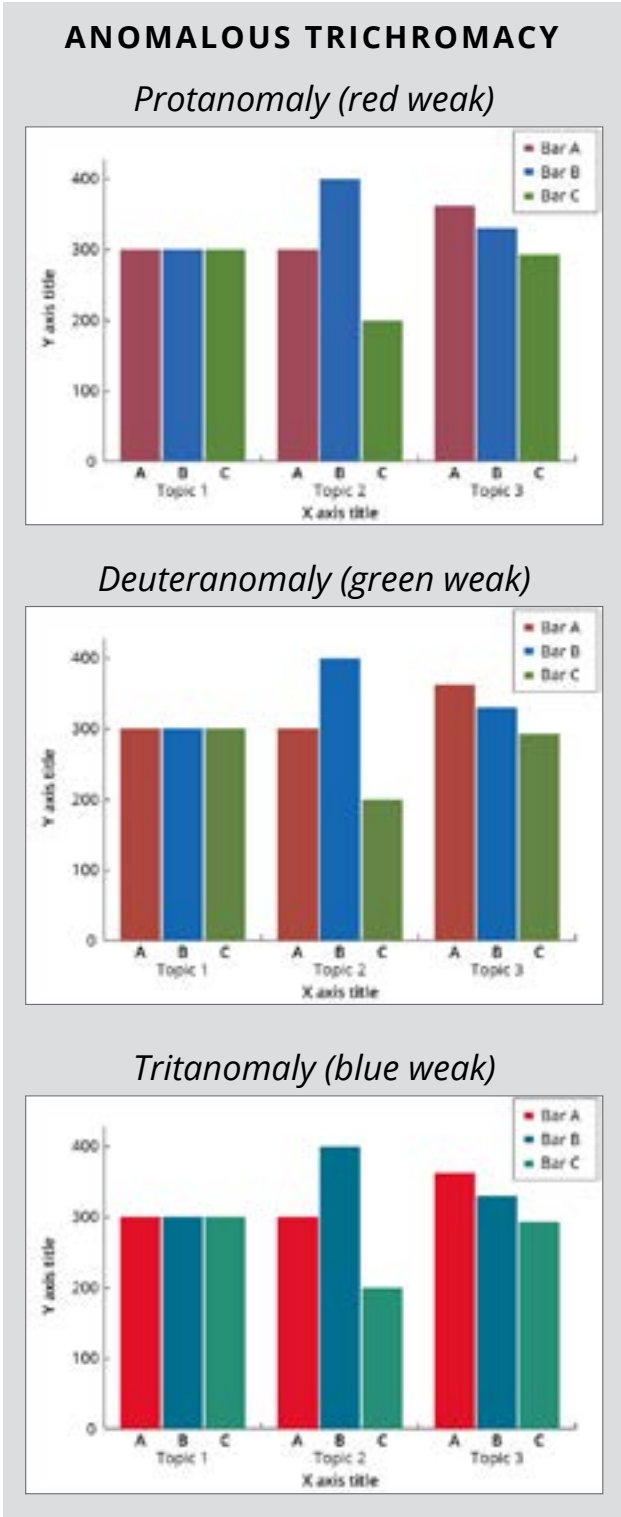
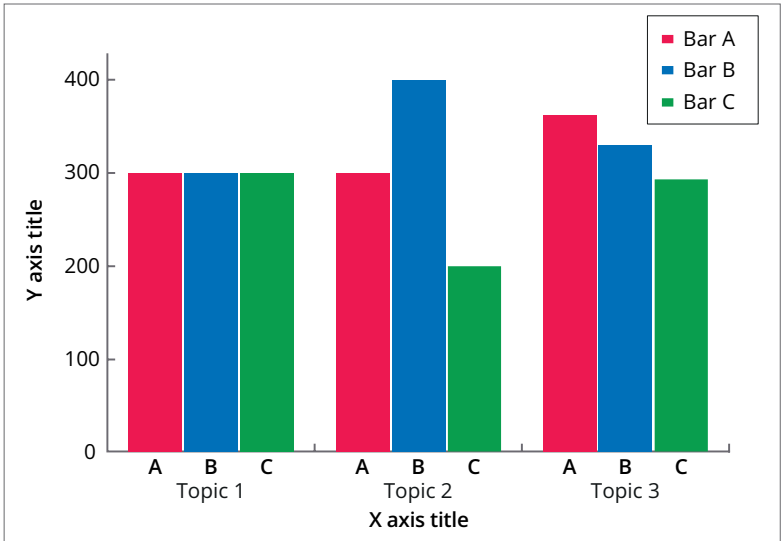
The accessibility of bar graphs can be improved by:

- labelling individual bars so the reader does not need to rely on perceiving colours in the legend
- using a colour palette with high contrast between shades (as opposed to using colours that are different hues but tonally similar)
- using patterns or textures with, or instead of, colour
- ensuring there is sufficient contrast between the bars, and between the bars and the background



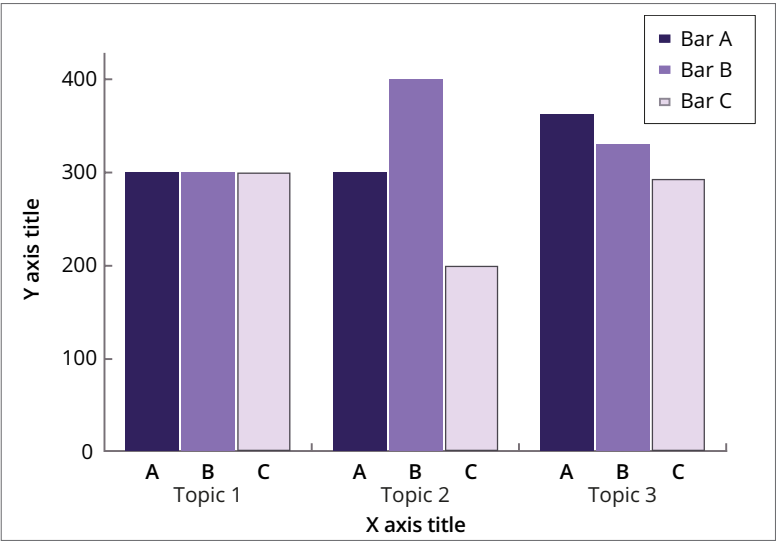
Option 1: Labels for each bar

Where possible, the bars should be labelled, so that a legend is not required to understand what the different bars represent.

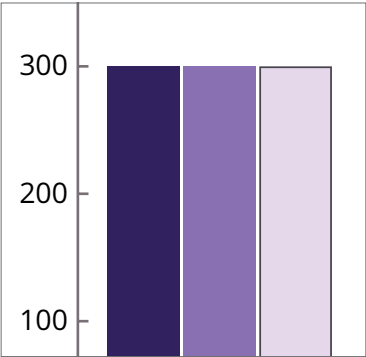


Option 2: Use a colour palette with high contrast

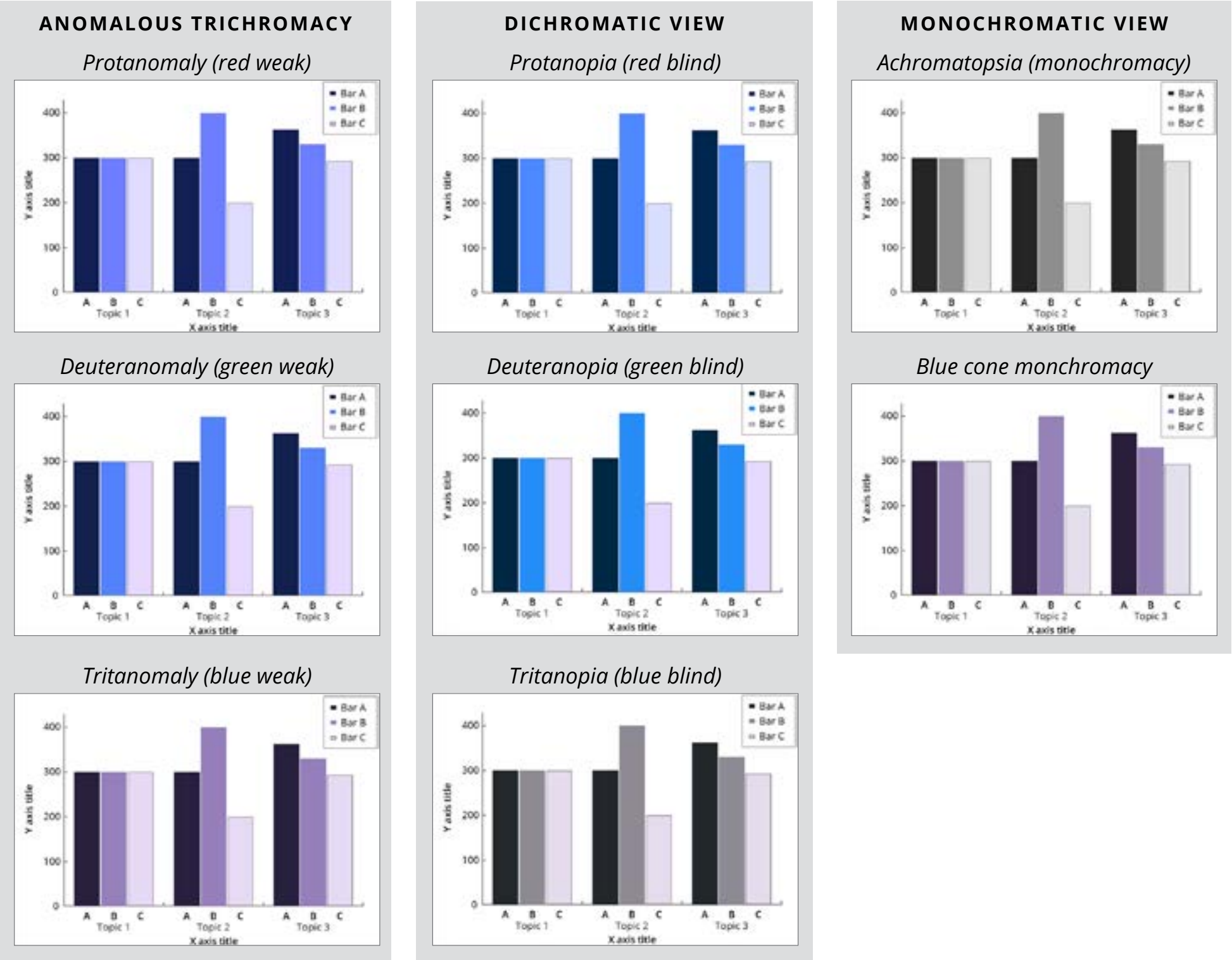
Increasing the contrast between the different colours used in the graph (e.g. using a dark, medium, and light shade as below), makes the different bars easier for most people to perceive. This option should be used in combination with another, such as labelling the bars, so that colour is not solely relied on to convey meaning.



Where any colours used do not have sufficient contrast with the background (e.g. the pale purple and the white background) a border in a shade with sufficient contrast should be used.

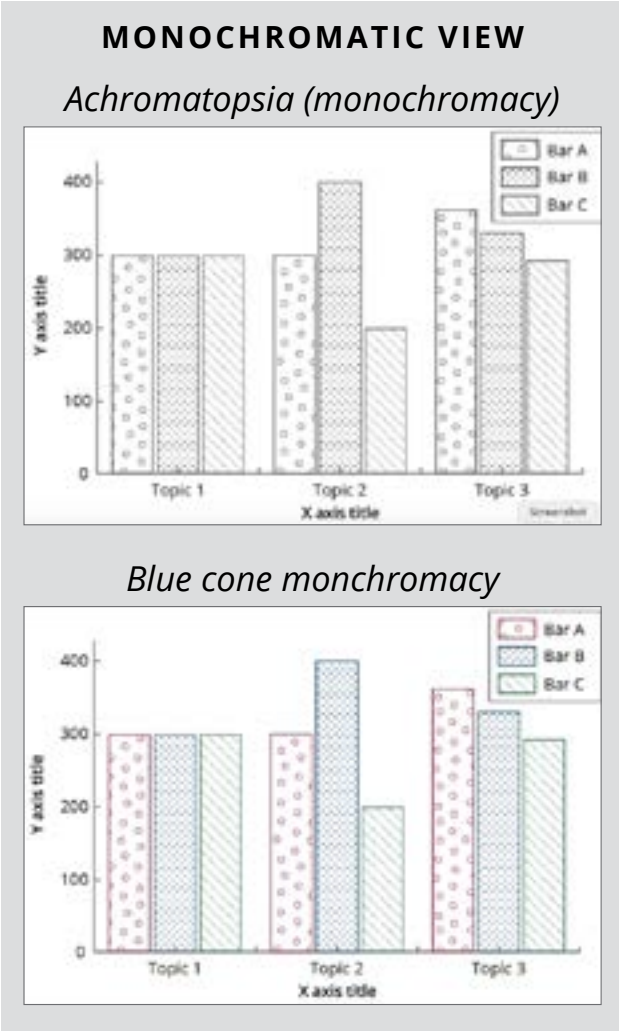
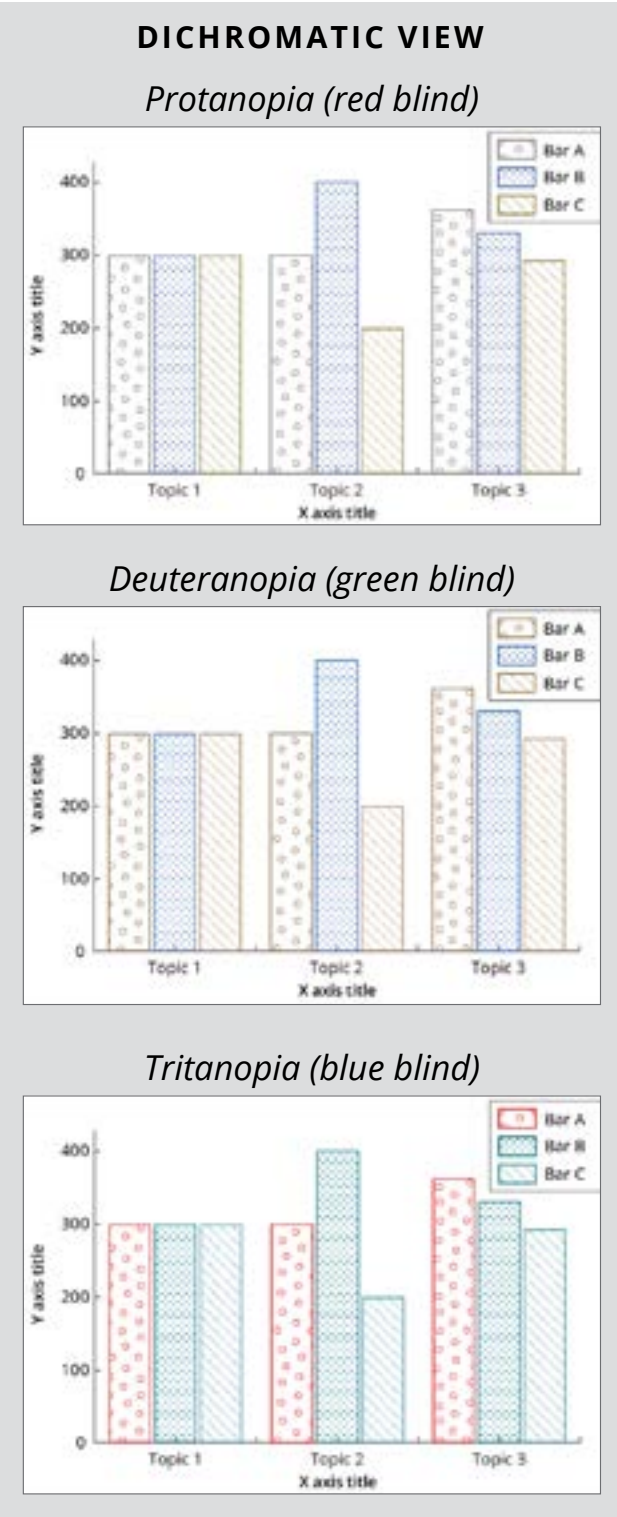
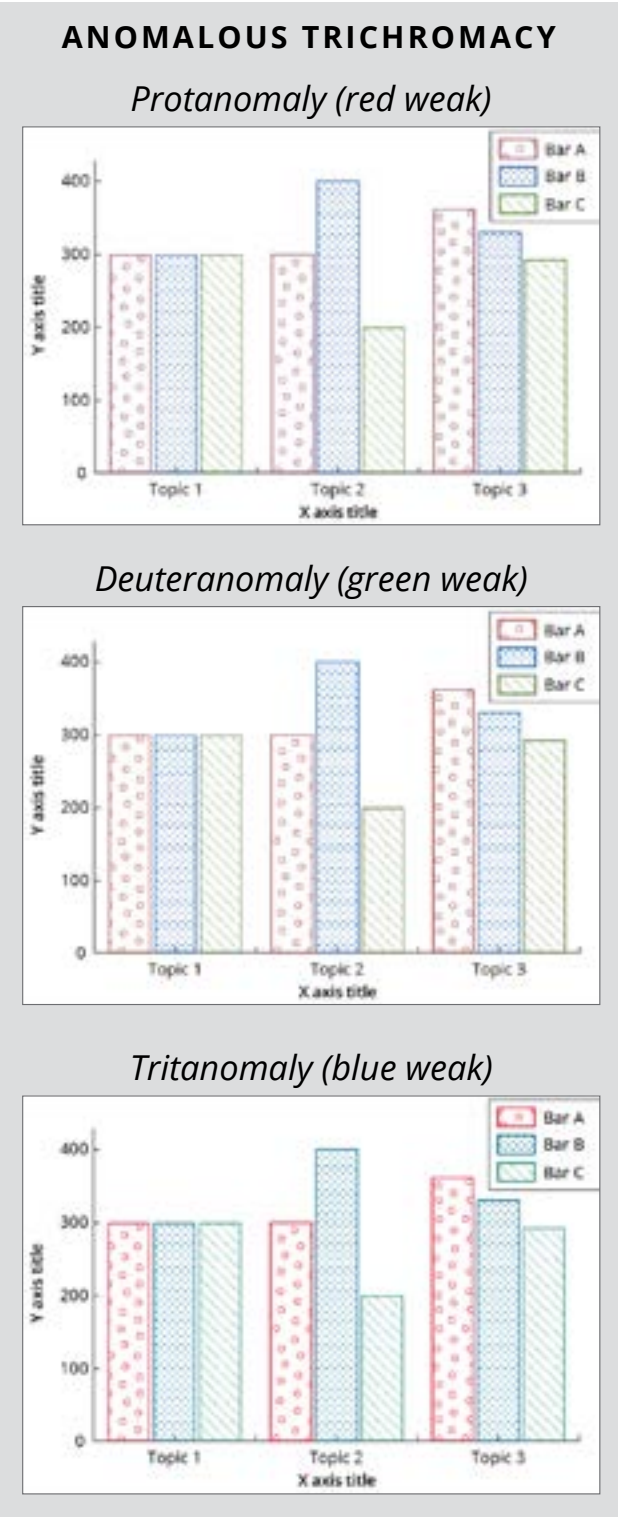
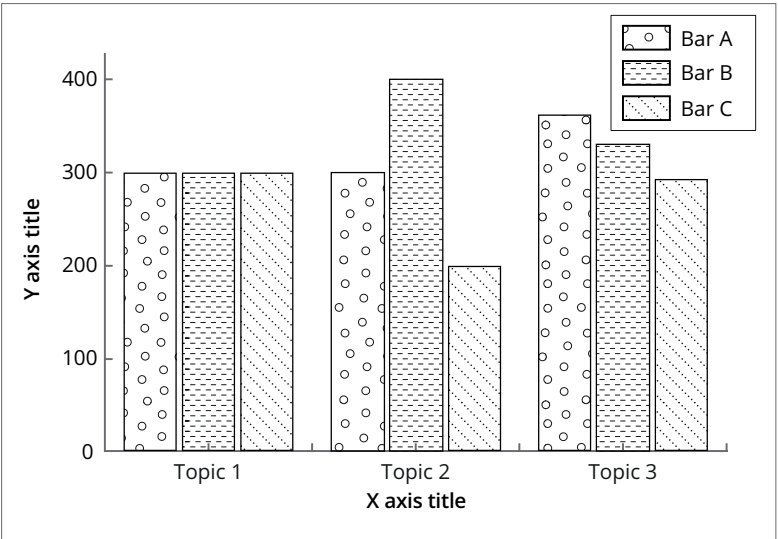
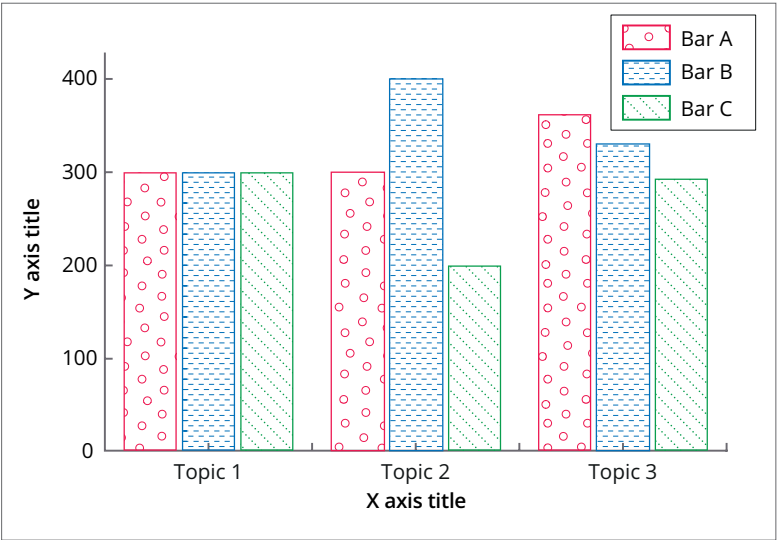


← The grey border around the pale purple ensures the required contrast ratio between the bar and background is met.



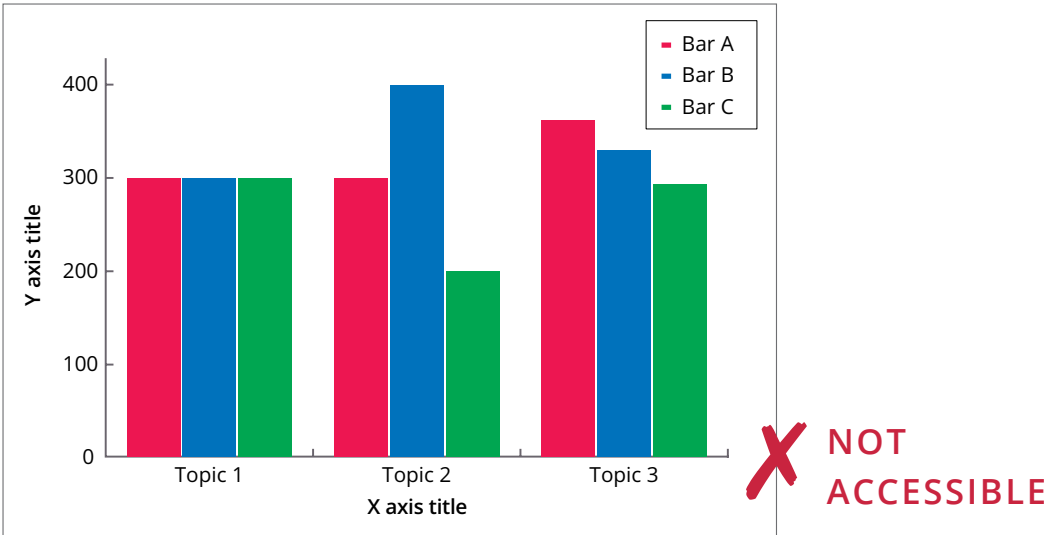
Option 3: Use patterns and textures

In this example, the bars are filled with different patterns or textures. This can be used in combination with colour, or simply in black and white. This option may be particularly useful where adding labels is not possible and a legend is required.

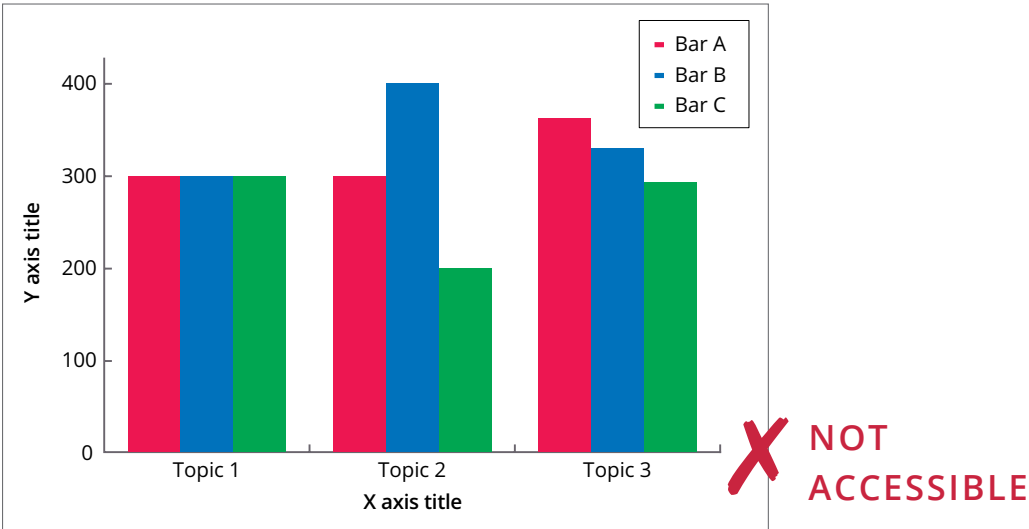


Cheat sheet

✗ Do not solely rely on colour to identify the meaning of the bars from the legend

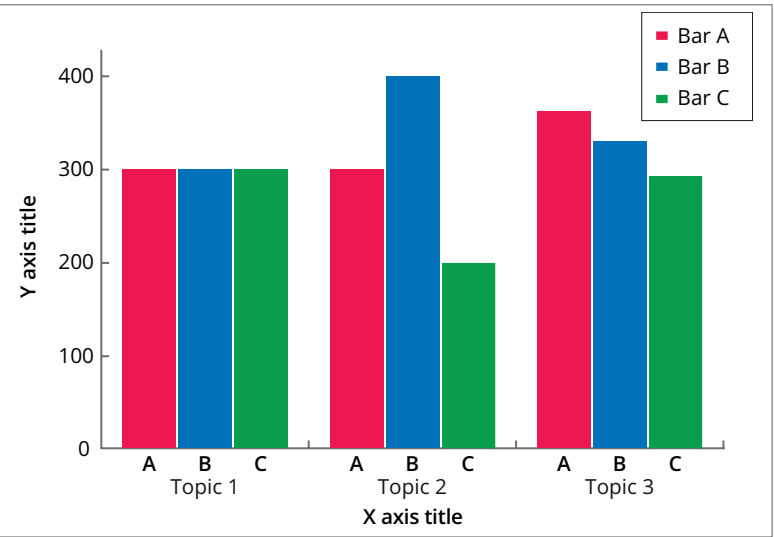
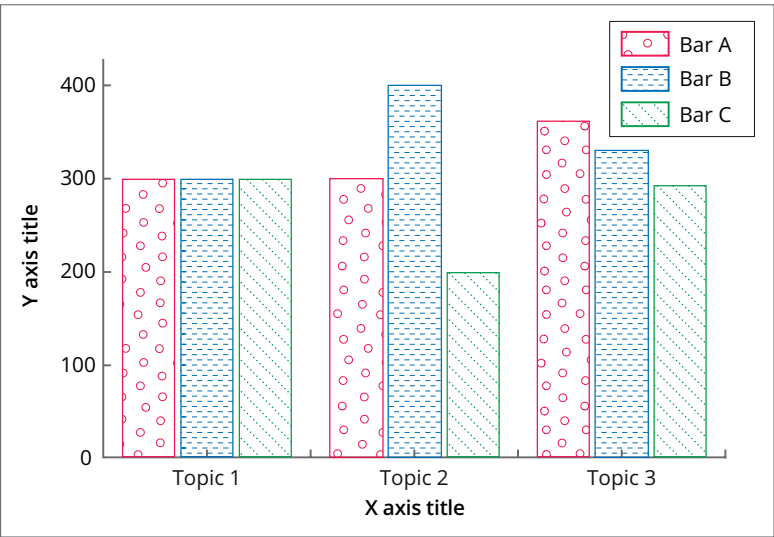
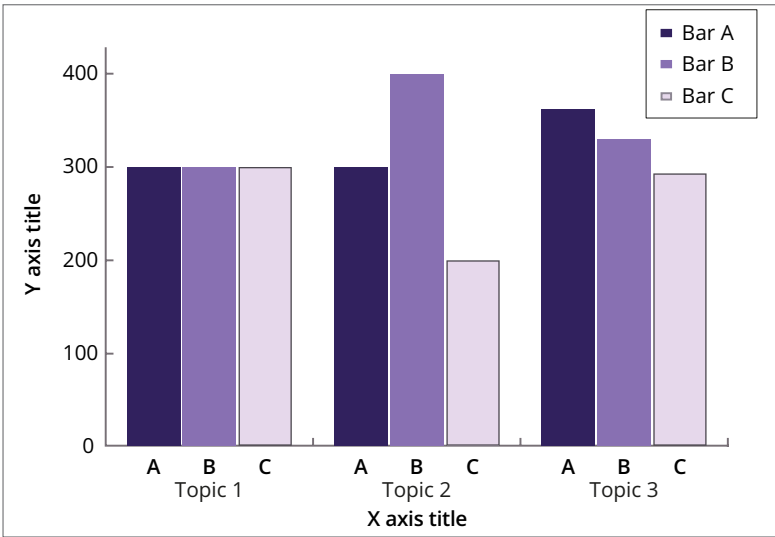


✗ Avoid bars touching—this makes it harder to differentiate when low contrast colours are used



The accessibility of grouped bar graphs can be improved by:

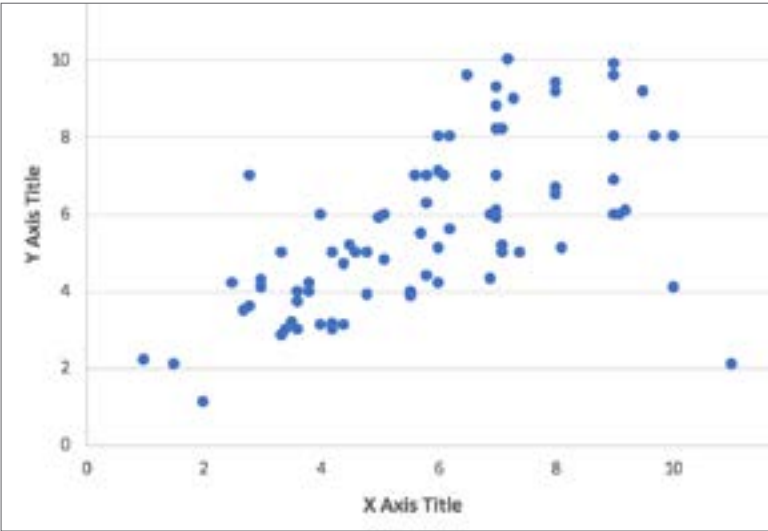
- ✓ having a label next to the relevant bar
- ✓ adding space or a border between each bar
- ✓ using textures/patterns for each bar
- ✓ using a colour palette with high tonal contrast
- ✓ using a border around each bar to ensure there is sufficient tonal contrast between the bar and the background (e.g. a light colour on a light background)



ACCESSIBLE

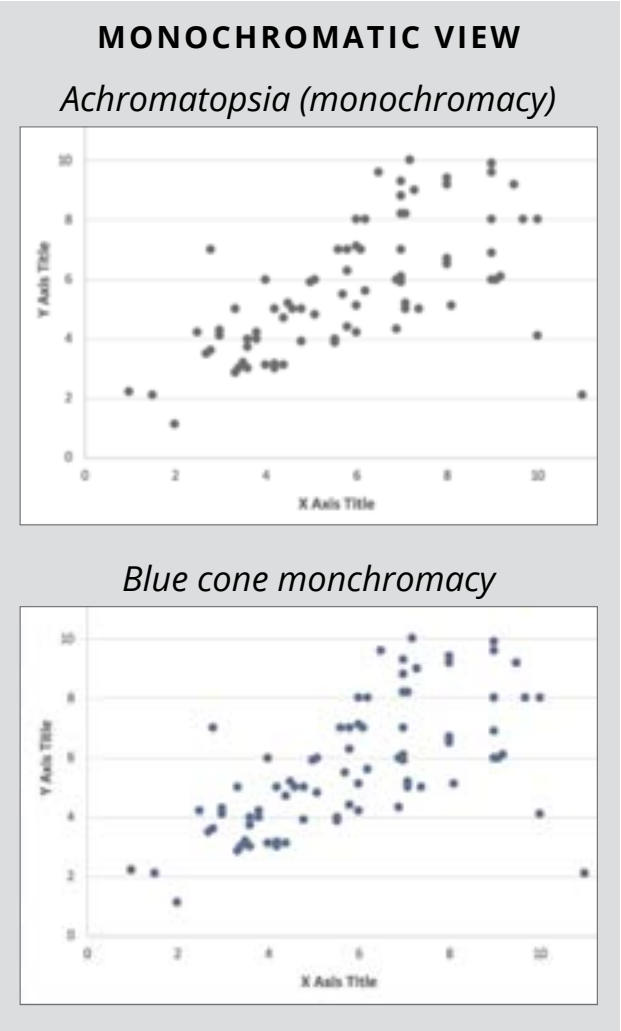
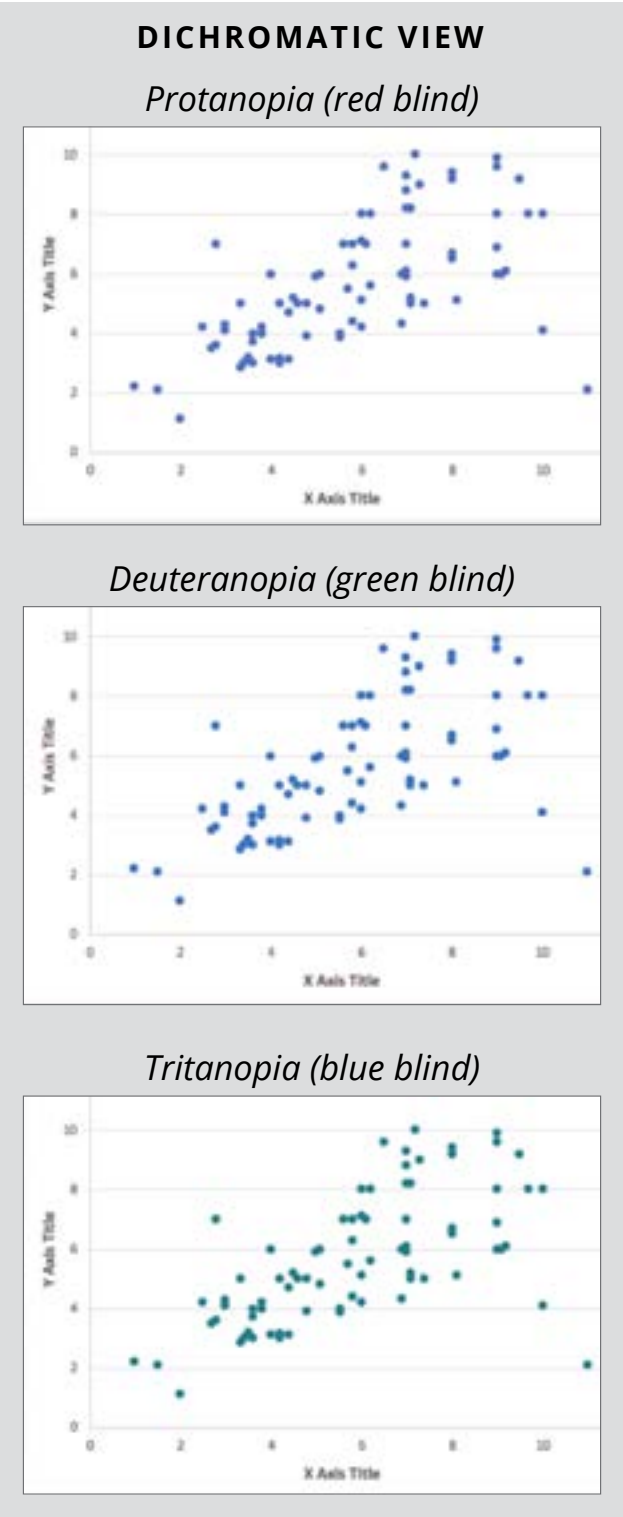
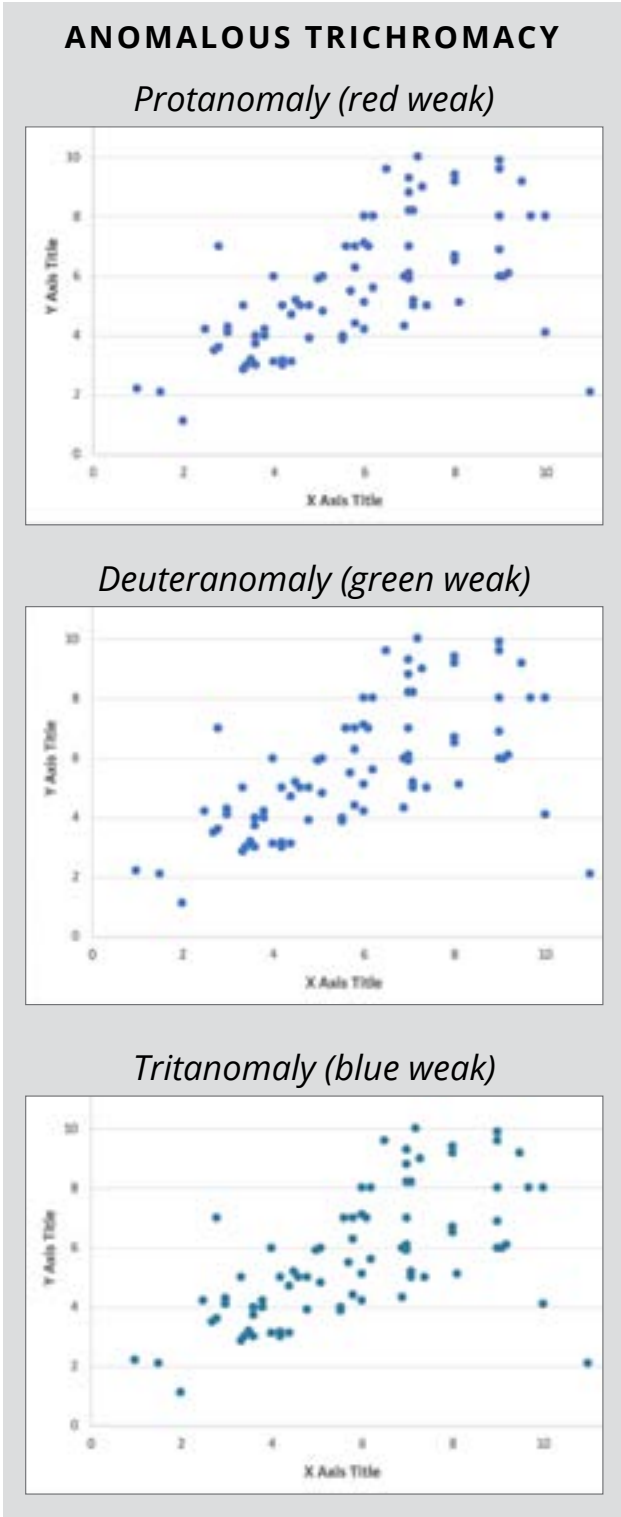
What do we need to consider for scatter plots?

A simple scatter plot is shown below, and the images on the right demonstrate how this graph may be perceived by readers with various forms of colour blindness.



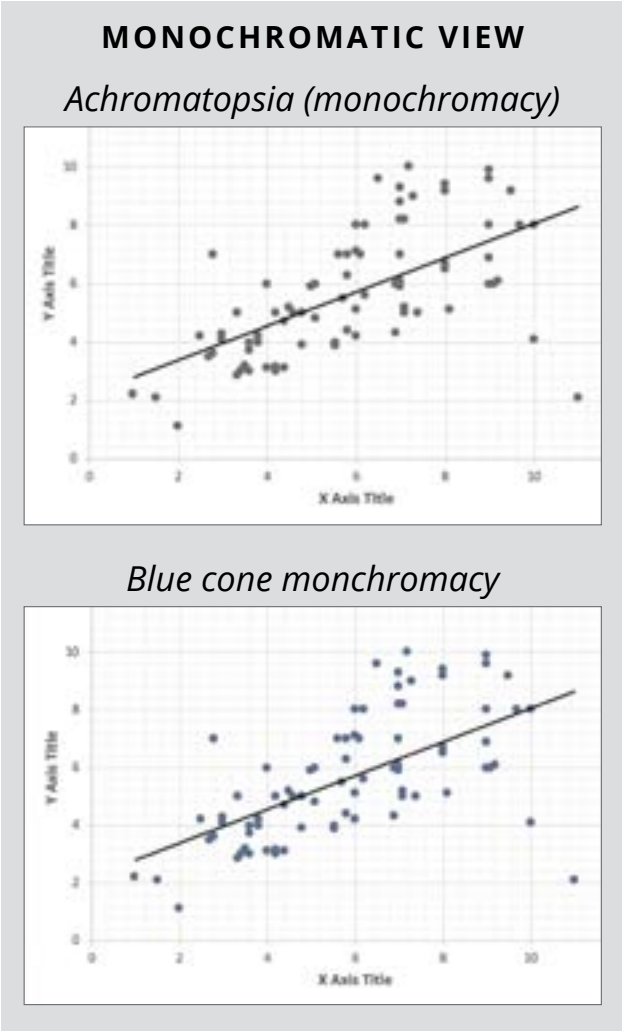
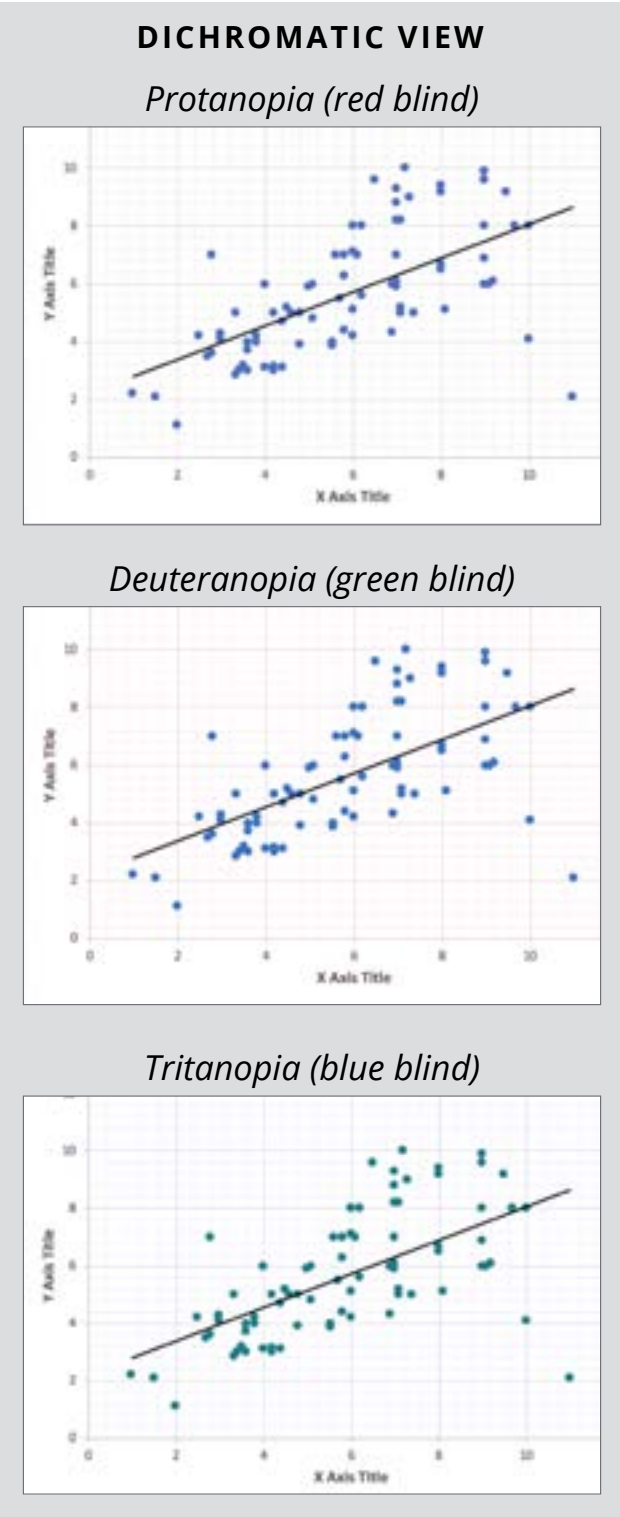
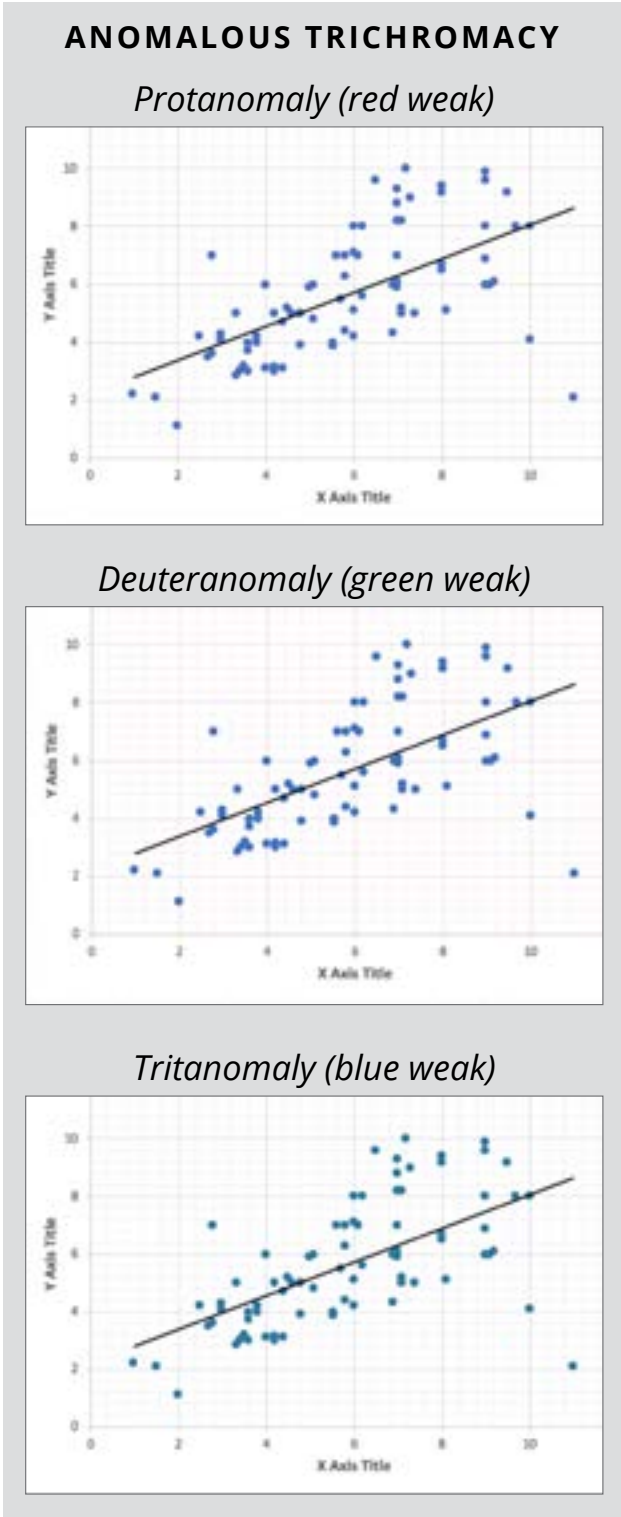
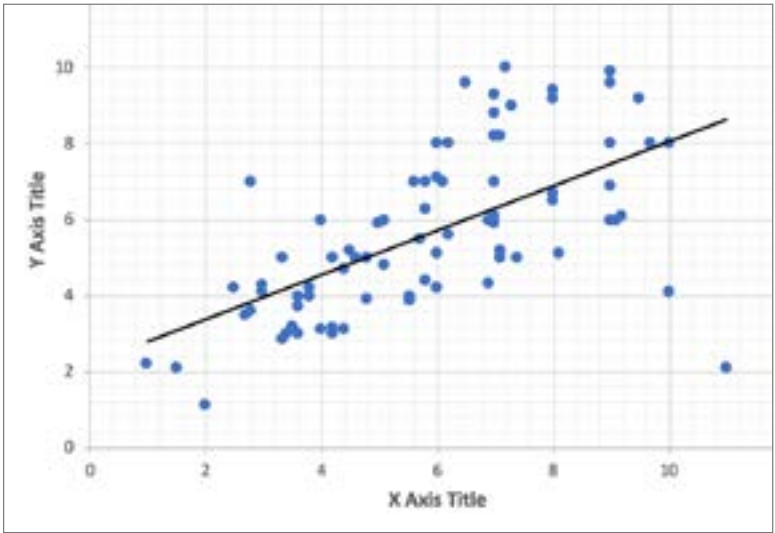
In most cases the reader may be able to see the correlation between the x and y axes, even if not perceiving the colour as intended. But, there are instances where the relationship isn't easily discernible, e.g. overlapping points. Therefore, in order to be accessible, the scatter graph should either:

- add a trend line where appropriate
- use transparency and/or a border to data points to prevent clustering
- introduce a categorical third value using shapes



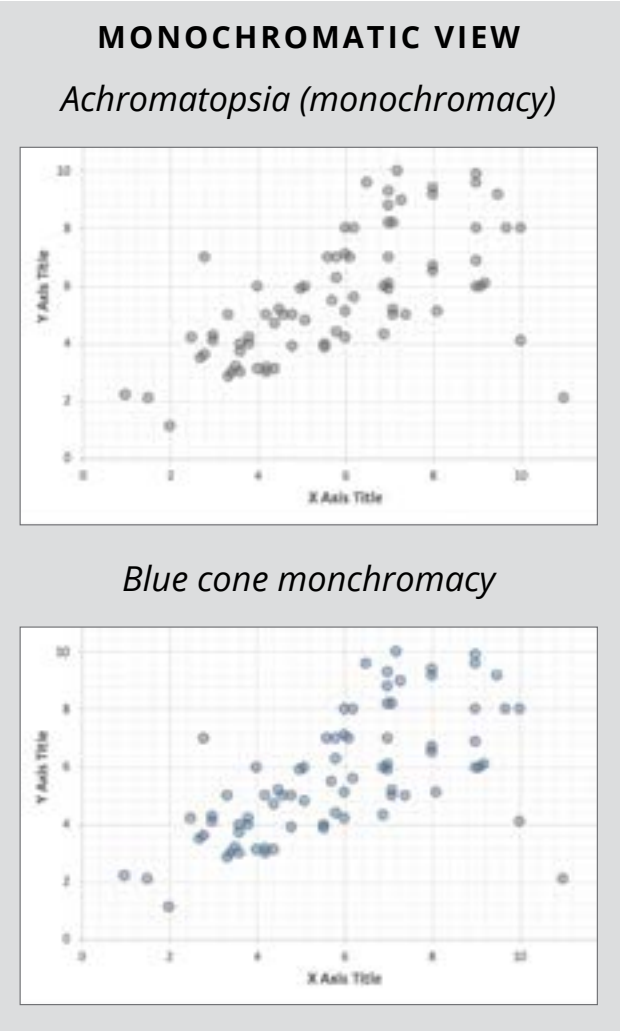
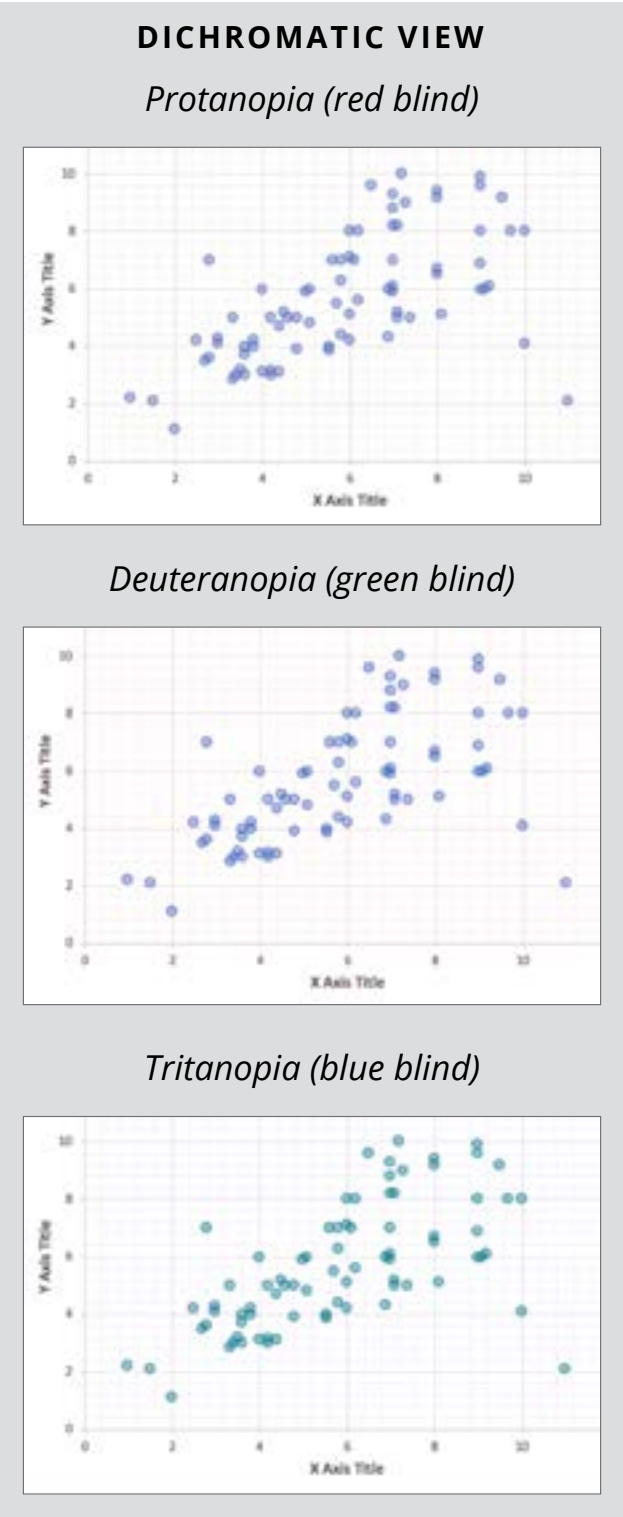
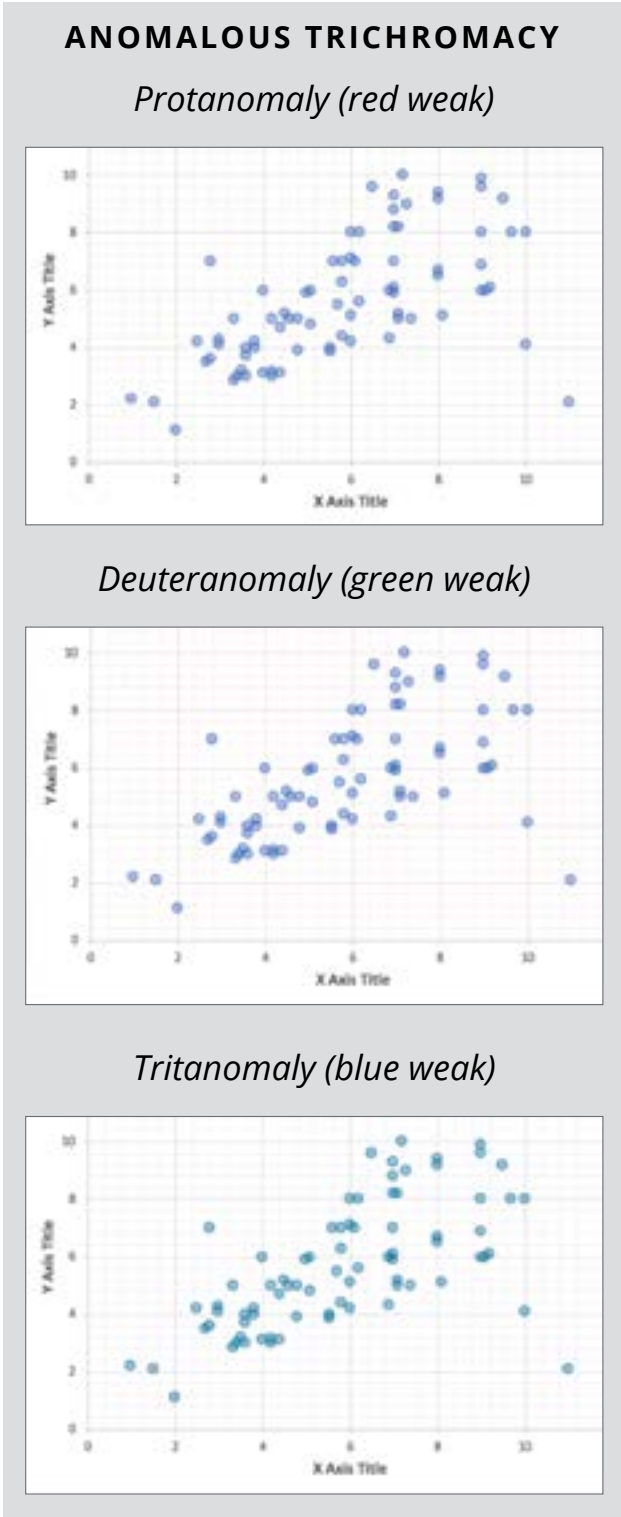
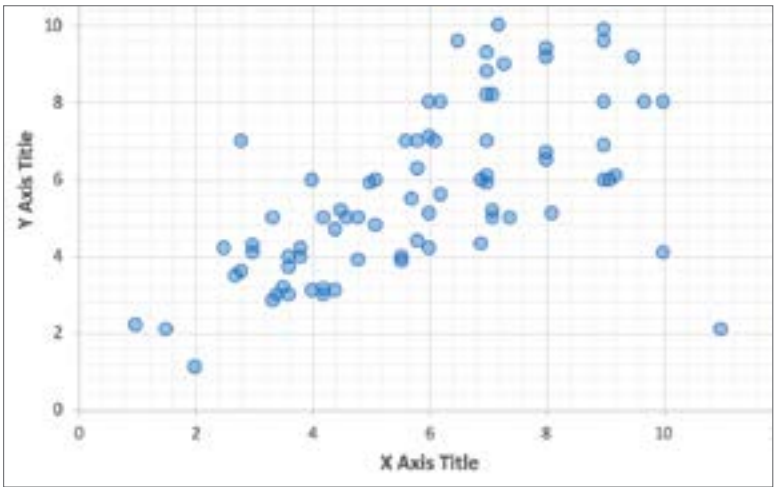
Option 1: Add a trend line where appropriate

Here the line graph has a trend line. A trend line is a straight line that best represents the points on a scatterplot that is used to show the pattern of the data. This line may show a positive trend or a negative trend, and if there are any unusual points that are affecting the computation of the trend.



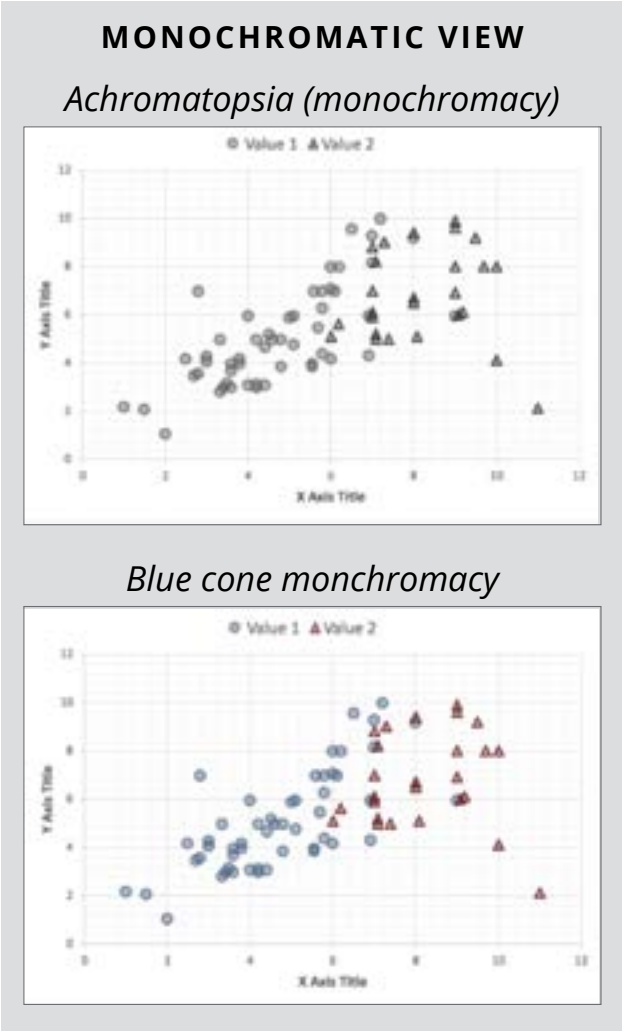
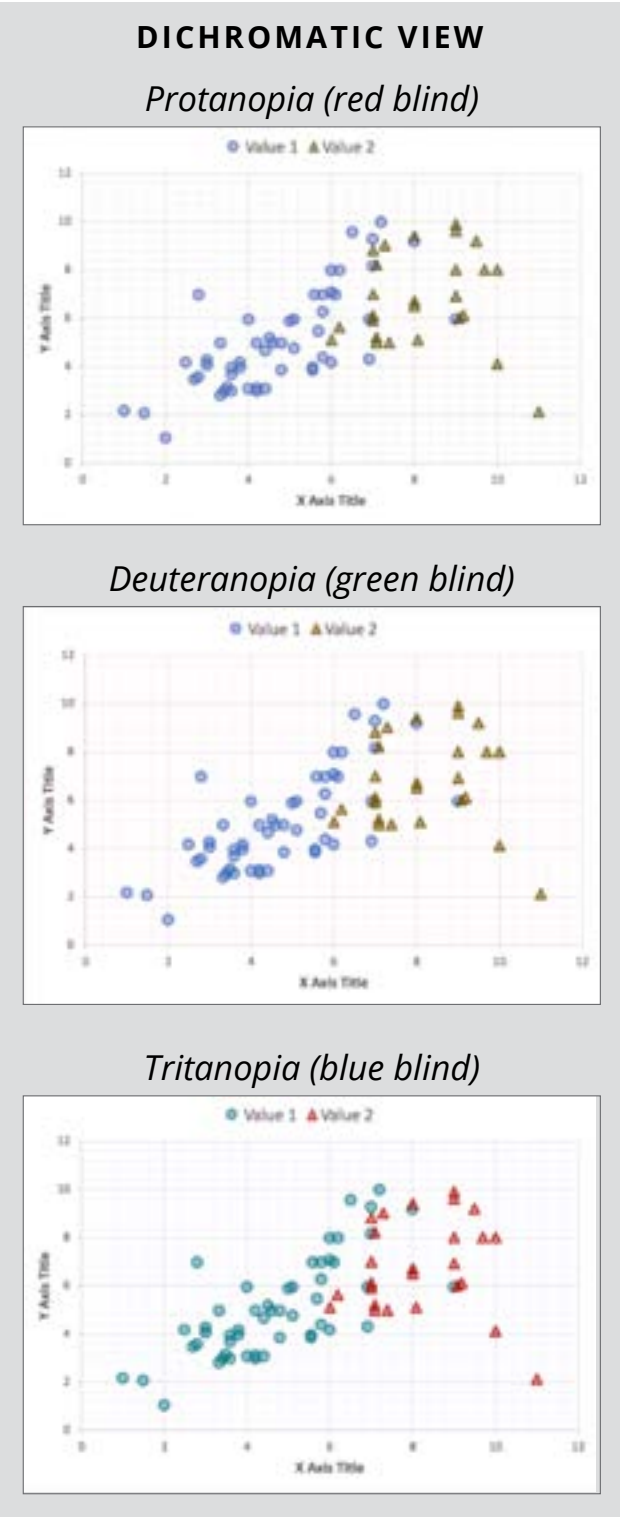
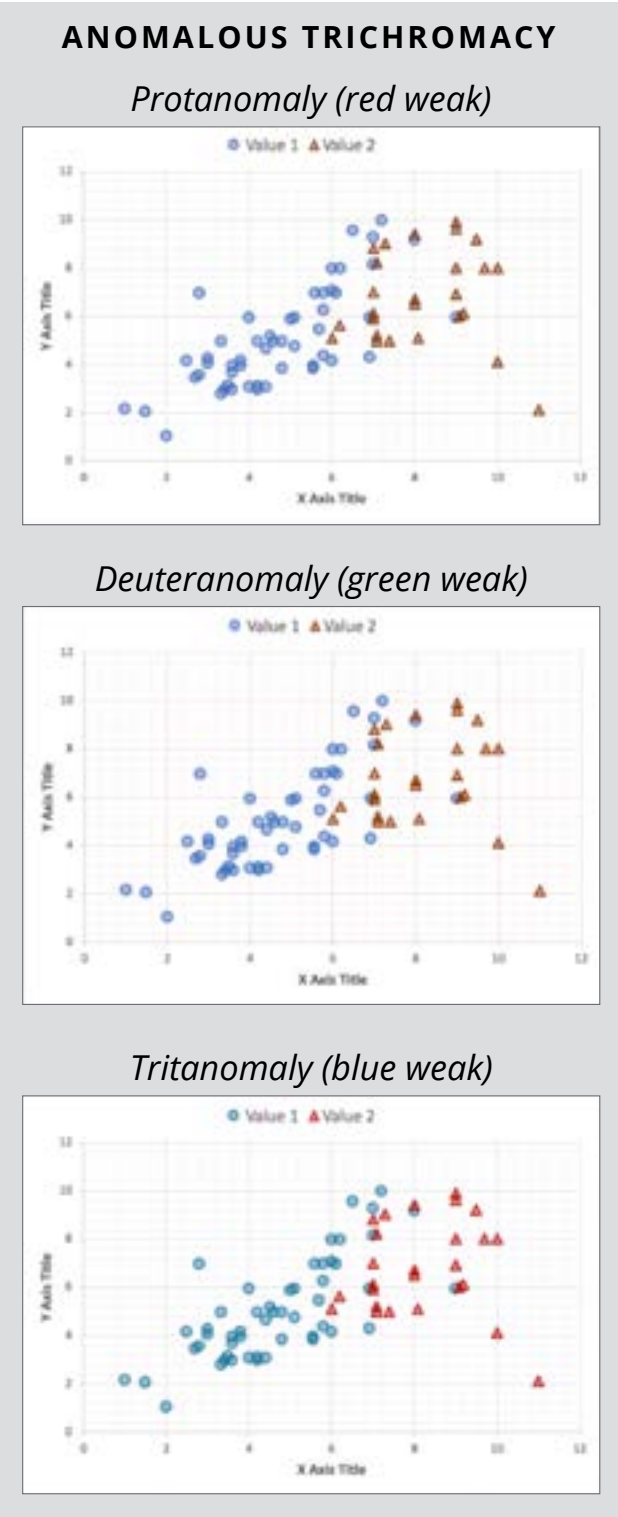
Option 2: Use transparency and/or a border on data points

Here the scatter plot has both transparency and a border applied to each data point—additional signals to differentiate between overlapping or clustered points.



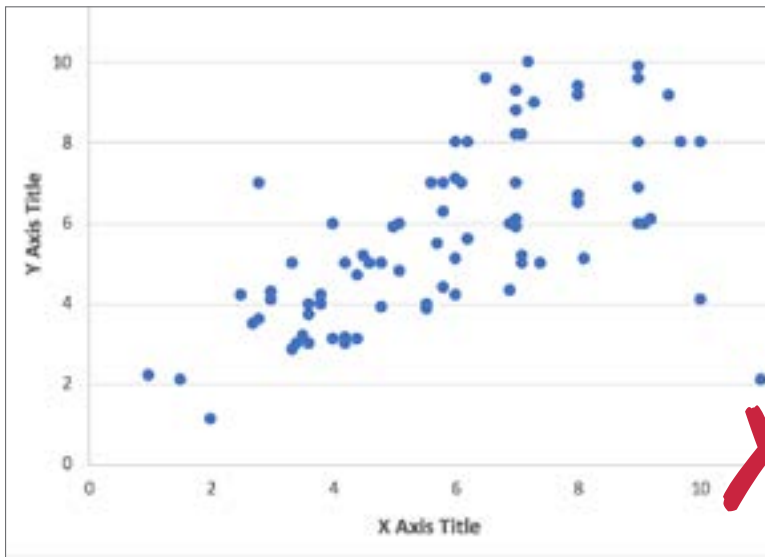
Option 3: Introduce a categorical third value using shapes

Here the scatter plot has transparency and a border applied to each data point, as well as the introduction of a new data point shape. This acts as additional signal to differentiate between a third category.



Cheat sheet

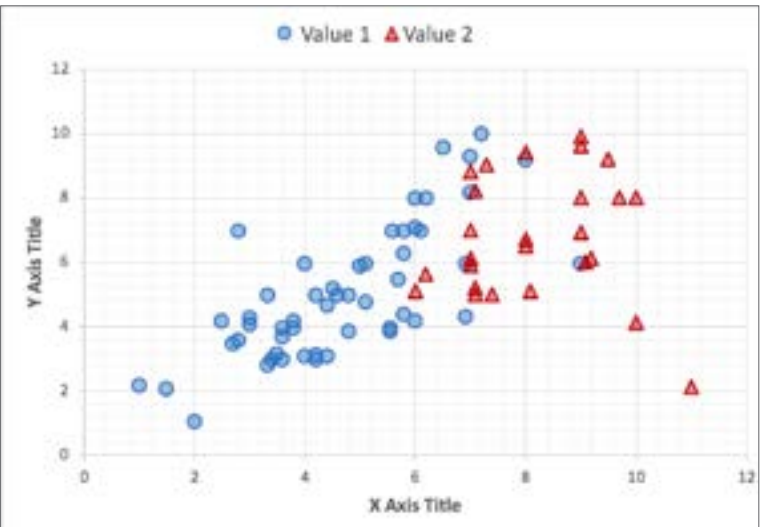
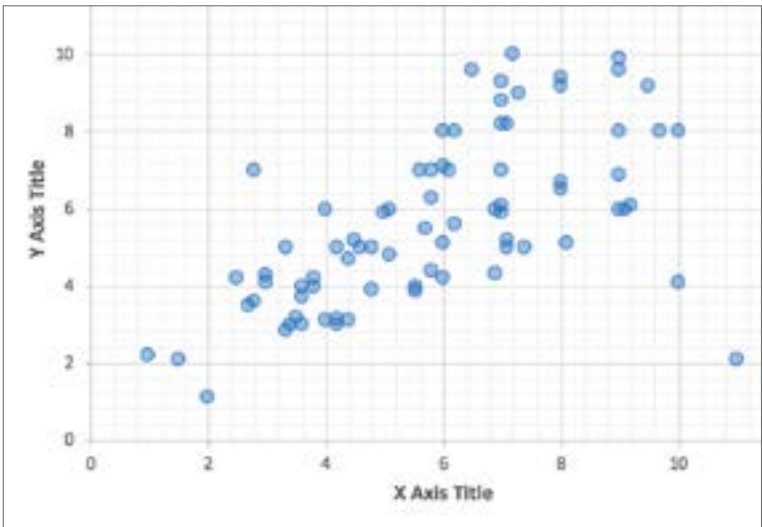
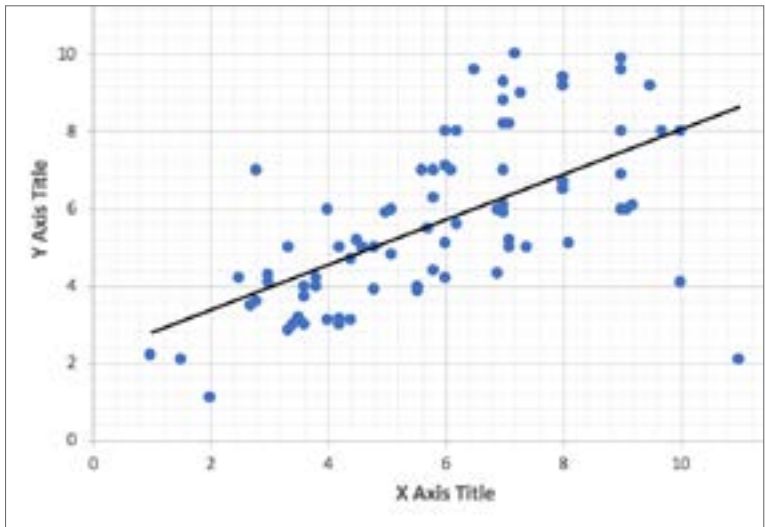
✗ Do not solely rely on colour to identify the relationship between the x and y axes



✗ NOT ACCESSIBLE

The data points on a scatter plot need to either:

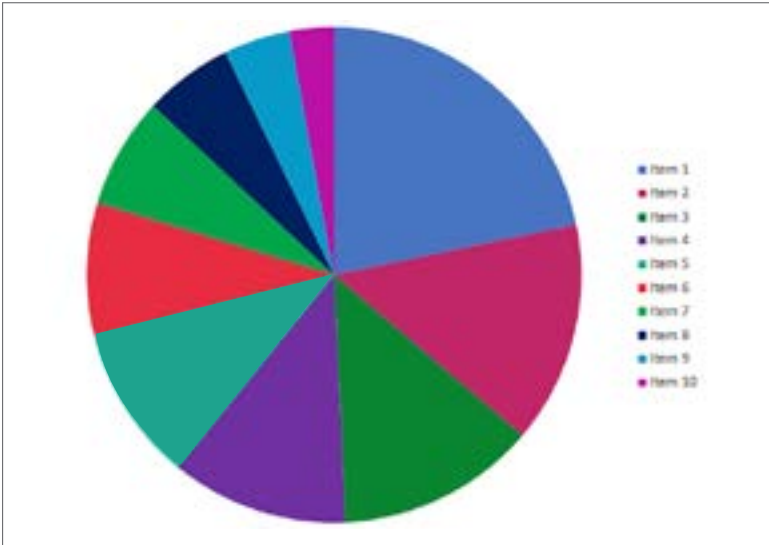
- ✓ use a trend line where appropriate
- ✓ use transparency and/or a border to data points to prevent clustering
- ✓ introduce a categorical third value using shapes



ACCESSIBLE

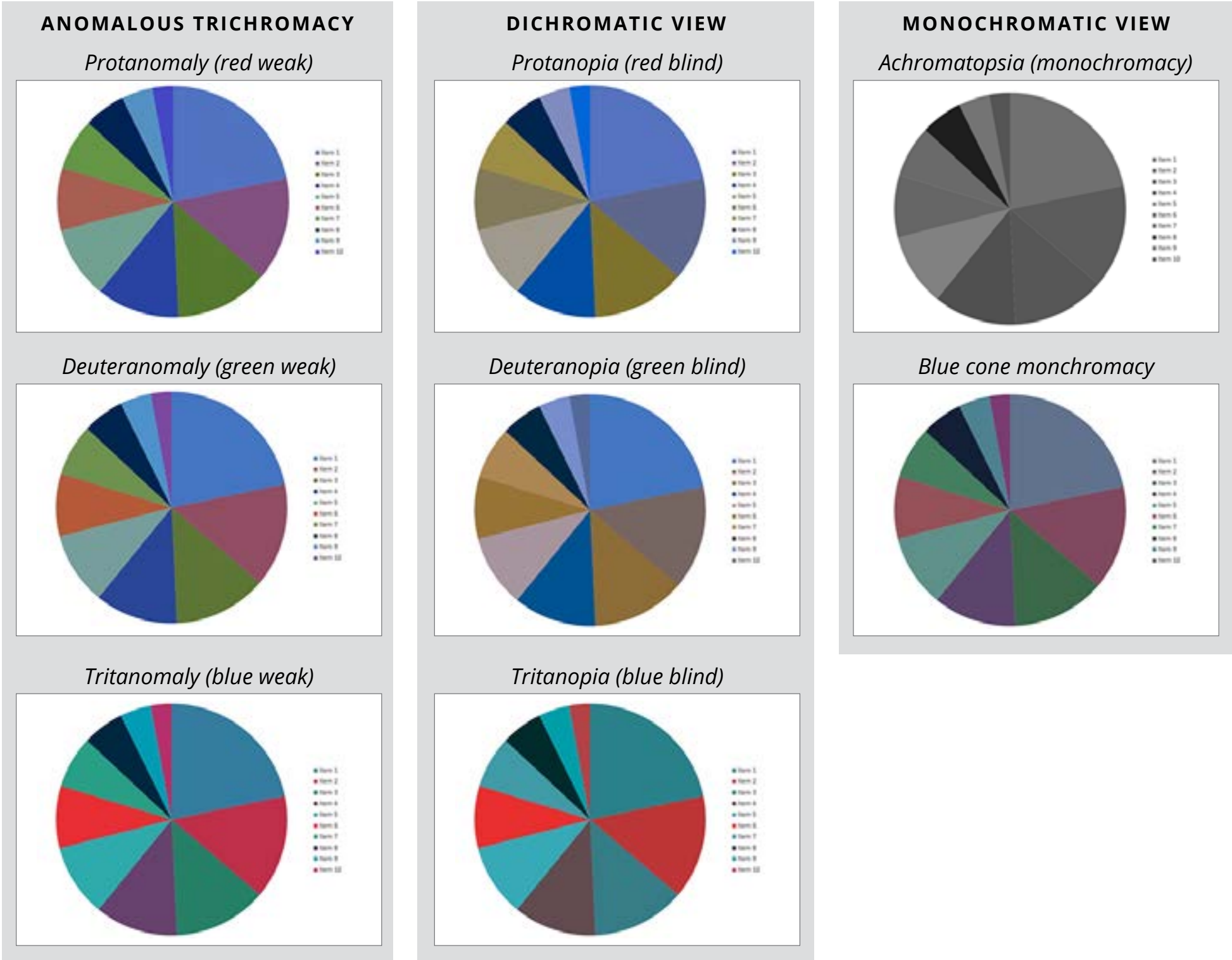
What do we need to consider for pie charts?

To understand a pie chart you have to discern each slice of the pie chart from the others. A simple pie chart is shown below, and the images on the right demonstrate how this graph may be perceived by readers with various forms of colour blindness.



In many cases the reader may be able to see the correlation between the parts of the pie chart. But, there are instances where the relationship isn't easily discernible, e.g. small "slices", or colours that are not perceived as intended. Therefore, in order to be accessible, the pie chart should either:

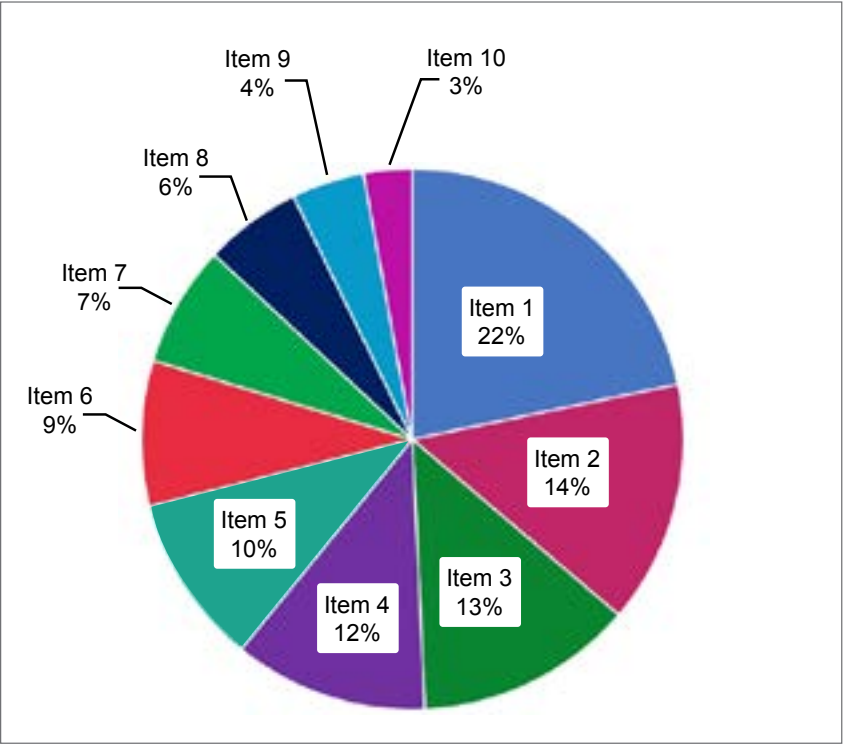
- add divider lines (and label the parts/slices)
- use patterns instead of just colour
- use a table instead of a pie chart, if differences are marginal



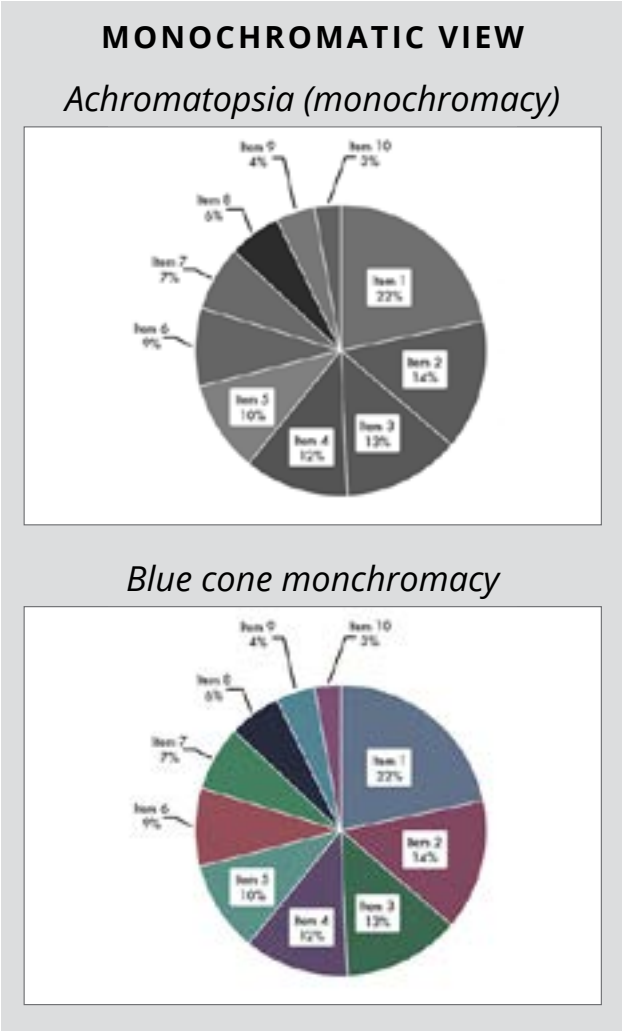
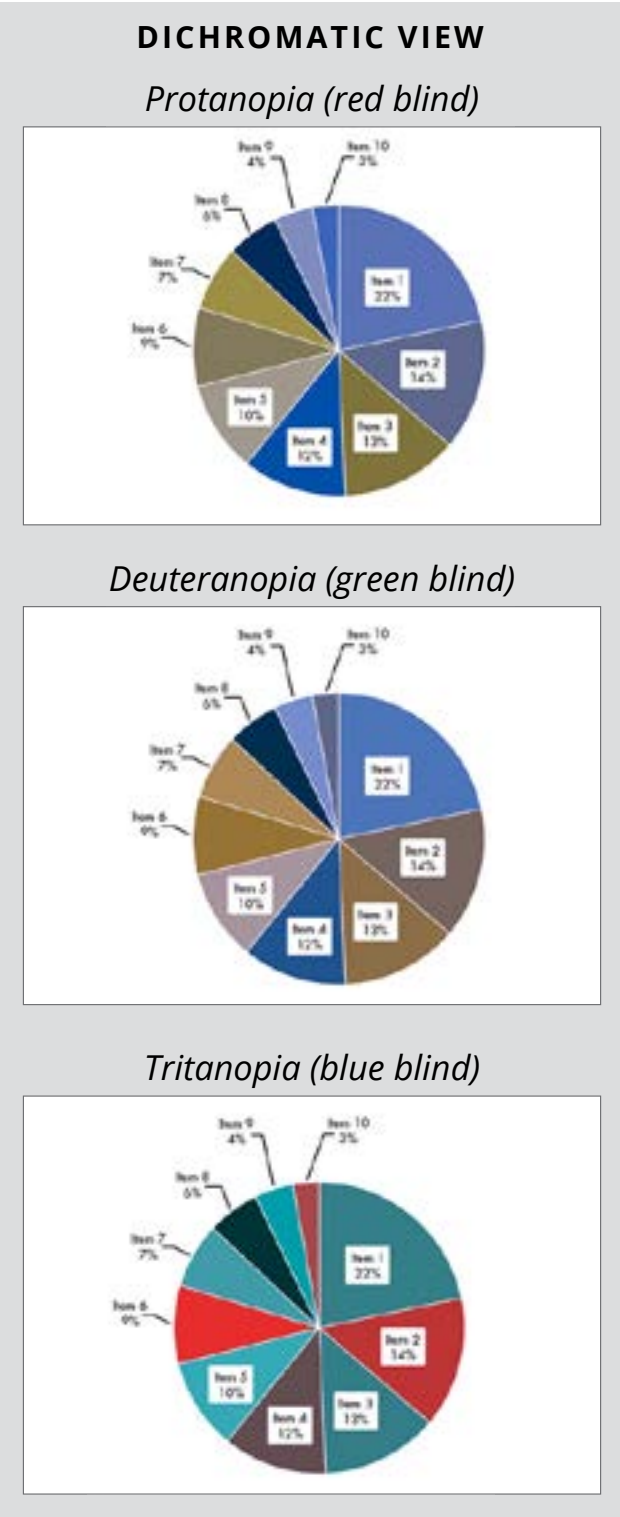
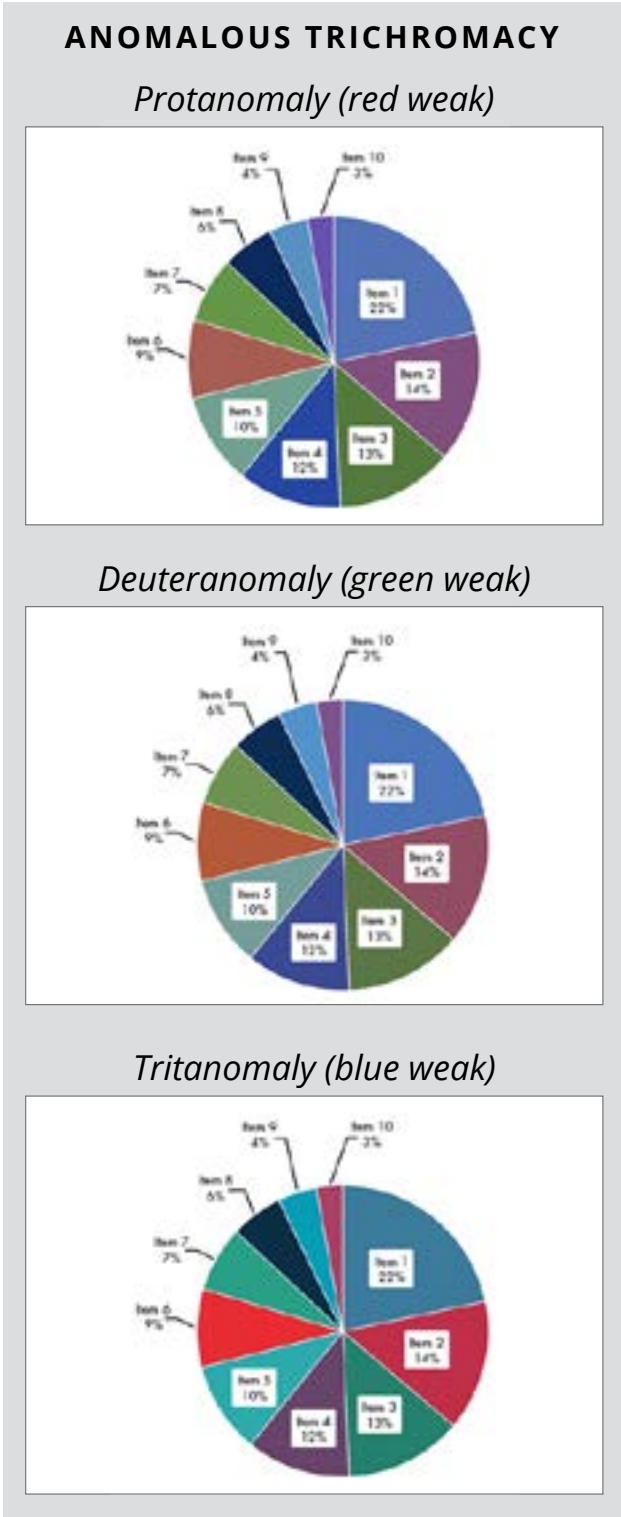
» The next pages show how these techniques ensure the content can be understood by a reader with any form of colour blindness.

Option 1: Add divider lines, and label the parts/slices

Here the pie chart has contrast divider lines, to separate any touching colours, as the colours are used to differentiate between categories. Additionally, both the category name and value are included within or near the “slice” or category to indicate the differences in data. A legend is not required.



N.B. Ensure there is sufficient colour contrast between the background, the slices, and the labels. In order to contrast on a white background, the non-text items (the slices) should have a contrast ratio of at least 3:1. Text should have a contrast ratio of 4.5:1. Under these circumstances, it is unlikely black text can sit directly on the slices. In the example above, the labels are encased in a white box, or if space does not allow, using a leader line.

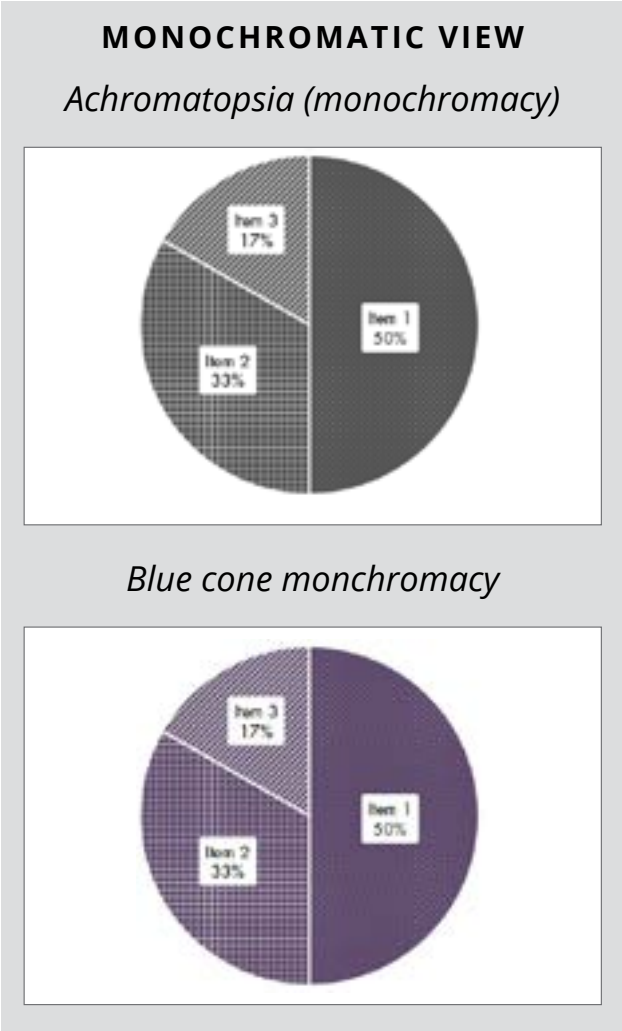
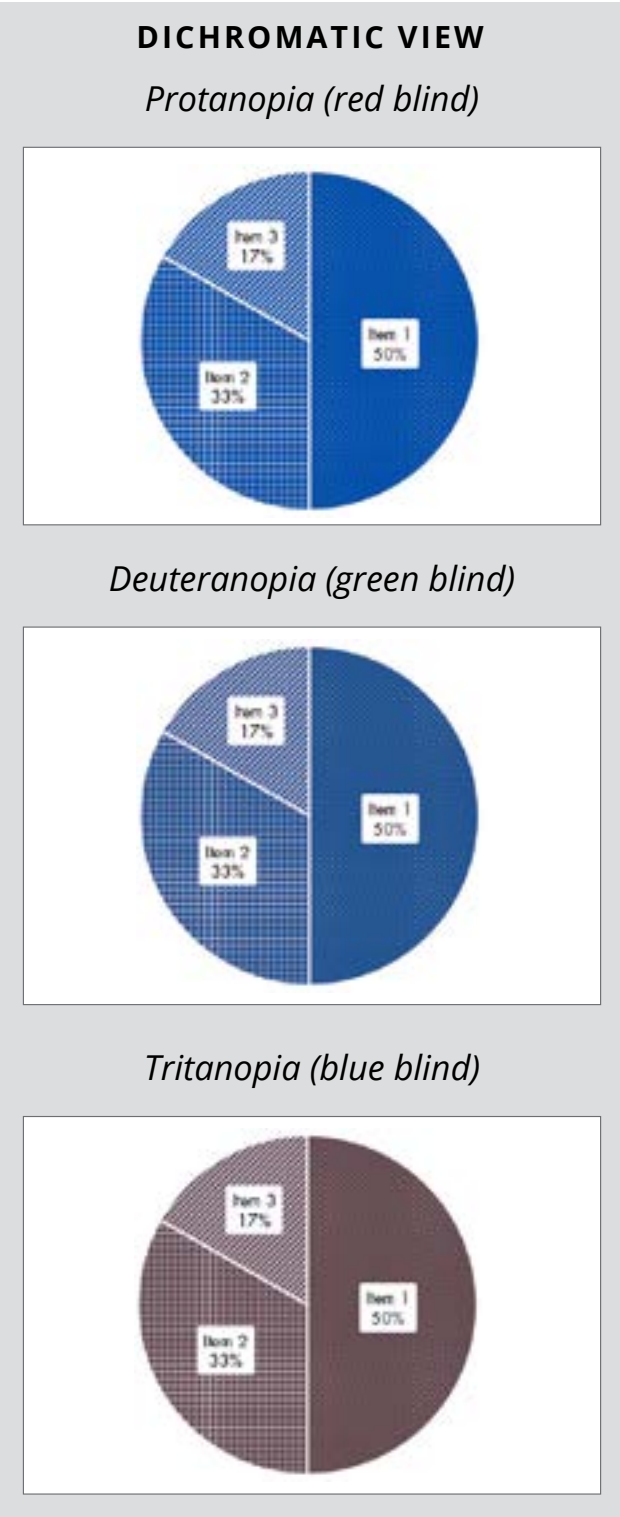
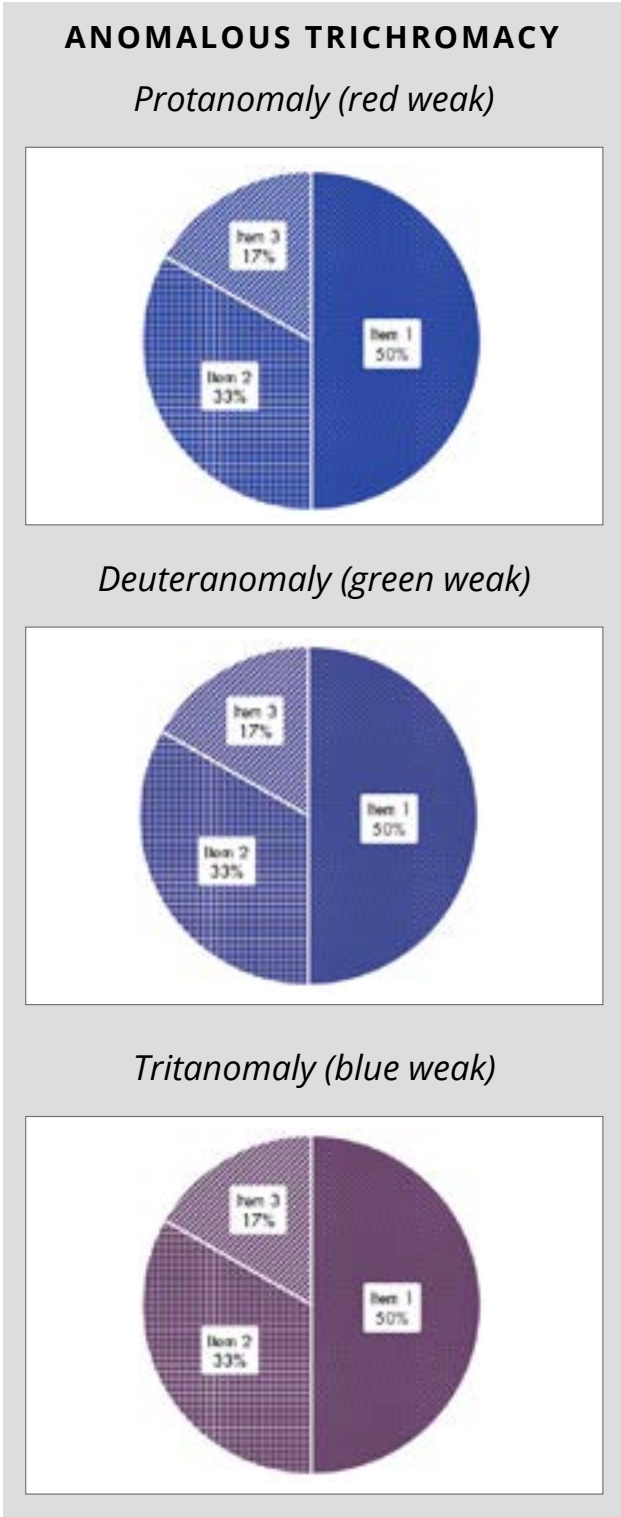


Option 2:
Use patterns instead of just colour

Here the pie chart uses patterns to differentiate between various groups of data, instead of using colour. Patterns and texture are well-established alternatives to colour coding areas. A legend could be used instead of data labels, if the chart is simple, but the patterns in the legend would need to be recognizable.



N.B. Patterns could pose a scalability challenge. When patterns are used, the sections should be outlined in a 1px stroke to ensure the patterns are recognizable.

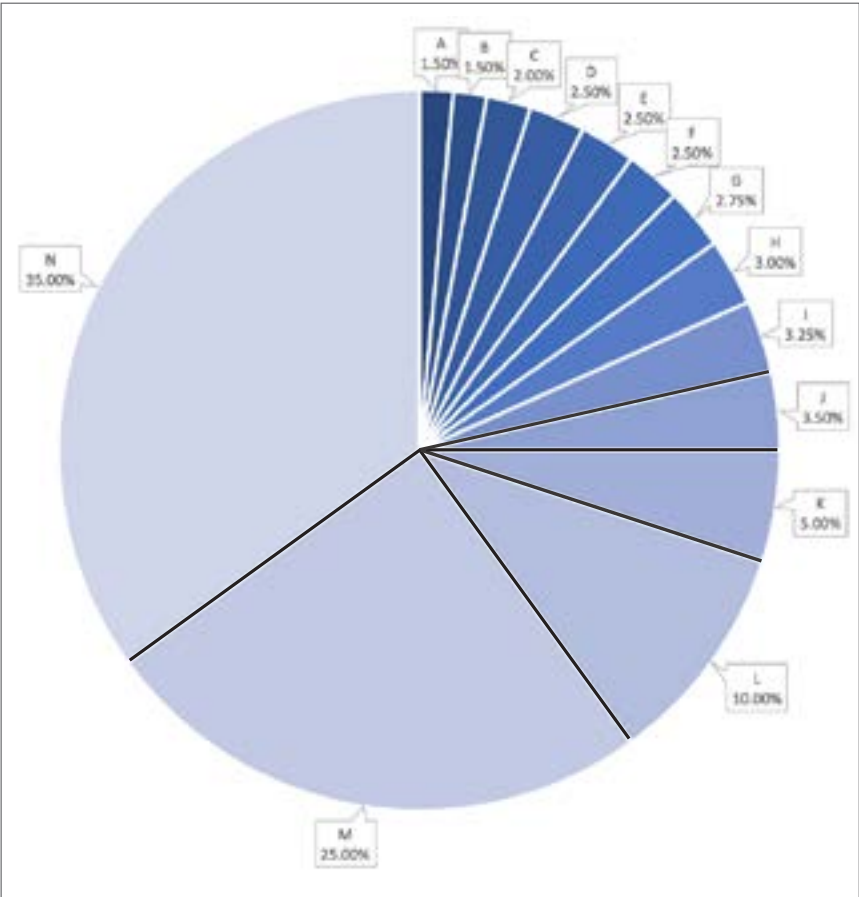


Option 3: Use a table instead of a pie chart, if differences are marginal

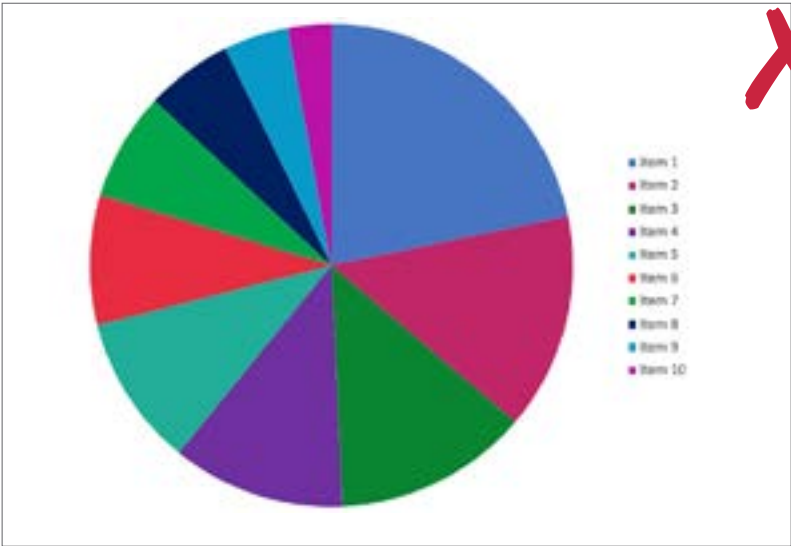
Here the pie chart shows several values with marginal difference in value. Even if the previous options were applied (divider lines, data labels, and patterns), it would be difficult to read, especially if the categories have long titles. In some cases, presenting raw numbers via a table will be easier for readers to comprehend.

TABLE TITLE

CATEGORY	VALUE
Category A	1.5
Category B	1.5
Category C	2.0
Category D	2.5
Category E	2.5
Category F	2.5
Category G	2.75
Category H	3.0
Category I	3.25
Category J	3.50
Category K	5.00
Category L	10.00
Category M	25.00
Category N	35.00



Cheat sheet



NOT ACCESSIBLE

Do not solely rely on colour to identify the differences between categories

The pie chart needs to either:

- ✓ use divider lines (and label the parts/slices)
- ✓ use patterns instead of only solid colour
- ✓ use a table instead of a pie chart, if differences in data values are marginal

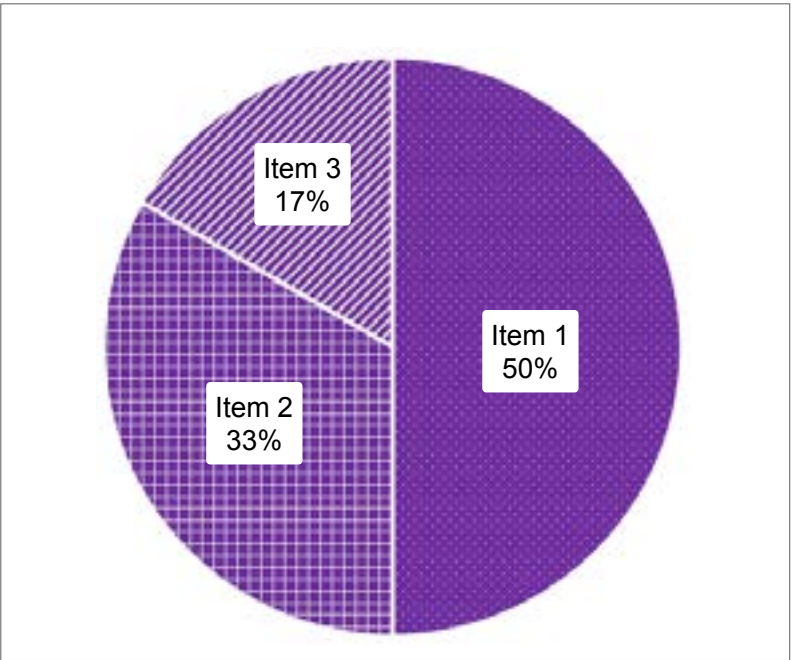
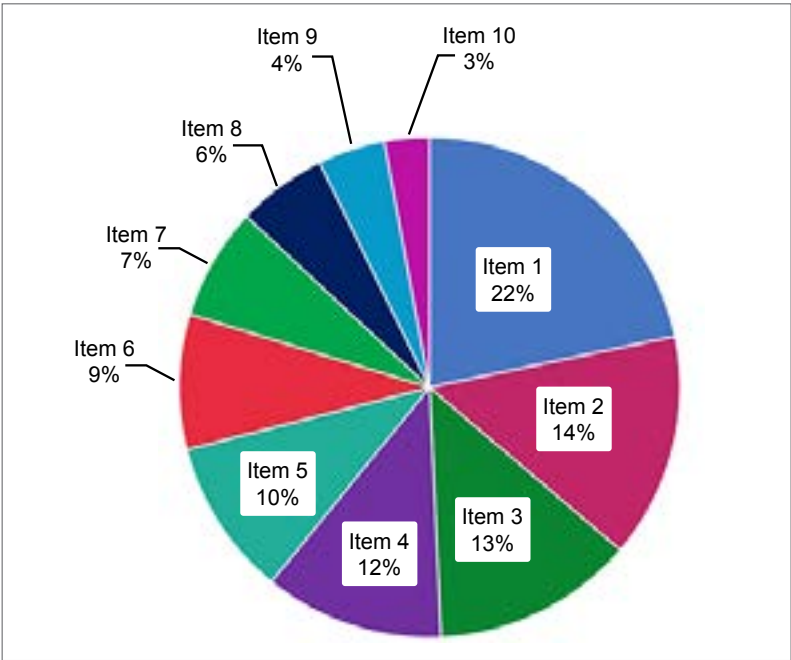


TABLE TITLE	
CATEGORY	VALUE
Category A	1.5
Category B	1.5
Category C	2.0
Category D	2.5
Category E	2.5
Category F	2.5
Category G	2.75
Category H	3.0
Category I	3.25
Category J	3.50

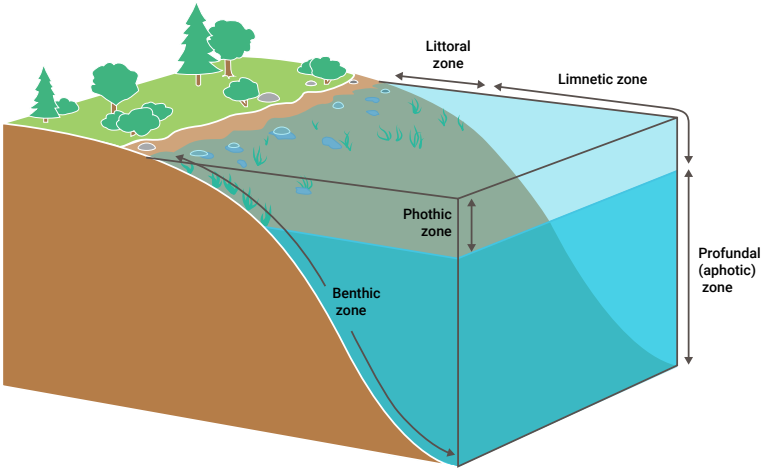
ACCESSIBLE

What do we need to consider for diagrams and figurative diagrams?

In both cases, it is important that:

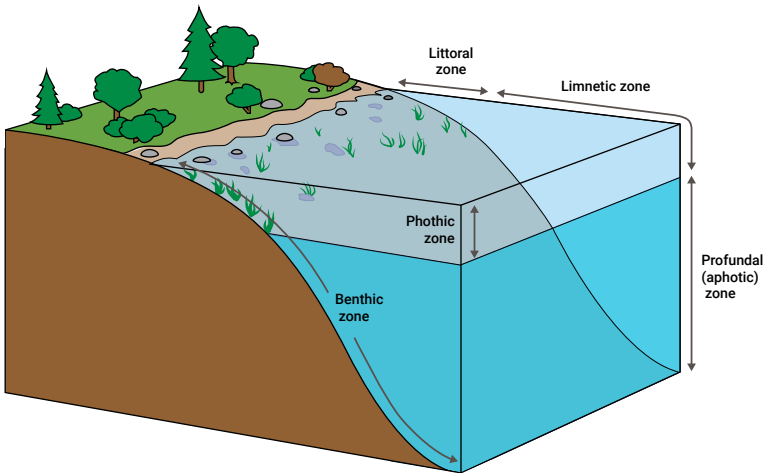
- meaning should not be communicated only through colour – if items need to be grouped, consider other ways of achieving this, such as layout/headings
- text labels and leader lines refer to areas of the diagram as necessary, and labelling is clear so that it contrasts sufficiently with any background – avoid placing labels across parts/edges of the diagram

X NOT ACCESSIBLE



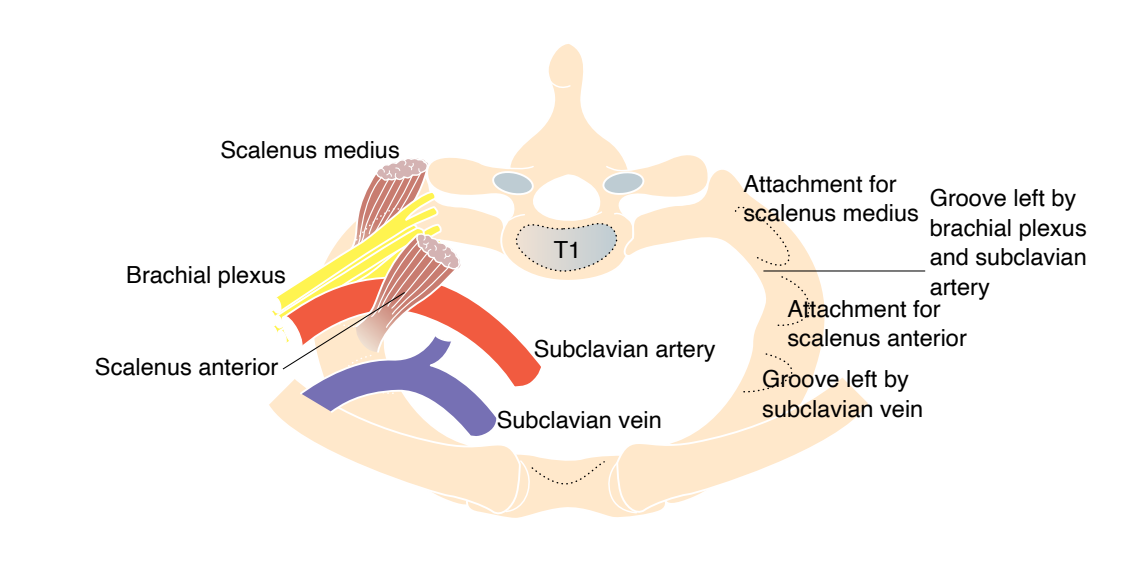
X colours don't contrast sufficiently

ACCESSIBLE

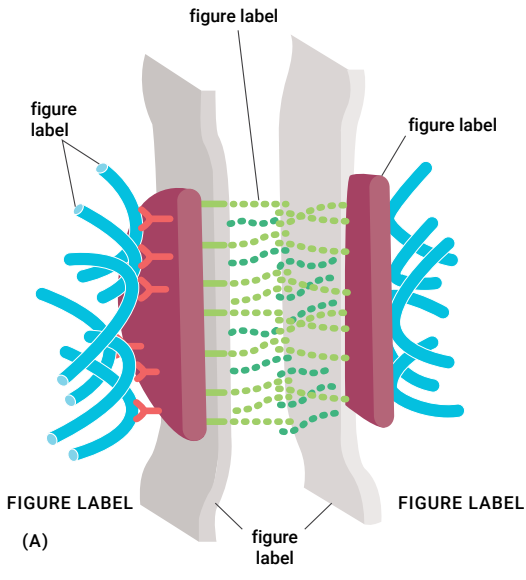


✓ use colours that give greater contrast and add outlines to define shapes

NOT ACCESSIBLE

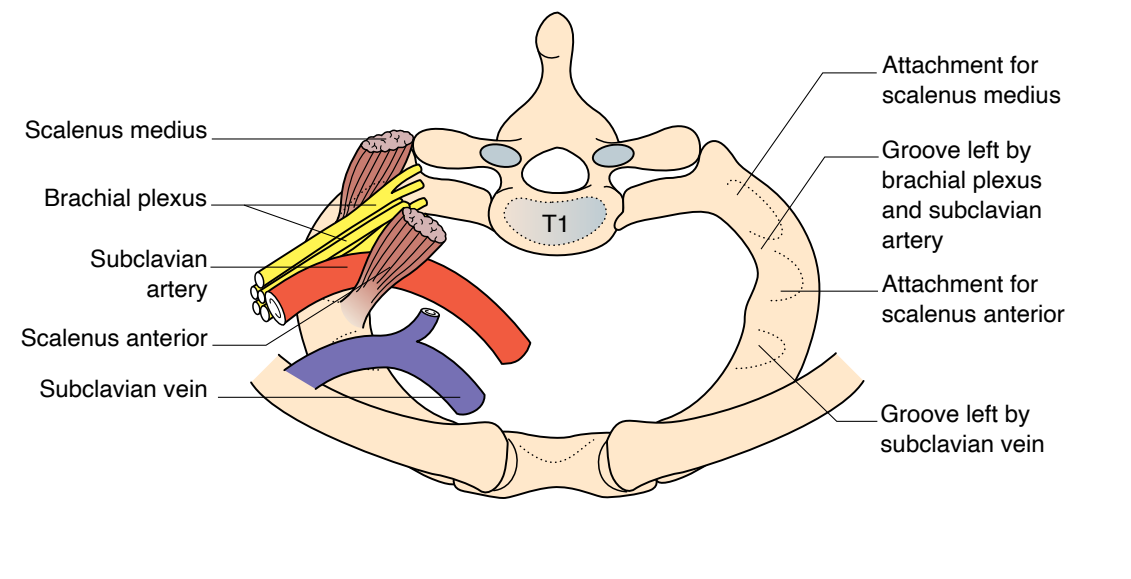


colours don't contrast sufficiently and labelling is unclear—it is difficult to tell what part of the diagram all the labels are referring to and the legibility of the text is impaired because it overlaps with the illustration

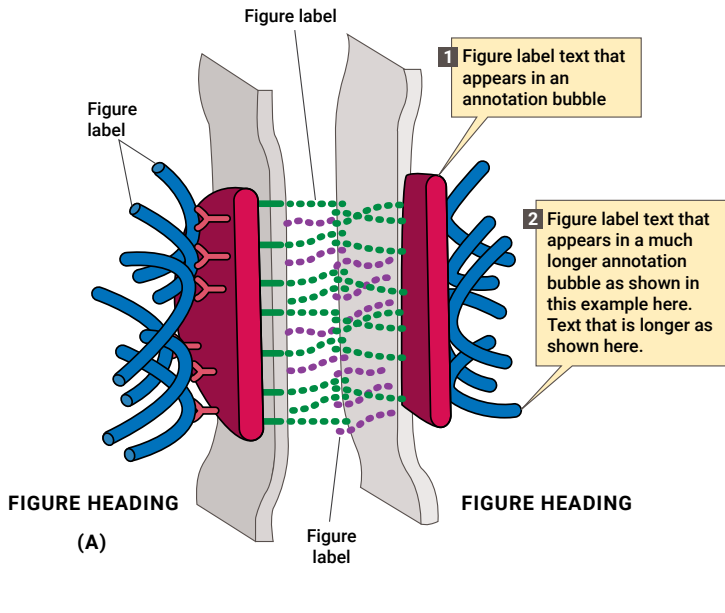


colours don't contrast sufficiently

ACCESSIBLE

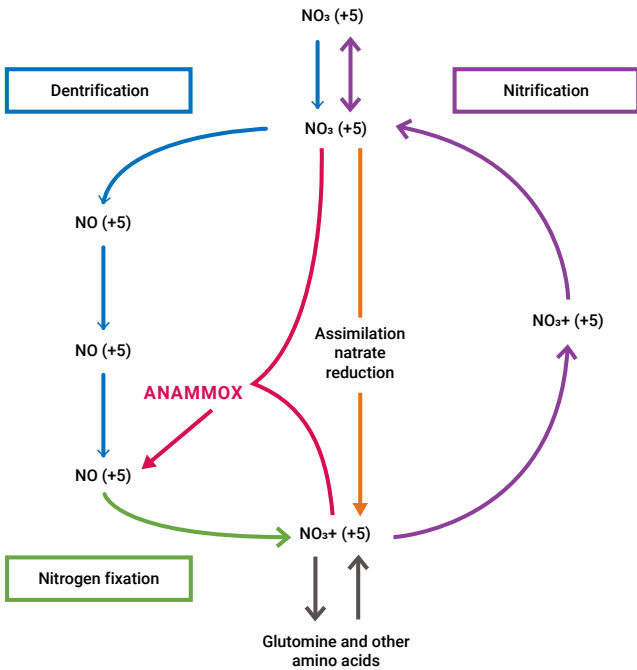


add outlines to define shapes and move labels so they are clear

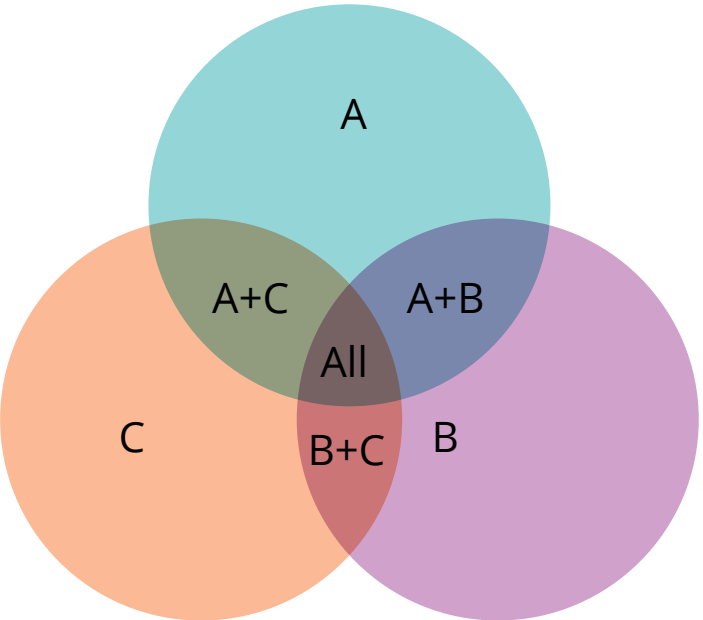


add outlines to define shapes and increase colour contrast

NOT ACCESSIBLE

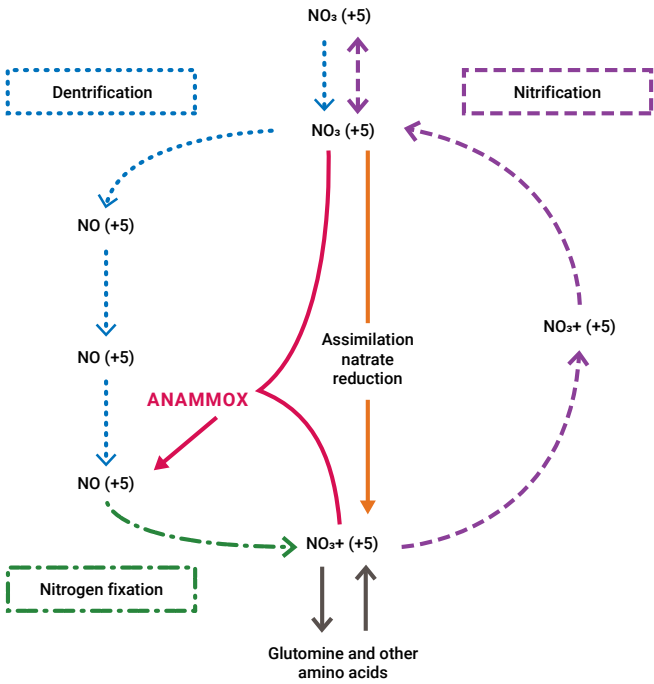


colour is used to link content to a label

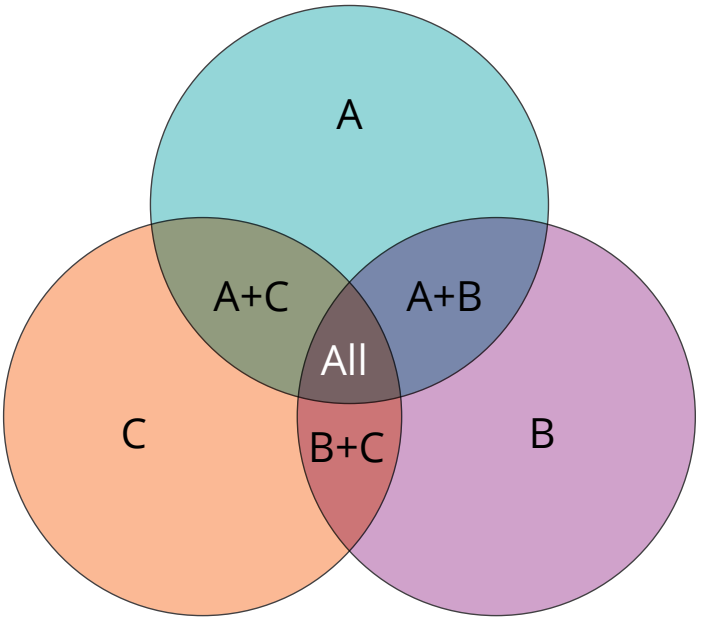


colours don't contrast sufficiently with each other or the background

ACCESSIBLE

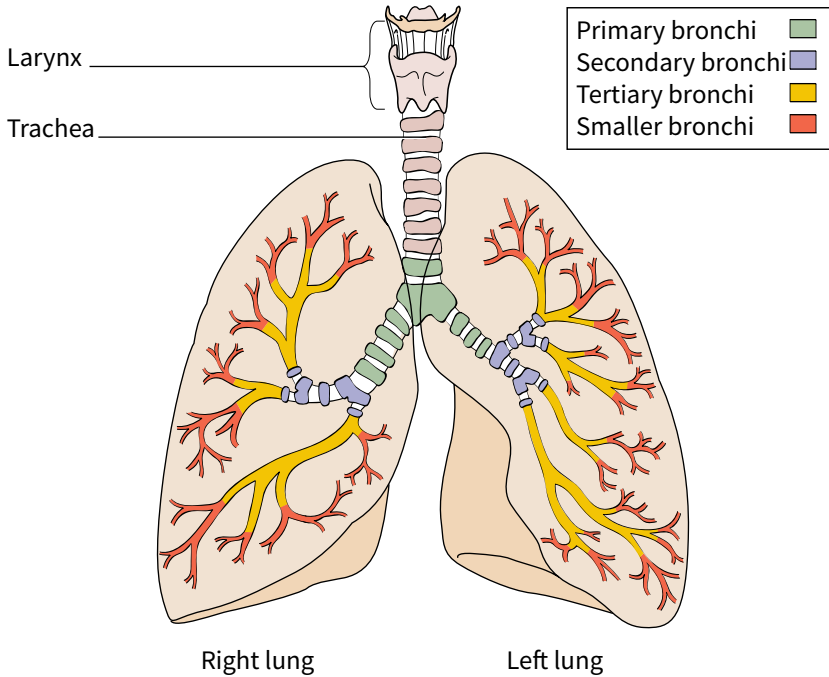


use varying stroke styles



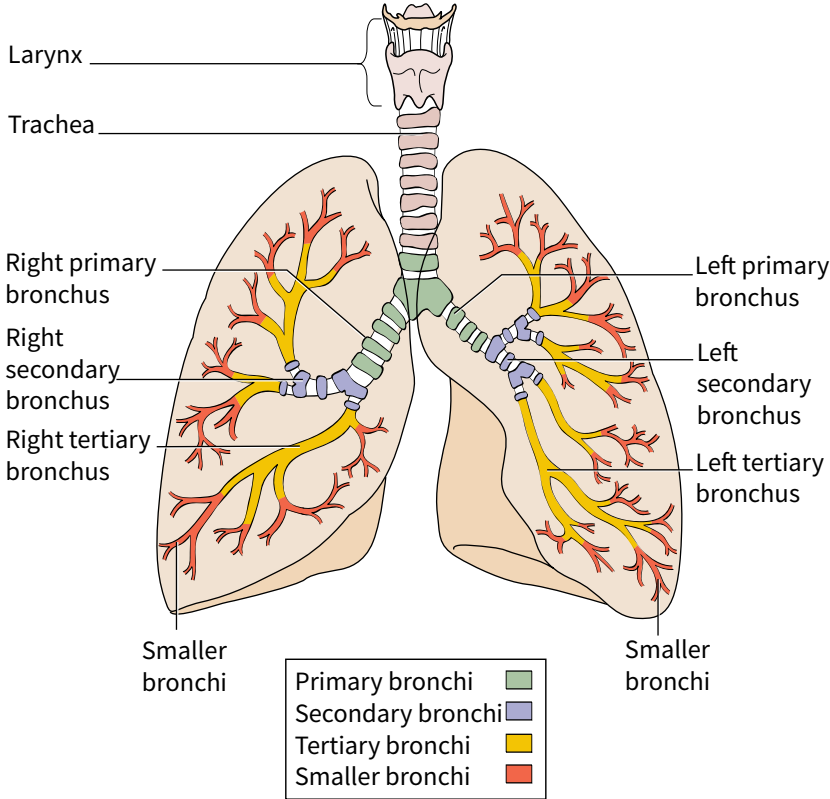
either increase the colour contrast, or add outlines to define the shapes

X NOT ACCESSIBLE

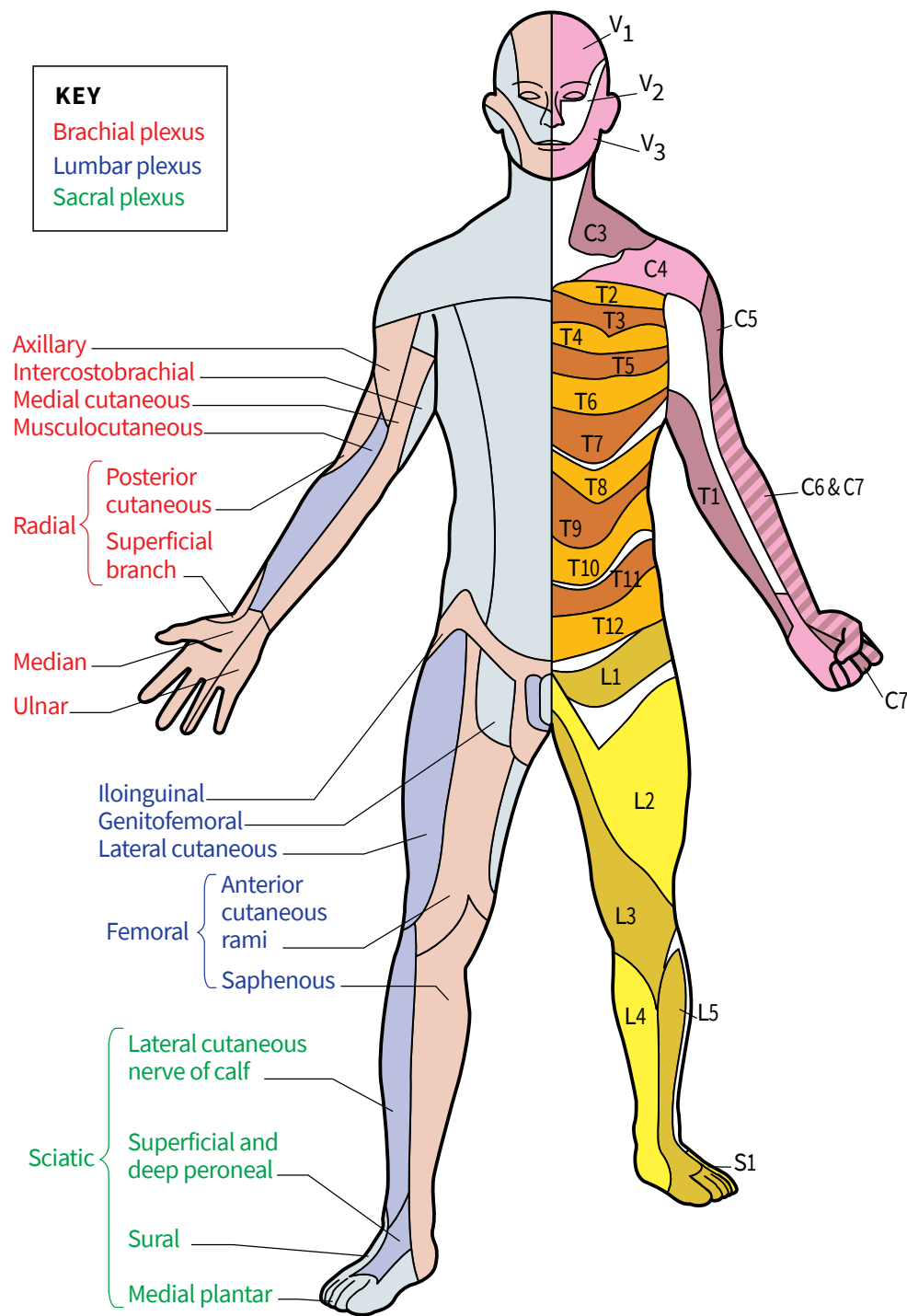


X colour is used to link the label to areas of the diagram

ACCESSIBLE

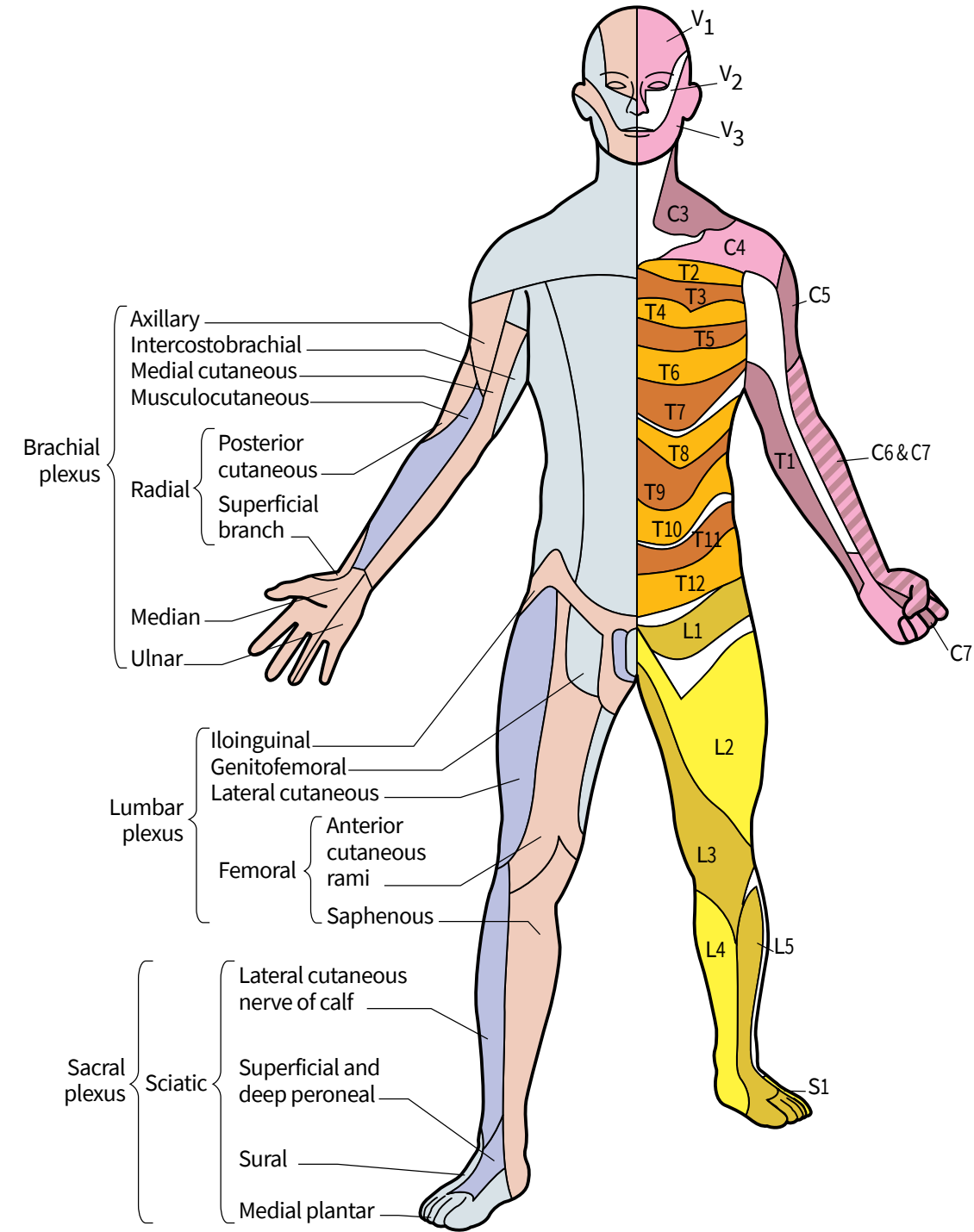


✓ use an alternative layout/labelling



X NOT ACCESSIBLE

X colour is used to group labels

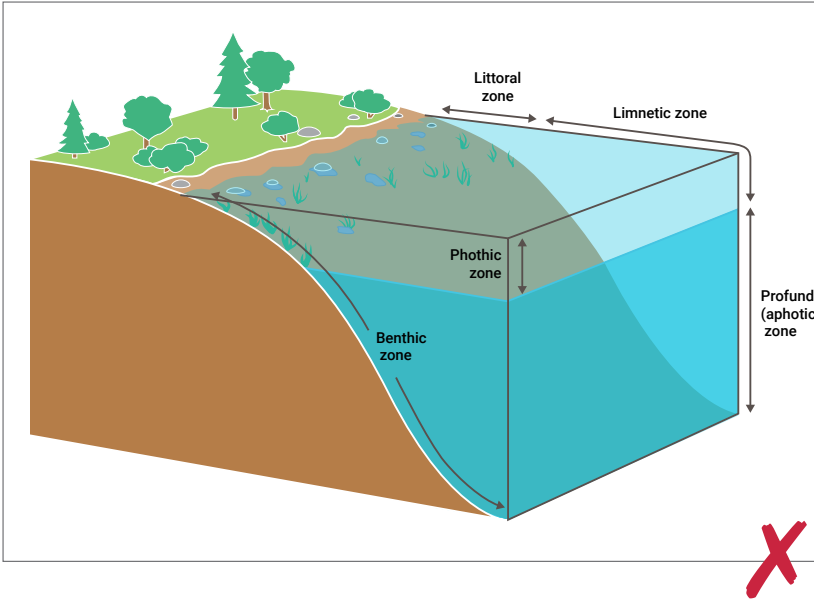


✓ ACCESSIBLE

✓ group the labels with headings

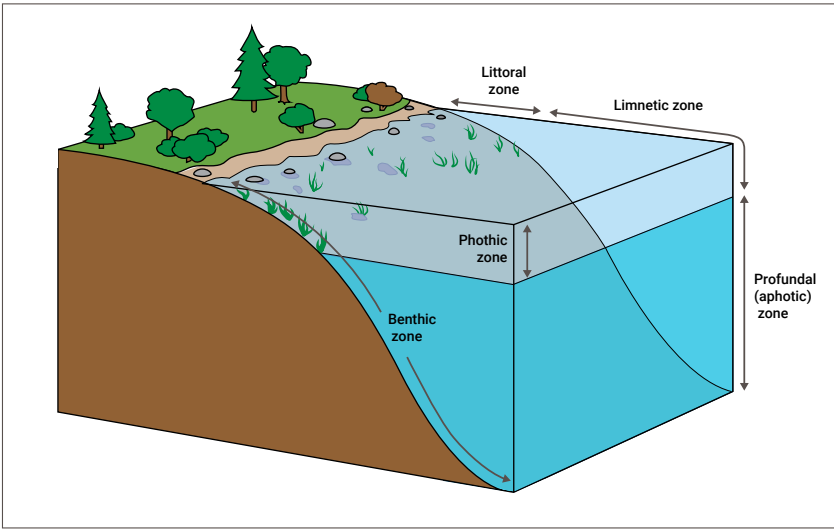
Cheat sheet

✗ Avoid using colours that do not meet contrast guidelines

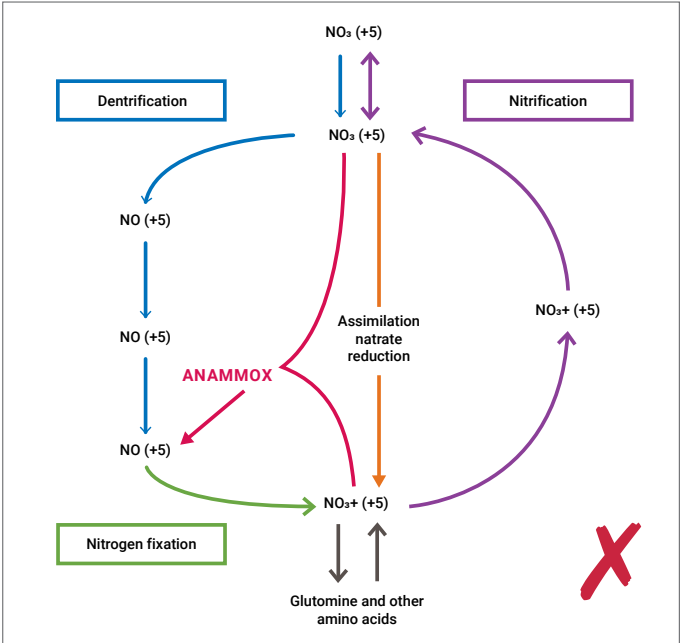


The accessibility of diagrams can be improved by:

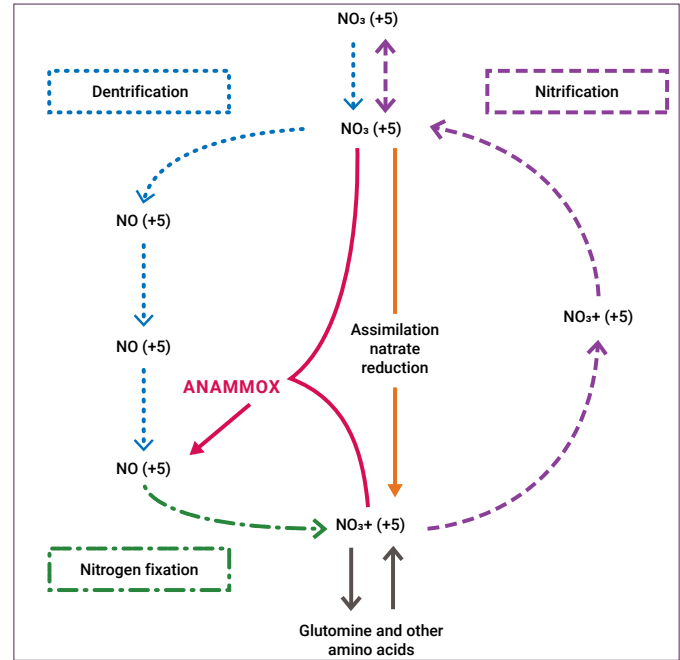
- ✓ using colours that give greater contrast and add outlines to define shapes
- ✓ using varying stroke styles



✗ Avoid relying on colour solely to link content to a



- ✓ using an alternative layout such as grouping or labels instead of relying on colour to link the content to areas of the diagram



✗ Avoid using colour to link the label to areas of the

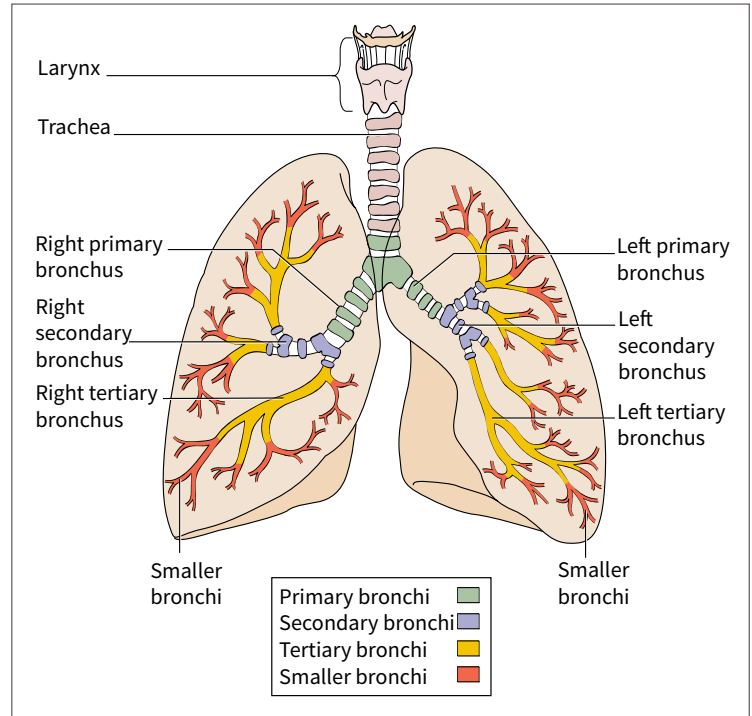
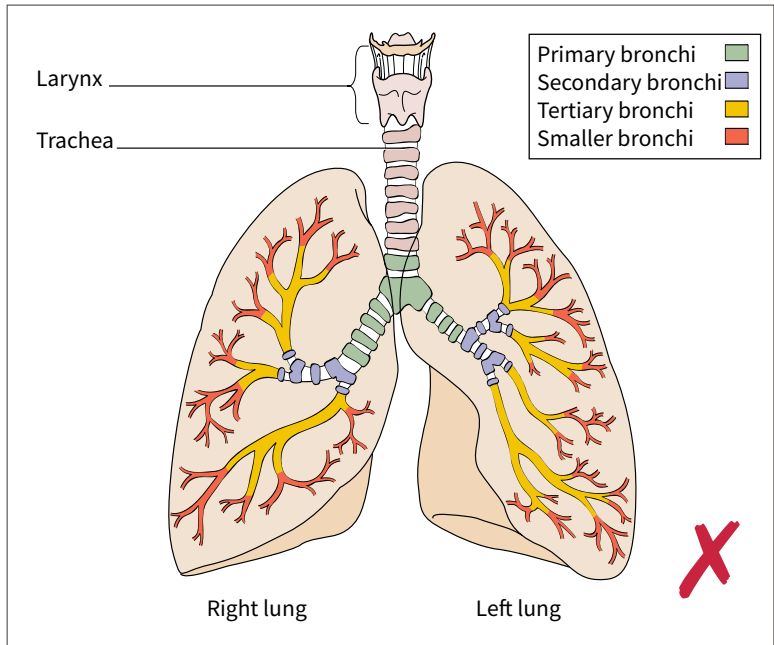


Figure accessibility and alt text

The aim of alt text is to provide concise and informative descriptions of your figure so that all readers have access to the same level of information and understanding, and that all can engage with and benefit from the visual elements integral to scholarly content.

Including alt text demonstrates a commitment to accessibility and enhances the overall impact and reach of your work. Good alt text ensures that individuals with visual impairments or those using screen readers can comprehend the content and context of your figures.

✓ ACCESSIBLE

ALT TEXT: Graphs and data on trem2 expression in brain tumours, with subfigures labelled from a to f, illustrating statistical analyses and gene expression profiles.

Alt Text Example 1

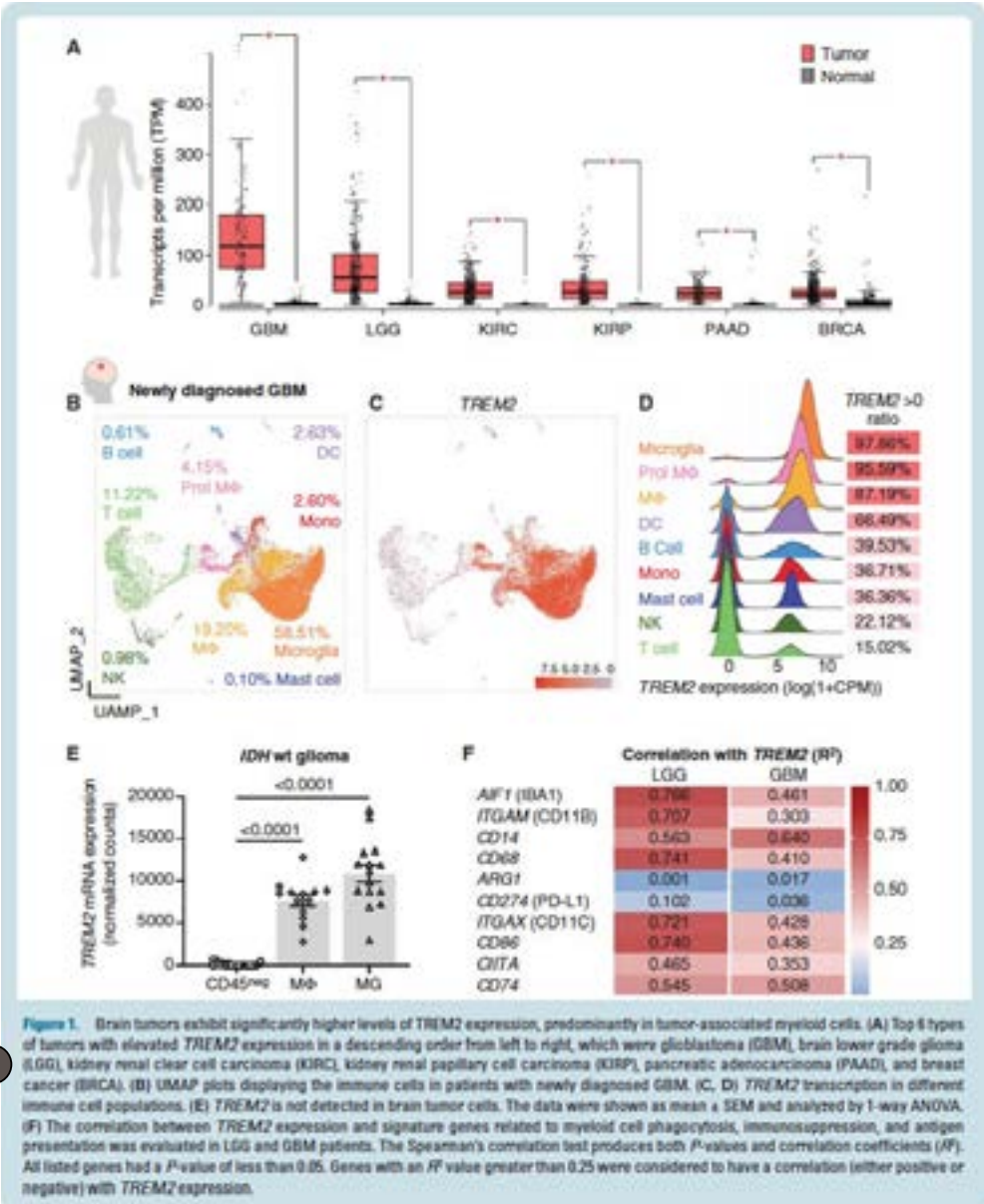


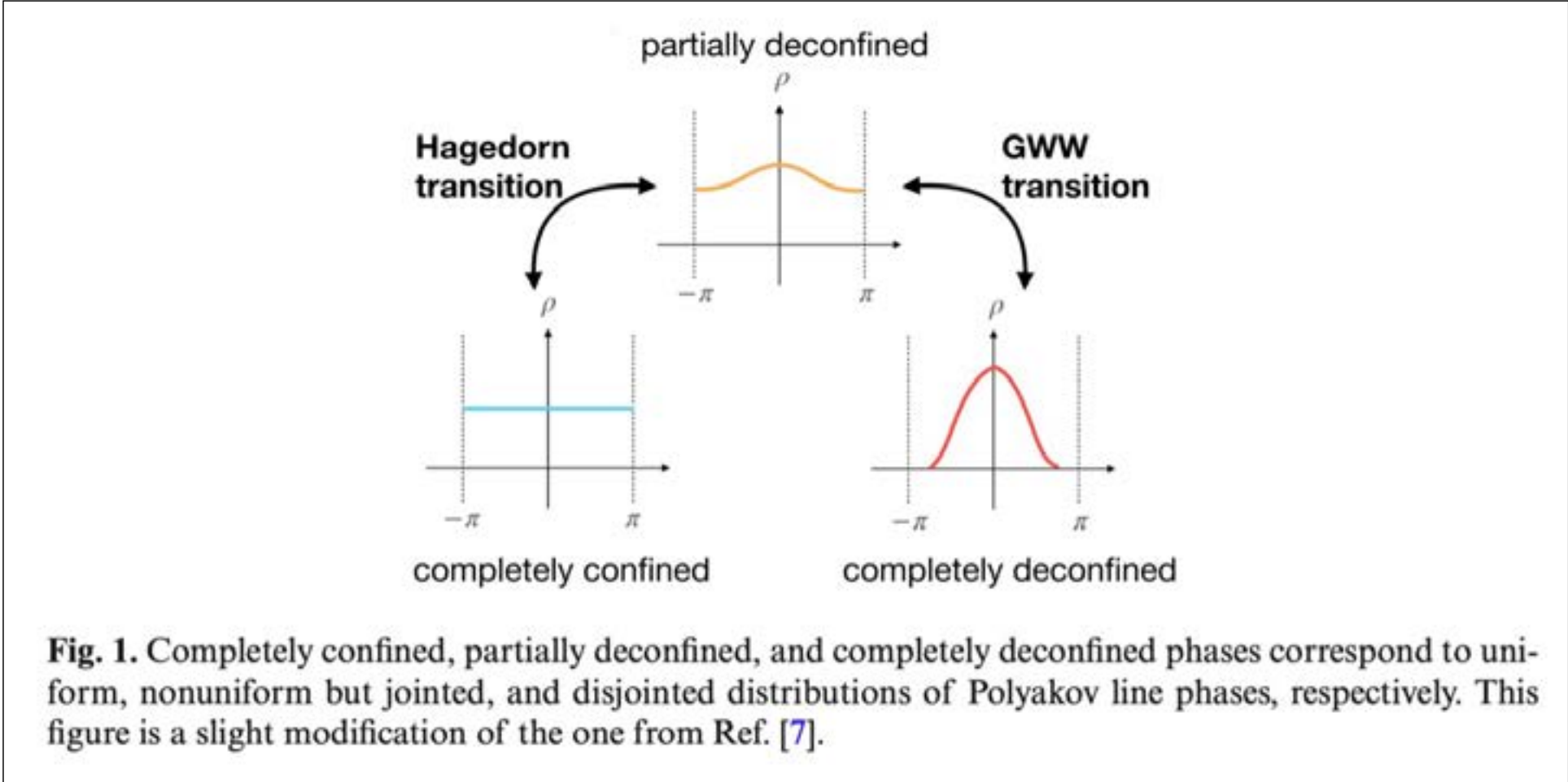
Figure 1. Brain tumors exhibit significantly higher levels of TREM2 expression, predominantly in tumor-associated myeloid cells. (A) Top 6 types of tumors with elevated *TREM2* expression in a descending order from left to right, which were glioblastoma (GBM), brain lower grade glioma (LGG), kidney renal clear cell carcinoma (KIRC), kidney renal papillary cell carcinoma (KIRP), pancreatic adenocarcinoma (PAAD), and breast cancer (BRCA). (B) UMAP plots displaying the immune cells in patients with newly diagnosed GBM. (C, D) *TREM2* transcription in different immune cell populations. (E) *TREM2* is not detected in brain tumor cells. The data were shown as mean $\pm$ SEM and analyzed by 1-way ANOVA. (F) The correlation between *TREM2* expression and signature genes related to myeloid cell phagocytosis, immunosuppression, and antigen presentation was evaluated in LGG and GBM patients. The Spearman's correlation test produces both *P*-values and correlation coefficients (*R*²). All listed genes had a *P*-value of less than 0.05. Genes with an *R*² value greater than 0.25 were considered to have a correlation (either positive or negative) with *TREM2* expression.

SOURCE: Zheng, J., Wang, L., Zhao, S., Zhang, W., Chang, Y., Bosco, D. B., Huang, T., Dheer, A., Gao, S., Xu, S., Ayasoufi, K., Al-Kharboosh, R., Qi, F., Xie, M., Johnson, A. J., Dong, H., Quiñones-Hinojosa, A., & Wu, L.-J. (2023). Trem2 mediates MHCII-associated CD4+ T-cell response against gliomas. *Neuro-Oncology*. <https://doi.org/10.1093/neuonc/noad214>

Good alt text should be objective and:

- **CONCISE:** Given the time-consuming nature of screen reader use, strive for alt text under 100 words, and preferably around 25 to 30 words.
- **CONSISTENT:** Maintain language consistency with the main body of text.
- **UNIQUE:** Avoid repeating information found in captions or surrounding text. For images entirely explained by captions or text, consider labeling them as decorative indicating that their purpose is primarily ornamental rather than conveying additional information.
- **CLEAR:** Spell out contractions, abbreviations, numbers, and non-Latin characters. Present information in a logical and consistent sequence.
- **RELEVANT:** Align descriptions with the image's context, intent, and the focus of the text, and title. Different purposes within a work may warrant distinct alt text descriptions.
- **SIMPLE:** Because screen readers don't interpret formatting, refrain from using formatting (e.g., bullet points) in alt text.
- **INCLUSIVE:** Ensure your text doesn't contain additional information that a sighted person would miss.
- **STANDALONE:** Screen readers indicate that alt text replaces an image, so avoid phrases like "Image of..."
- **COMPLETE:** Conclude alt text with a full stop/period, allowing for a pause before the screen reader continues.

Alt Text Example 2



ACCESSIBLE

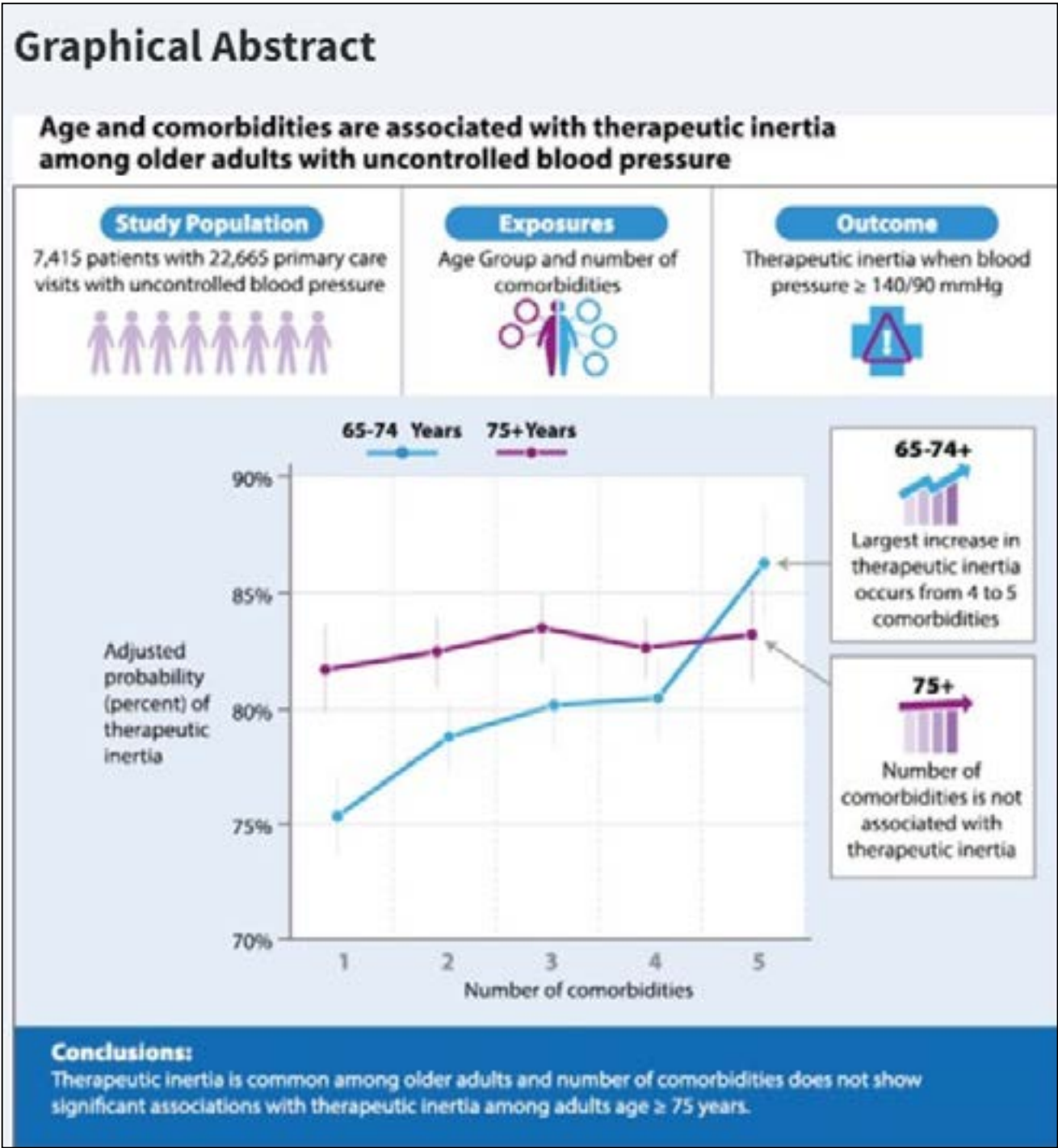
ALT TEXT: Graphical representation of three phases—completely confined, partially deconfined, and completely deconfined—depicting Polyakov line phases with corresponding transition profiles.

SOURCE: Hanada, M., & Watanabe, H. (2024). On thermal transition in QCD. Progress of Theoretical and Experimental Physics, 2024(4). <https://doi.org/10.1093/ptep/ptae033>

Additional considerations:

- **Grouped images and multi-panel figures**
When images are grouped, provide alt text for individual figures rather than a collective description.
- **Punctuation**
Screen readers are unable to interpret all types of punctuation, so only use the following: periods, commas, semi-colons, colons, parentheses, brackets, quotation mark, and sashes.
- **Art works**
Although starting with a generic “picture of” or “image of” is redundant, it can be beneficial to specify a particular type of image in art works (sculpture, painting, drawing, etc.).
- **Graphs**
Summarize the graph by emphasizing key details, trends, and relationships in the data. If significant, explicitly list underlying values in a logical order. Avoid redundancy if key values have already been discussed in the text. Use full units like “kilometers” for accurate pronunciation by screen readers.

Alt Text Example 3



✓ ACCESSIBLE

ALT TEXT: Graphical abstract showing study results where therapeutic inertia in blood pressure management differs by age group in older adults, with no significant association after age 75.

SOURCE: Kristen John, Jan Stenum, Cheng-Chuan Chiang, Margaret A French, Christopher Kim, John Manor, Matthew A Statton, Kendra M Cherry-Allen, Ryan T Roemmich, Accuracy of Video-Based Gait Analysis Using Pose Estimation During Treadmill Walking Versus Overground Walking in Persons After Stroke, Physical Therapy, Volume 104, Issue 2, February 2024, pzad121, <https://doi.org/10.1093/ptj/pzad121>

To Submit Alt Text

To submit alt text with your article, include it alongside figure captions as in the example below, related to the image in Example 4.

Figure 1 caption: L'Exposition Manet au Musée de l'Orangerie in 'Une Rétrospective d'Édouard Manet a Été Inaugurée Hier au Musée de l'Orangerie', L'Excelsior, 19 June 1932. (Photo: Bibliothèque Nationale de France.).

Alt text: Decorative photograph of a vintage French journal cover featuring an article from 1932.

Alt Text Example 4



SOURCE: Published in Michela Passini, A Microhistory of Heritage Creation Processes: The Impressionists Exhibited at the Orangerie (1930–1937), Oxford Art Journal, Volume 46, Issue 2, August 2023, Pages 241-259, <https://doi.org/10.1093/oxartj/kcad023>

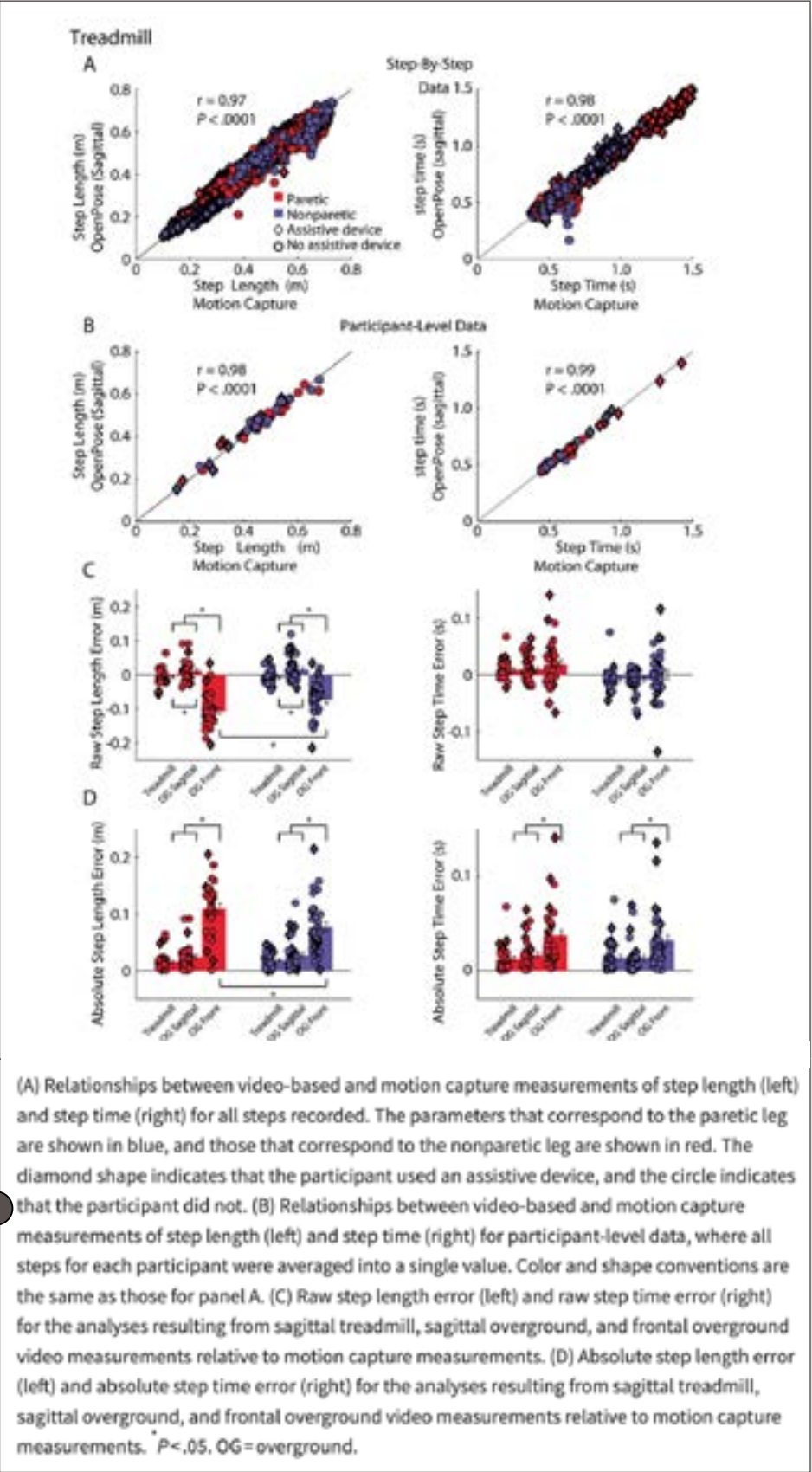


ALT TEXT: Graphs comparing biomechanical data from treadmill vs. overground walking, showing steps, device use, and stride length correlations, with detailed annotations and statistical values.

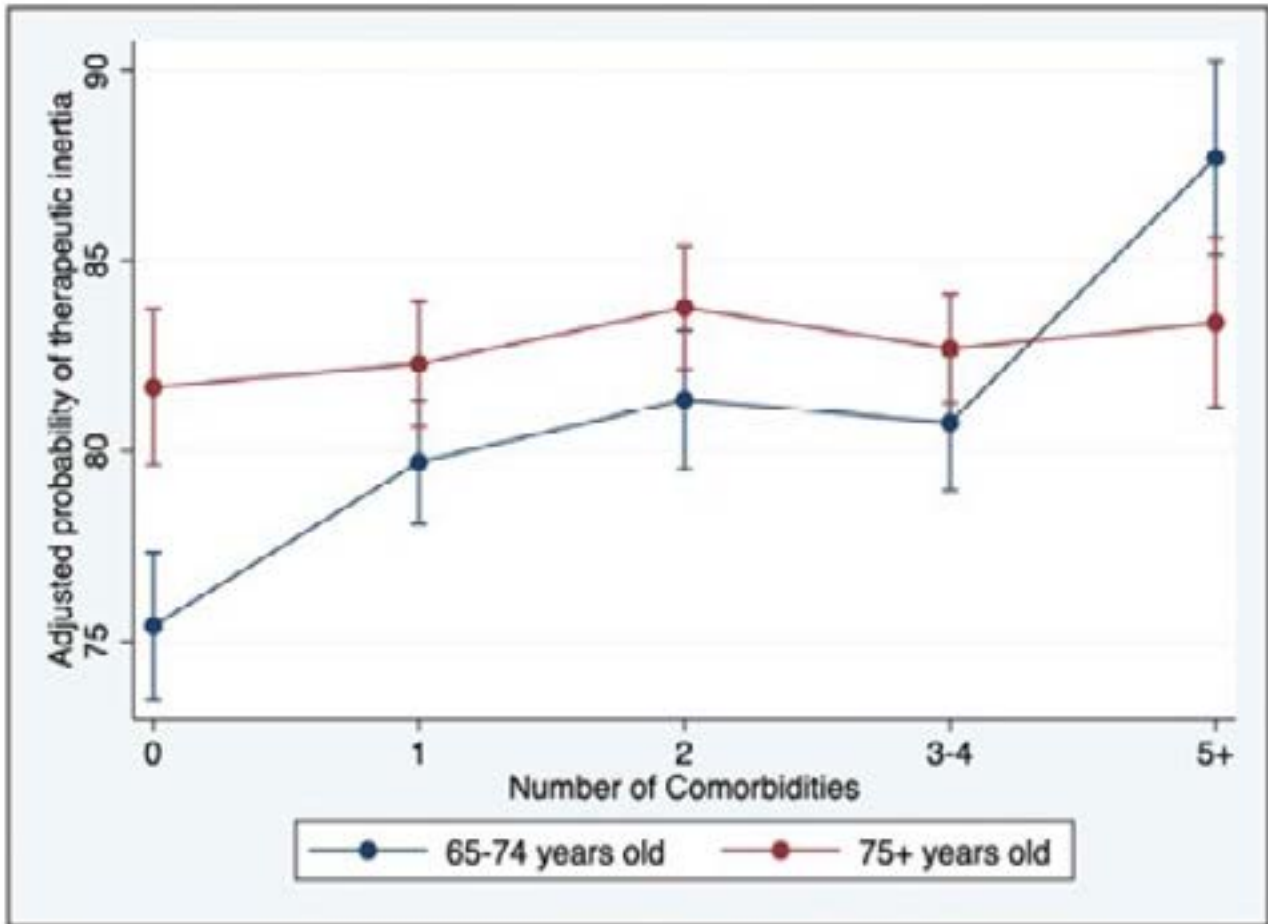
Alt Text Example 5

(A) Relationships between video-based and motion capture measurements of step length (left) and step time (right) for all steps recorded. The parameters that correspond to the paretic leg are shown in blue, and those that correspond to the nonparetic leg are shown in red. The diamond shape indicates that the participant used an assistive device, and the circle indicates that the participant did not. (B) Relationships between video-based and motion capture measurements of step length (left) and step time (right) for participant-level data, where all steps for each participant were averaged into a single value. Color and shape conventions are the same as those for panel A. (C) Raw step length error (left) and raw step time error (right) for the analyses resulting from sagittal treadmill, sagittal overground, and frontal overground video measurements relative to motion capture measurements. (D) Absolute step length error (left) and absolute step time error (right) for the analyses resulting from sagittal treadmill, sagittal overground, and frontal overground video measurements relative to motion capture measurements. * $P < .05$. OG = overground.

SOURCE: Kristen John, Jan Stenum, Cheng-Chuan Chiang, Margaret A French, Christopher Kim, John Manor, Matthew A Statton, Kendra M Cherry-Allen, Ryan T Roemmich, Accuracy of Video-Based Gait Analysis Using Pose Estimation During Treadmill Walking Versus Overground Walking in Persons After Stroke, Physical Therapy, Volume 104, Issue 2, February 2024, pzad121, <https://doi.org/10.1093/ptj/pzad121>



Alt Text Example 6



Open in new tab Download slide

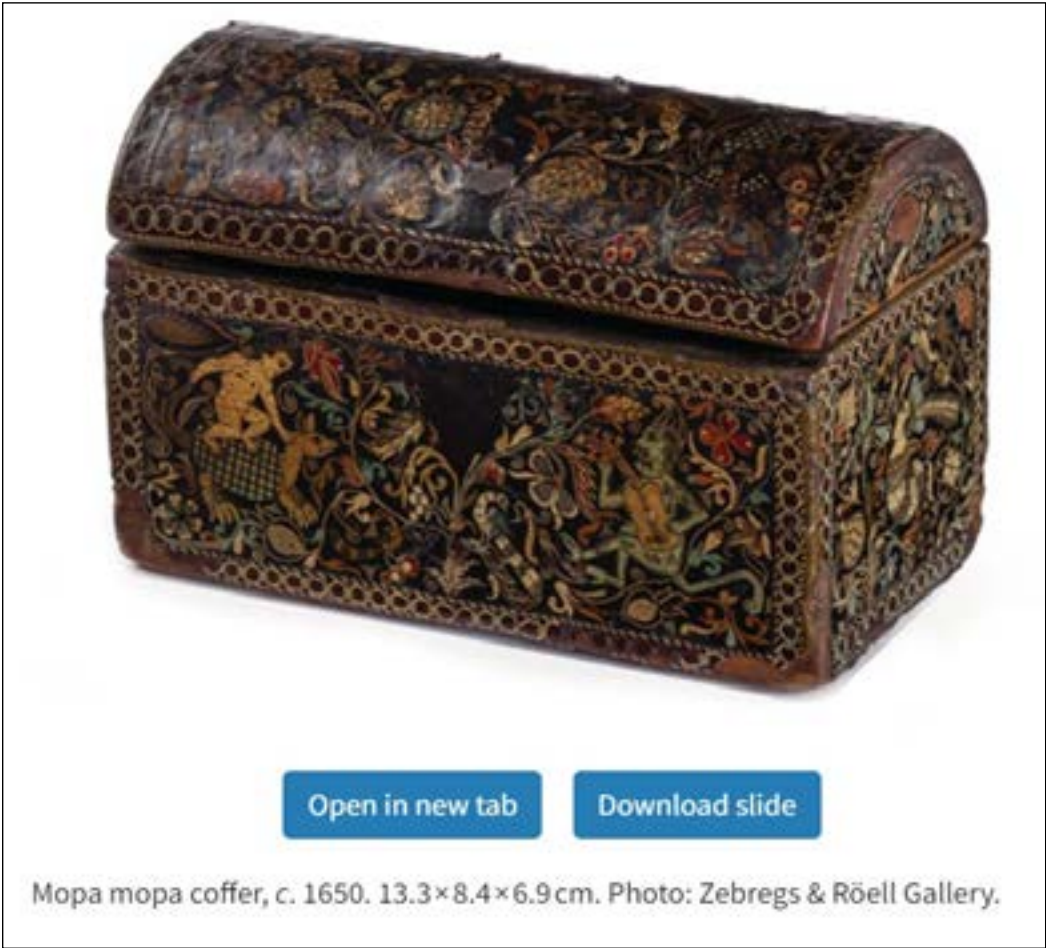
Adjusted probability of therapeutic inertia by age group and by number of comorbidities during 22,665 primary care visits with uncontrolled BP defined as SBP $\geq$ 140 mm Hg and/or DBP $\geq$ 90 mm Hg among 7,415 patients with a hypertension diagnosis. Probabilities were calculated using marginal effects after using generalized linear mixed-effects models with a binomial distribution stratified by age group (65–74 and $\geq$ 75 years) and adjusted for self-reported race and ethnicity, body mass index, smoking status, clinic site, average time between visits, insurance status, marital status, classes of antihypertensive medications, hypertension severity at the clinic visit, and patient random effects. Abbreviations: BP, blood pressure; DBP, diastolic blood pressure; SBP, systolic blood pressure.

✓ ACCESSIBLE

ALT TEXT: Chart depicting the percentage of therapeutic inertia in hypertension by age and number of comorbidities, showing higher inertia in older age groups and with more comorbidities.

SOURCE: Grant T Hiura, Talar W Markossian, Beatrice D Probst, Mohammad Samie Tootooni, Gregory Wozniak, Michael Rakotz, Holly J Kramer, Age and Comorbidities Are Associated With Therapeutic Inertia Among Older Adults With Uncontrolled Blood Pressure, *American Journal of Hypertension*, Volume 37, Issue 4, April 2024, Pages 280–289, <https://doi.org/10.1093/ajh/hpad108>

Alt Text Example 7

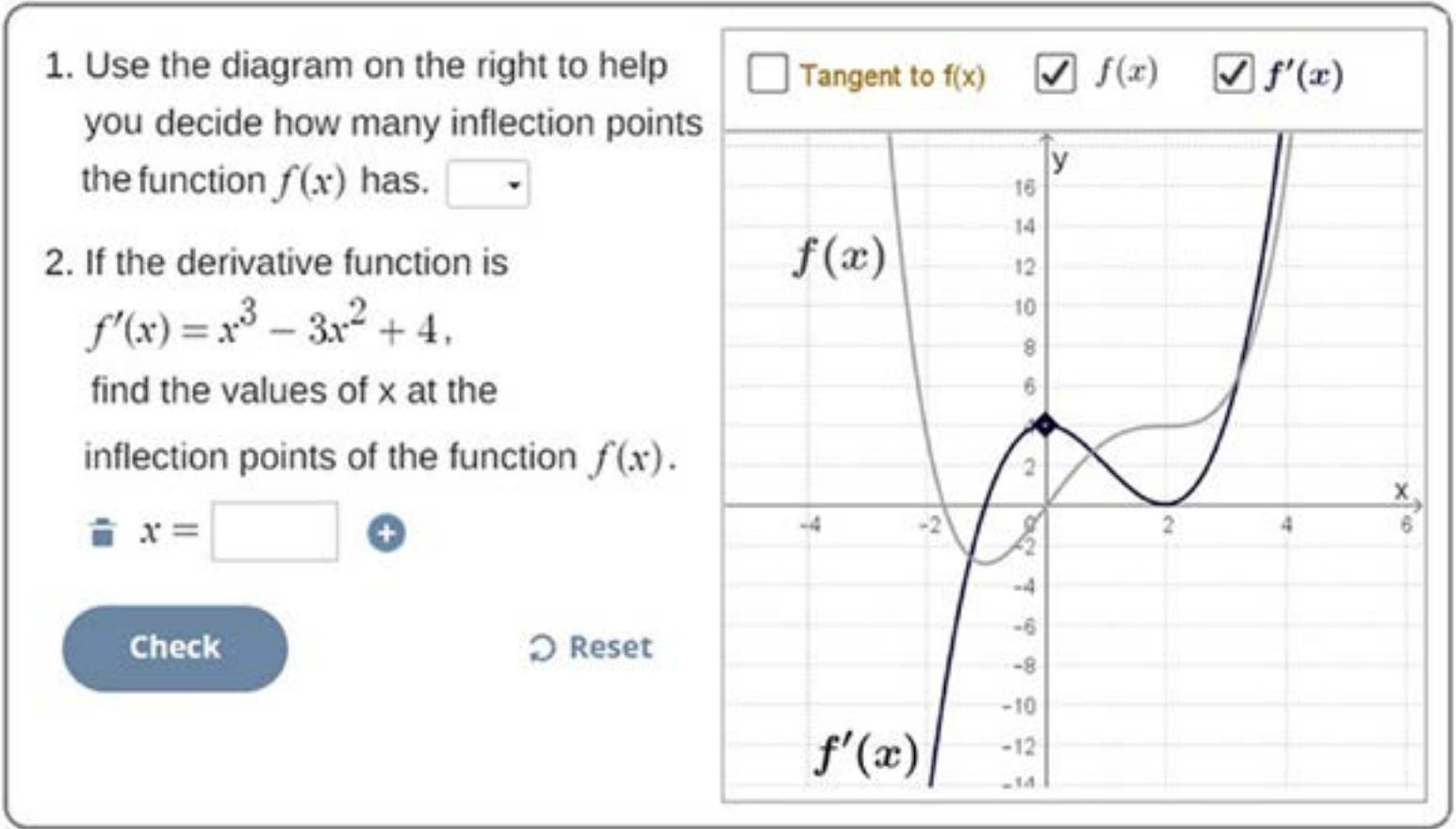


✓ ACCESSIBLE

ALT TEXT: An intricately decorated antique chest with floral and animal motifs, featuring rich colors and a curved lid.

SOURCE: Catalina Ospina, Indigenous Knowledge and Women’s Cosmetics: Mopa Mopa in the Colonial Northern Andes, Art History, Volume 46, Issue 5, November 2023, Pages 918–945, <https://doi.org/10.1111/1467-8365.12756>

Alt Text Example 8



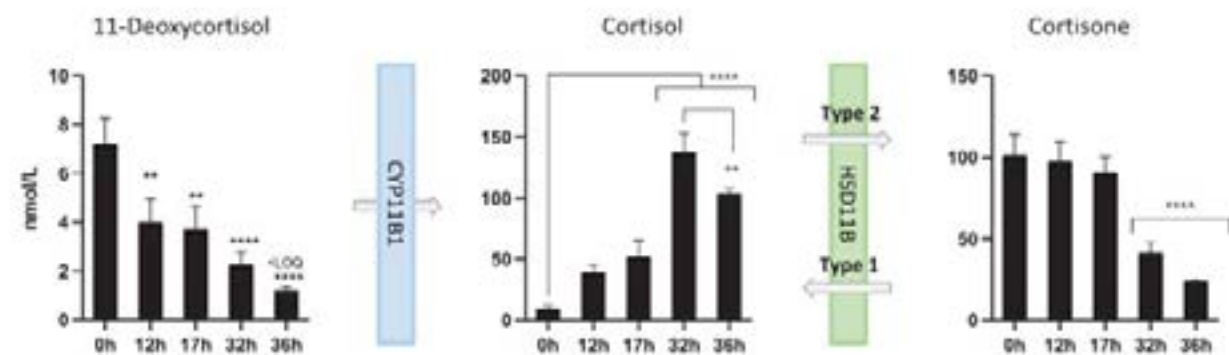
✓ ACCESSIBLE

ALT TEXT: Educational diagram showing a function f(x) graph with a tangent line to help find inflection points, accompanied by an interactive calculation tool on left.

SOURCE: Anatoli Kouropatov, Regina Ovodenko, An explorative digital tool as a pathway to meaning: the case of the inflection point, Teaching Mathematics and its Applications: An International Journal of the IMA, Volume 41, Issue 2, June 2022, Pages 142–166, <https://doi.org/10.1093/teamat/hrac007>

Alt Text Example 9

Figure 2. Concentration of 11-deoxycortisol, cortisol, and cortisone measured in follicular fluid in nmol/l (nM) for each of the 5 measured time point. All data points were expressed as average with SEMs. Significant values are indicated by an asterisk: **P < 0.01, ****P < 0.0001.



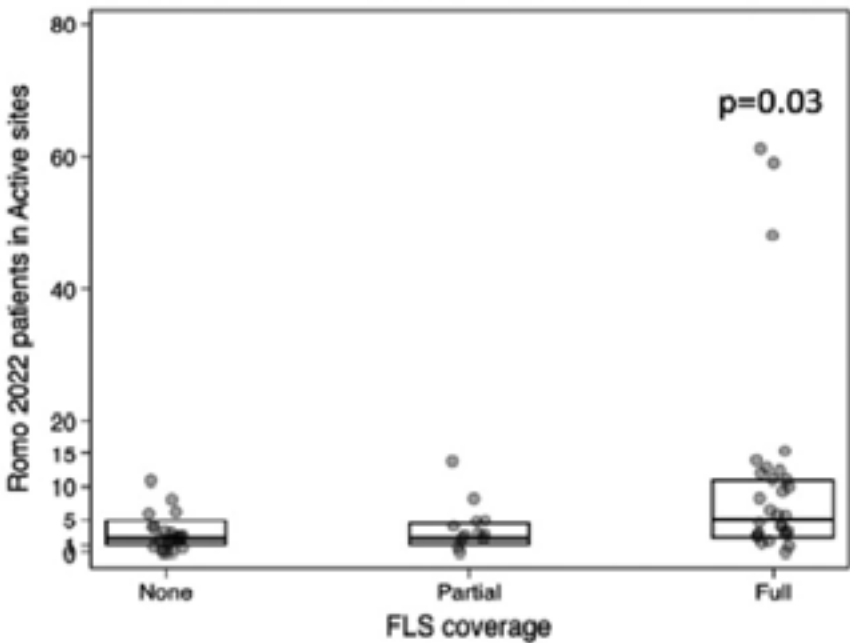
ACCESSIBLE

ALT TEXT: Bar graph showing concentrations of 11-deoxycortisol and cortisol in follicular fluid at different time points, with significance markers.

SOURCE: Hanada, M., & Watanabe, H. (2024). On thermal transition in QCD. Progress of Theoretical and Experimental Physics, 2024(4). <https://doi.org/10.1093/ptep/ptae033>

Alt Text Example 10

Association between FLS status and number of romosozumab prescription in 2022

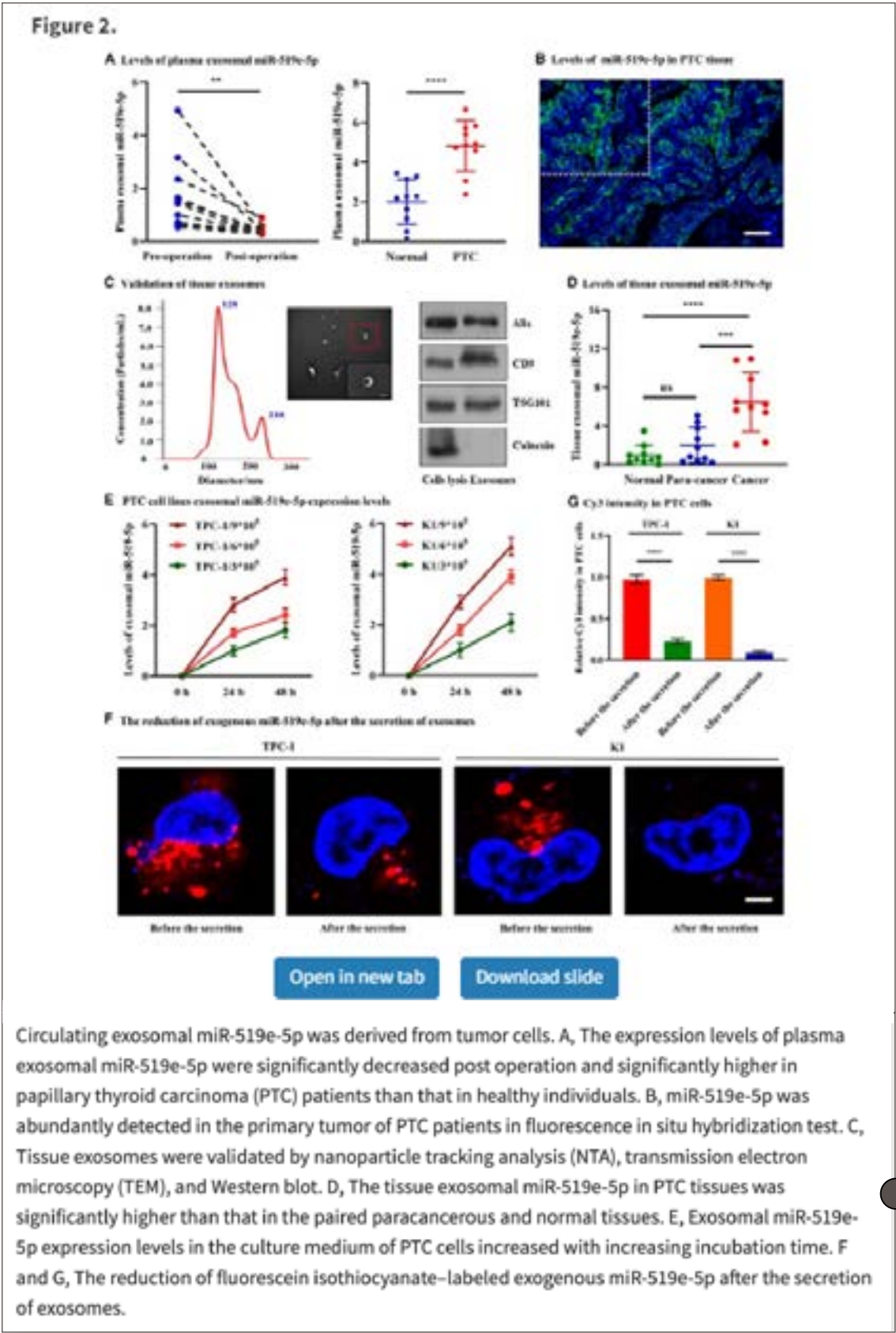


ACCESSIBLE

ALT TEXT: Graph depicting the relationship between fls coverage levels and number of romosozumab prescriptions in 2022, illustrated with box plots and individual data points.

SOURCE: Muhammad K Javaid, Tim Jones, Stephn Hepworth, P078 United Kingdom rates and determinants of romosozumab prescriptions, a need for greater action, Rheumatology, Volume 63, Issue Supplement_1, April 2024, keae163.119, <https://doi.org/10.1093/rheumatology/keae163.119>

Alt Text Example 11



ACCESSIBLE

ALT TEXT: Graphs and microscopic images labelled A to F. A is graph plotting particle count vs. diameter, microscopy image of exosomes, and a western blot analysis for various markers. B is a heatmap showing data across 14 plasma samples with red, green, and black color coding, indicating varying levels of measured variables. C is a Dot plot displaying pathway enrichment differences, with pathways labeled on the y-axis and logp values on the x-axis, using a color gradient to denote significance levels. D is Bar graph comparing levels of exosomal mirnas between pic-m0 and pic-m1 groups, with statistical significance indicators. E is Graph showing roc curves for exosomal mirnas, with lines representing different mirnas annotated with auc values, and a diagonal reference line indicating chance-level performance. F is fluorescence microscopy images showing association of exosomal mir-519c-5p with circulating tumor cells (ctcs), labeled with dapi, cd45, and epcam markers.

Circulating exosomal miR-519e-5p was derived from tumor cells. A, The expression levels of plasma exosomal miR-519e-5p were significantly decreased post operation and significantly higher in papillary thyroid carcinoma (PTC) patients than that in healthy individuals. B, miR-519e-5p was abundantly detected in the primary tumor of PTC patients in fluorescence in situ hybridization test. C, Tissue exosomes were validated by nanoparticle tracking analysis (NTA), transmission electron microscopy (TEM), and Western blot. D, The tissue exosomal miR-519e-5p in PTC tissues was significantly higher than that in the paired paracancerous and normal tissues. E, Exosomal miR-519e-5p expression levels in the culture medium of PTC cells increased with increasing incubation time. F and G, The reduction of fluorescein isothiocyanate-labeled exogenous miR-519e-5p after the secretion of exosomes.

SOURCE: Genpeng Li, Wenjie Chen, Ke Jiang, Jing Huang, Jinjing Zhong, Xiaowei Liu, Tao Wei, Rixiang Gong, Zhihui Li, Jingqiang Zhu, Hubing Shi, Jianyong Lei, Exosome-mediated Delivery of miR-519e-5p Promotes Malignant Tumor Phenotype and CD8+ T-Cell Exhaustion in Metastatic PTC, The Journal of Clinical Endocrinology & Metabolism, 2023;, dgad725, <https://doi.org/10.1210/clinem/dgad725>